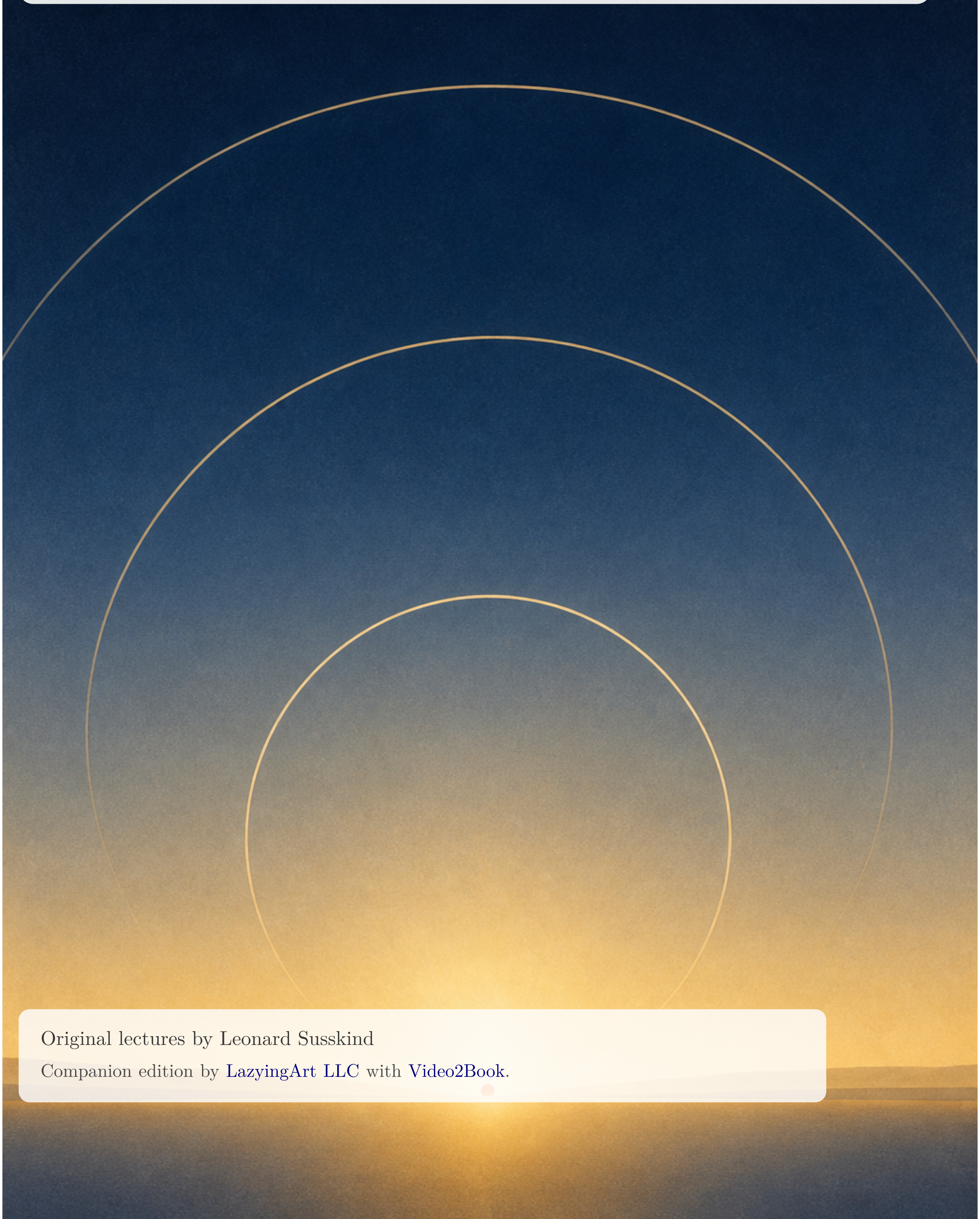
Cosmology

Core track, 2013 Winter Theoretical Minimum



Original lectures by Leonard Susskind

Companion edition by LazyingArt LLC with Video2Book.

About These Notes

These independently edited companion notes follow lectures by Leonard Susskind. They were reconstructed from machine transcripts, subtitles, and selected blackboard frames, then checked against the available source material. They are not an original manuscript by Professor Susskind and have not been reviewed or endorsed by him.

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Chapter 1

The Expanding Newtonian Universe

Overview. Modern cosmology begins when the universe itself is treated as a physical system. Starting from large-scale isotropy and homogeneity, we introduce a comoving grid and encode its motion in a single scale factor $a(t)$. Newton's shell theorem then turns the gravitational many-body problem into an equation for $a(t)$. Energy conservation distinguishes recollapsing, critical, and eternally expanding universes, and the critical matter-dominated case gives the first Friedmann equation and the law $a(t) \propto t^{2/3}$.

1.1 When Cosmology Became Physics

Cosmology is ancient as a subject, but quantitative cosmology is young. For the present purpose there is no need to begin with the Greeks. The useful historical starting points are Hubble's discovery of cosmic expansion in the second quarter of the twentieth century and, later, the recognition of the roughly three-kelvin microwave background as a remnant of a hot early universe.

Before measurements became precise, much of astronomy necessarily resembled natural history. One found an unusual star, classified a peculiar galaxy, measured what could be measured, and assembled a catalogue of objects. Those objects were certainly physical systems: they carried angular momentum, contained chemical elements, and obeyed mechanics. The decisive change was to treat the *whole universe* as one physical system, governed by equations accurate enough to confront observation.

That is the subject here. We shall not merely list the contents of the universe; we shall ask for its variables, its motion, and its dynamical law. Equations are not an ornament added after the cosmology. They are the means by which the universe becomes a physics problem.

Historically, the expanding universe was understood in the language of general relativity. Logically, however, much of the first argument is Newtonian. A seventeenth-century physicist who had guessed the right large-scale motion could have derived a remarkably good model. The point of beginning with Newton is not to replace relativity, but to expose the mechanics that relativity later reorganizes geometrically.

1.2 Isotropy, Homogeneity, and the Cosmological Principle

The first step is observational. On sufficiently large scales the universe is approximately *isotropic*: after averaging over patches of sky and looking beyond the foreground of the Milky Way, it looks much the same in every direction. This is not a statement of exact uniformity. Looking directly at a nearby star is different from looking into an empty patch, and the distribution of matter contains real structure. Isotropy is an averaged large-scale statement.

Homogeneity is a different statement. It says that the averaged universe is the same from place to place. An observer many galaxies away should see, after the same large-scale averaging, approximately the same kind of universe that we see.

For the first approximation, a galaxy may be treated simply as a mass point. The visible universe contains roughly

$$N_{\text{gal}} \sim 10^{11} \tag{1.1}$$

galaxies, each with of order 10^{11} stars. This gives approximately 10^{22} stars. If one uses ten planets per star as a deliberately rough order-of-magnitude estimate, one arrives at 10^{23} planets, amusingly close to a mole of planets. The numerical estimate is not part of the dynamics, but it fixes the scale of the system being idealized.

> **Question for the class.** How many galaxies are there within the observable universe, to order of magnitude?
>
> **Response.** About one hundred billion, or 10^{11}. With roughly 10^{11} stars per galaxy, that means approximately 10^{22} stars in the visible universe. These are useful scale estimates to keep in mind even though the calculation below replaces each galaxy by a single mass point. [1]

Why should isotropy around us suggest homogeneity? Imagine an isotropic distribution that is *not* homogeneous. Its density would have to vary with distance from us while remaining independent of direction: in effect, it would form a pattern of concentric shells about our position. An observer displaced from the common center would no longer see isotropy. Thus there are two possibilities:

1. we occupy a distinguished center about which the universe has been arranged; or
2. the large-scale distribution is approximately homogeneous, so no occupied point is distinguished.

The second possibility is the economical one. Isotropy, together with the refusal to assign ourselves a privileged cosmic position, leads to homogeneity.

> **Question for the class.** What form could an isotropic but inhomogeneous distribution take?
>
> **Response.** Its variation could depend only on distance from us, producing a shell-like pattern centered on our position. Move the observer away from that center and the pattern ceases to be isotropic. Unless our location is special, isotropy therefore points to homogeneity. [2]

[1] Lecture timestamp: 00:06:08–00:07:05.
[2] Lecture timestamp: 00:07:06–00:08:54.

The combined assumption is called the *cosmological principle*: on sufficiently large scales, the universe is spatially homogeneous and isotropic. Calling it a principle does not prove it. Historically, cosmologists adopted the assumption before observations justified it strongly. Its standing comes from empirical success.

Nor is homogeneity exact. Galaxies form clusters and superclusters, and there have long been discussions of structures extending across very large fractions of the visible sky. The appropriate claim is that fluctuations average away when the averaging region is sufficiently large—of order hundreds of millions to roughly a billion light-years in this introductory discussion. The scale may be refined by observation; the method is to coarse-grain until local structure no longer dominates the description.

> **Question from the audience.** Does the uniformity of the cosmic microwave background provide evidence for the cosmological principle?
>
> **Response.** Yes. The cosmological principle was proposed before there was strong observational warrant for it, but the microwave background later showed that the primordial universe was extraordinarily smooth. Galaxies and clusters grew from small departures from that smooth state; they do not erase the large-scale evidence for it. [3]

1.3 A Grid That Moves with the Galaxies

Once the observational approximation is chosen, we must choose variables. One classroom answer to the question "What is the first step in formulating a physics problem?" is to know the variables; the traditional answer is to sharpen one's pencil. Here the important variable is found by choosing coordinates well.

Imagine a three-dimensional grid carrying labels x, y, z. These labels need not be distances measured in meters. They are coordinate numbers. Choose the grid so that, after small peculiar motions have been averaged away, every galaxy remains at a fixed grid point. The galaxies are then *comoving*.

Such a choice would be useless if galaxies moved randomly. Observation instead reveals a coherent large-scale motion. Neighboring grid cells may expand or contract, but the same expansion or contraction occurs everywhere. All physical distances can therefore be written as a common time-dependent scale multiplied by a fixed coordinate separation.

For two galaxies A and B,

$$\begin{aligned} d_{AB}(t) &= a(t)\,\Delta\chi_{AB}, \\ \Delta\chi_{AB} &= \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2 + (z_A - z_B)^2}. \end{aligned} \tag{1.2}$$

If they differ only in the x-coordinate, this reduces to

$$d_{AB}(t) = a(t)\,\Delta x_{AB}. \tag{1.3}$$

The function $a(t)$ is the *scale factor*. Its absolute normalization is conventional: multiplying all comoving coordinates by a constant and dividing a by the same constant changes no physical distance. Ratios such as $a(t_2)/a(t_1)$, and derivatives divided by a, carry the physical information.

[3] Lecture timestamp: 00:10:53–00:11:43.

Differentiate equation (1.2). The comoving separation is fixed, so

$$v_{AB} = \dot{a}\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \tag{1.4}$$

Dividing by the physical distance gives

$$\frac{v_{AB}}{d_{AB}} = \frac{\dot{a}}{a} \equiv H(t), \tag{1.5}$$

or

$$\boxed{v_{AB} = H(t)d_{AB}.} \tag{1.6}$$

This is Hubble's law in the ideal homogeneous flow.

The name "Hubble constant" can mislead. At one cosmic time the same value of H applies at every place in the homogeneous model, but $H(t)$ generally changes with time. Spatial universality is not temporal constancy.

> **Question from the audience.** By writing every distance as one common $a(t)$ times a fixed coordinate separation, have we already assumed Hubble's law?
>
> **Response.** The ansatz is certainly motivated by Hubble's observation; it would not be guessed without evidence for coherent recession. Its value is to organize that evidence. Instead of assigning an independent velocity to every galaxy, one describes a single expanding grid. Once that description is adopted, the proportionality $v = Hd$ follows by differentiation. [4]

The same motion admits two verbal descriptions. One may say that galaxies move apart through space, or that galaxies remain at fixed comoving positions while the grid—and hence physical space between them—expands. At this stage these are two coordinate descriptions of the same mathematics. General relativity will make the expanding-space language especially natural, but no observation can distinguish two descriptions that assign the same physical separations at every time.

1.4 Mass Bookkeeping in a Comoving Cell

Before introducing gravity, keep track of mass. Consider a small coordinate cell with coordinate volume

$$\Delta V_{\text{com}} = \Delta x\,\Delta y\,\Delta z. \tag{1.7}$$

It must still be large enough to contain many galaxies, so that local fluctuations may be averaged. Let ν denote mass per unit comoving coordinate volume. The mass in the cell is

$$\Delta M = \nu\,\Delta x\,\Delta y\,\Delta z. \tag{1.8}$$

Because the cell moves with the matter, no galaxy crosses its boundary in the ideal flow, and ΔM is constant.

Each physical edge is longer than its coordinate edge by a factor $a(t)$. The physical volume is therefore

$$\Delta V_{\text{phys}} = a^3(t)\,\Delta x\,\Delta y\,\Delta z. \tag{1.9}$$

[4] Lecture timestamp: 00:24:20–00:25:33.

The physical mass density is

$$\boxed{\rho(t) = \frac{\Delta M}{\Delta V_{\text{phys}}} = \frac{\nu}{a^3(t)}.} \tag{1.10}$$

Expansion dilutes pressureless matter as a^{-3}; contraction concentrates it by the same law.

Up to this point nothing dynamical has been assumed. We have used Euclidean geometry, a kinematic ansatz, and conservation of mass in a comoving cell. Now gravity enters.

1.5 Now Enters Newton

A tempting but incorrect picture says that an infinite uniform distribution of matter should remain static because the pull from one side is canceled by the pull from the other. The flaw is that this informal cancellation does not supply a consistent equation for relative acceleration. Newton's shell theorem gives the correct organization of the force.

Choose any galaxy as an origin O, and consider a galaxy A at fixed comoving radius

$$R = \sqrt{x^2 + y^2 + z^2}. \tag{1.11}$$

Its physical radius, radial velocity, and radial acceleration are

$$d = aR, \qquad \dot{d} = \dot{a}R, \qquad \ddot{d} = \ddot{a}R. \tag{1.12}$$

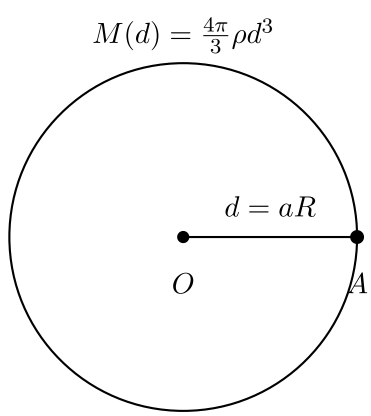

Figure 1.1: The Newtonian reduction. For the acceleration of galaxy A, the exterior shells cancel and the mass inside radius d acts as though concentrated at O.

For a spherically symmetric mass distribution, the net force from matter outside the sphere through A vanishes. The matter inside acts as though it were concentrated at the center. The enclosed mass is

$$M(d) = \frac{4\pi}{3}\rho d^3. \tag{1.13}$$

Newton's inverse-square law then gives the acceleration of A:

$$\ddot{d} = -\frac{GM(d)}{d^2}. \tag{1.14}$$

Using equation (1.12),

$$\ddot{a}R = -\frac{G}{d^2}\left(\frac{4\pi}{3}\rho d^3\right) \tag{1.15}$$

$$= -\frac{4\pi G}{3}\rho d \tag{1.16}$$

$$= -\frac{4\pi G}{3}\rho a R. \tag{1.17}$$

The comoving radius cancels, leaving

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\rho.} \tag{1.18}$$

This cancellation is the consistency test. Every comoving galaxy, regardless of R, obeys the same equation for the same scale factor.

Substituting equation (1.10) gives either of the equivalent forms

$$\frac{\ddot{a}}{a} = -\frac{4\pi G\nu}{3a^3}, \qquad \ddot{a} = -\frac{4\pi G\nu}{3a^2}. \tag{1.19}$$

The first form displays fractional acceleration; the second isolates $\ddot{a}$. Keeping a on the left is a convention, not a change of physics.

Question from the audience. The origin was arbitrary, yet every force in the calculation points toward it. How can a different choice of origin give the same universe?

Response. The equation concerns relative motion. Repeating the calculation around any other galaxy gives the same equation because homogeneity makes the enclosed mass depend only on physical radius. If one transforms to an accelerating frame attached to another galaxy while retaining forces referred to the first origin, the corresponding inertial force must also be included. Choosing a central rest frame merely avoids carrying that extra bookkeeping. The invariant result is equation (1.18), not a distinguished center. [5]

Question from the audience. Would the result survive if the comoving mass density ν varied from place to place?

Response. Not in this one-function form. The clean cancellation depends on $M \propto d^3$ with the same proportionality everywhere. If ν varied appreciably across the averaging scale, different regions would require different local dynamics and a single homogeneous scale factor would no longer describe them. Homogeneity is doing essential work. [6]

Question from the audience. Why write $\ddot{a}/a$ rather than move the factor a to the right?

Response. Either form is correct. Cosmology conventionally writes the fractional acceleration $\ddot{a}/a$, partly because it can be compared directly with the fractional expansion rate $H = \dot{a}/a$. Multiplying through by a gives the identical differential equation. [7]

[5] Lecture timestamp: 00:43:56–00:46:58.
[6] Lecture timestamp: 00:47:05–00:47:51.
[7] Lecture timestamp: 00:50:36–00:51:29.

Equation (1.18) immediately rules out a static homogeneous Newtonian universe filled with ordinary matter. A static scale factor would have $\ddot{a} = 0$, whereas positive density makes the right-hand side negative. This does *not* yet tell us whether the universe is expanding or contracting. It tells us only the sign of the acceleration.

1.6 Negative Acceleration Is Not Contraction

The distinction between velocity and acceleration is elementary but decisive. A Ferrari moving forward on the Autobahn can have positive velocity while braking with negative acceleration. Likewise, a universe can have

$$\dot{a} > 0, \qquad \ddot{a} < 0 : \tag{1.20}$$

it is expanding, but the expansion is slowing.

To decide whether the motion eventually reverses, compare the cosmological problem with a test particle moving radially in the gravitational field of the Earth. Let x now denote the particle's physical distance from a central mass M; it is unrelated to the earlier comoving coordinate. Newton's equation is

$$\ddot{x} = -\frac{GM}{x^2}. \tag{1.21}$$

The acceleration always points inward. A particle thrown outward may nevertheless continue outward for a long time, or forever. A particle already moving inward acquires an increasingly negative velocity. The force law alone does not decide which motion occurs; initial conditions do.

Question from the audience. Does $\ddot{a} < 0$ mean that the universe is contracting?

Response. No. Contraction means $\dot{a} < 0$. Negative $\ddot{a}$ means only that $\dot{a}$ decreases with time. If $\dot{a}$ begins positive, gravity slows the expansion; whether it reaches zero depends on the initial expansion speed, exactly as the fate of an outward-moving projectile depends on whether it is below or above escape velocity. [8]

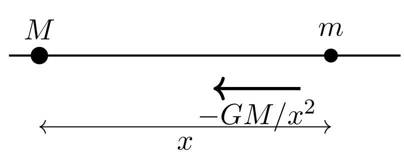

Figure 1.2: The radial one-particle problem used to separate the sign of acceleration from the direction and fate of motion.

Energy makes the alternatives transparent. The conserved mechanical energy is

$$E = \frac{1}{2}m\dot{x}^2 - \frac{GMm}{x}. \tag{1.22}$$

The potential energy is proportional to $1/x$, not $1/x^2$; differentiating $-GMm/x$ produces the inverse-square force.

[8] Lecture timestamp: 00:51:30–00:56:18.

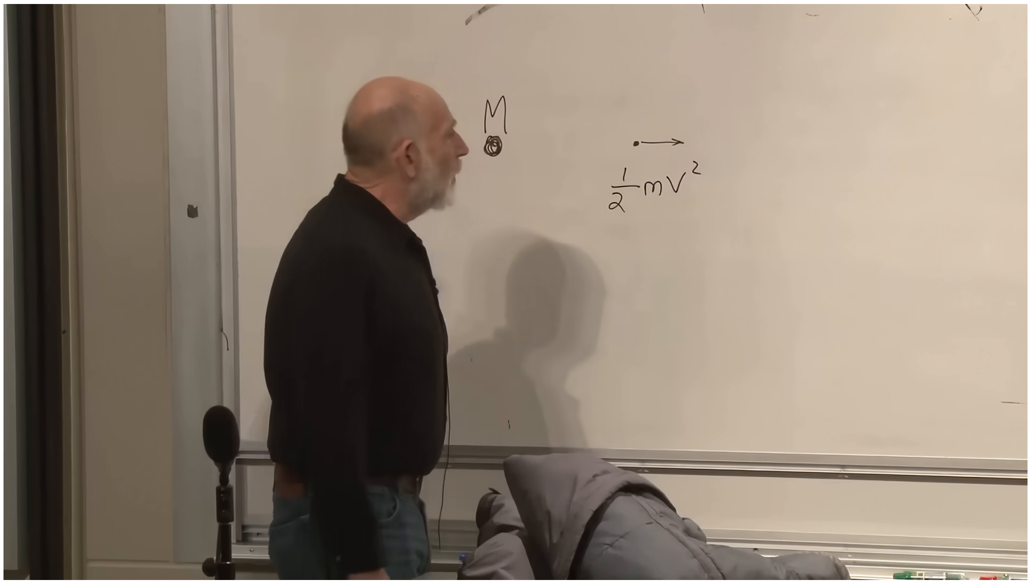

Figure 1.3: The board at the beginning of the energy argument: a moving test mass and its kinetic term, just before the gravitational potential energy is restored.

Lecture frame: 00:58:57.

If $E < 0$, an outward-moving particle cannot reach infinity, where its potential energy tends to zero and its kinetic energy is nonnegative. It must stop at finite x and return. If $E > 0$, it cannot stop, because at a turning point the kinetic energy would vanish and the remaining potential energy would be negative, contradicting conservation of a positive E. The boundary is

$$E = 0, \tag{1.23}$$

which defines escape velocity:

$$\frac{1}{2}mv_{\text{esc}}^2 - \frac{GMm}{x} = 0, \qquad \boxed{v_{\text{esc}}^2 = \frac{2GM}{x}.} \tag{1.24}$$

Exactly at escape velocity the particle slows indefinitely. Its speed tends to zero only as $x \to \infty$; it never reaches a finite turning point. This edge case will be the simplest cosmological model.

1.7 The Cosmological Energy Equation

Return to a galaxy at fixed comoving radius R. By the shell theorem its radial motion is identical to that of a particle moving outside a point mass equal to the mass enclosed by its comoving sphere.

The enclosed mass does not increase as the physical sphere expands. The sphere is comoving, so it always contains the same galaxies:

$$M = \frac{4\pi}{3}\rho(aR)^3 \tag{1.25}$$

$$= \frac{4\pi}{3}\frac{\nu}{a^3}a^3R^3 \tag{1.26}$$

$$= \frac{4\pi}{3}\nu R^3. \tag{1.27}$$

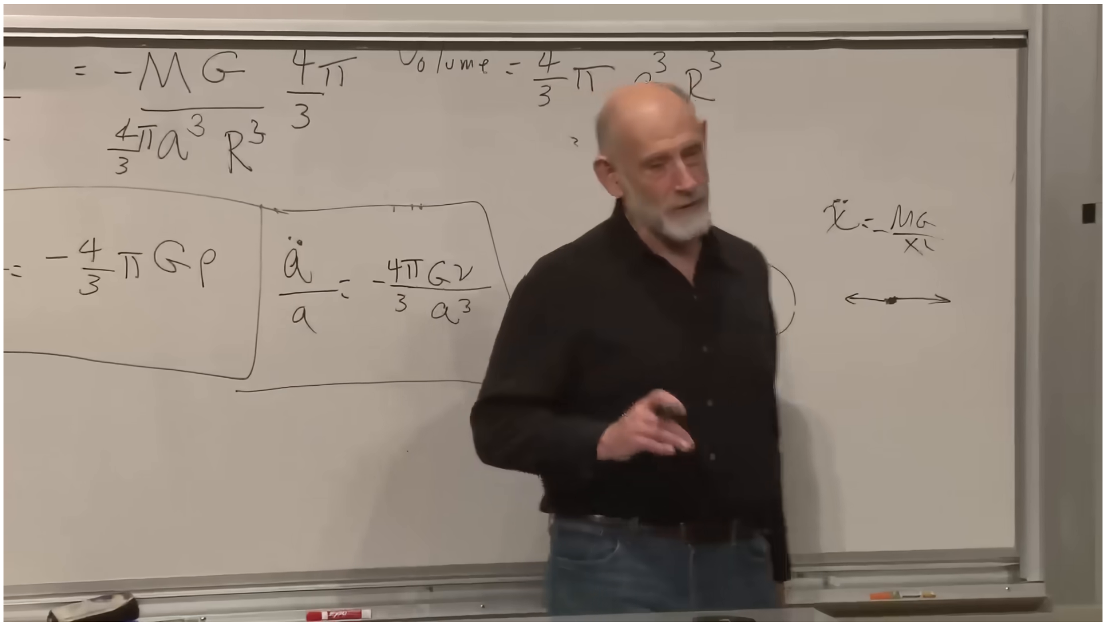

Figure 1.4: The zero-energy boundary taking shape on the board. The earlier cosmological acceleration equations remain boxed at left while the radial escape problem is developed at right.

Lecture frame: 01:01:37.

The first expression appears time dependent only until density dilution is included. The factors of a^3 cancel.

> **Question from the audience.** As the sphere expands, does it enclose more mass?
>
> **Response.** No. Its physical radius grows, but its comoving boundary moves with the galaxies. No matter crosses that boundary in the ideal homogeneous flow, and equation (1.27) shows the cancellation explicitly. [9]

The galaxy's physical speed is $\dot{d} = \dot{a}R$, and its distance from the chosen origin is $d = aR$. Its conserved Newtonian energy is therefore

$$\boxed{E = \frac{1}{2}m\dot{a}^2R^2 - \frac{GMm}{aR}.} \tag{1.28}$$

The three particle-energy cases become three cosmological cases:

$E < 0$**:**
: the initial expansion is below escape velocity; $a(t)$ reaches a maximum and the universe recollapses.

$E = 0$**:**
: the universe is exactly critical; expansion slows forever and approaches zero speed only asymptotically.

[9] Lecture timestamp: 01:04:37–01:05:20.

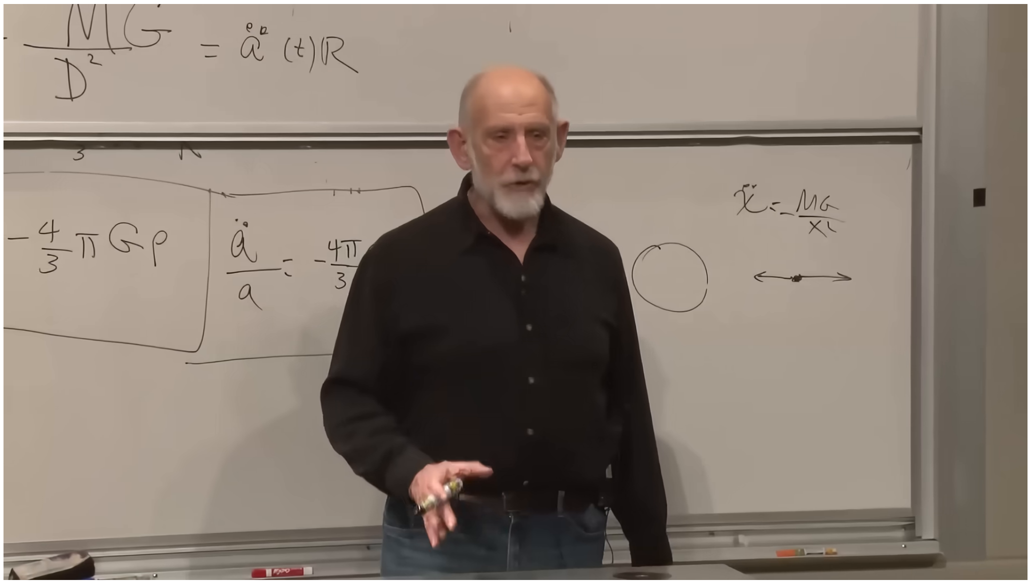

Figure 1.5: The inverse-square force and radial-particle sketch used when the one-particle energy argument is transferred back to cosmology.

Lecture frame: 01:04:14.

$E > 0$**:**
: the initial expansion is above escape velocity; the universe expands forever with nonzero asymptotic recession speed in the Newtonian model.

We begin with the edge $E = 0$. Setting equation (1.28) to zero, canceling m, and dividing by $a^2R^2/2$ gives

$$\frac{1}{2}\dot{a}^2R^2 = \frac{GM}{aR}, \tag{1.29}$$

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{2GM}{a^3R^3}. \tag{1.30}$$

Since

$$\rho = \frac{M}{(4\pi/3)a^3R^3}, \tag{1.31}$$

equation (1.30) becomes

$$\boxed{H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho.} \tag{1.32}$$

This is the first Friedmann equation for a spatially flat, matter-only universe with zero cosmological constant, obtained here from Newtonian energy conservation.

Question for the class. What happens to a particle launched at exactly escape velocity?

Response. It slows without ever reversing. Its speed approaches zero only at infinite distance. The zero-energy universe behaves in the same way: H decreases, but $\dot{a}$ remains positive and no finite-time turnaround occurs. [10]

[10] Lecture timestamp: 01:10:45–01:12:10.

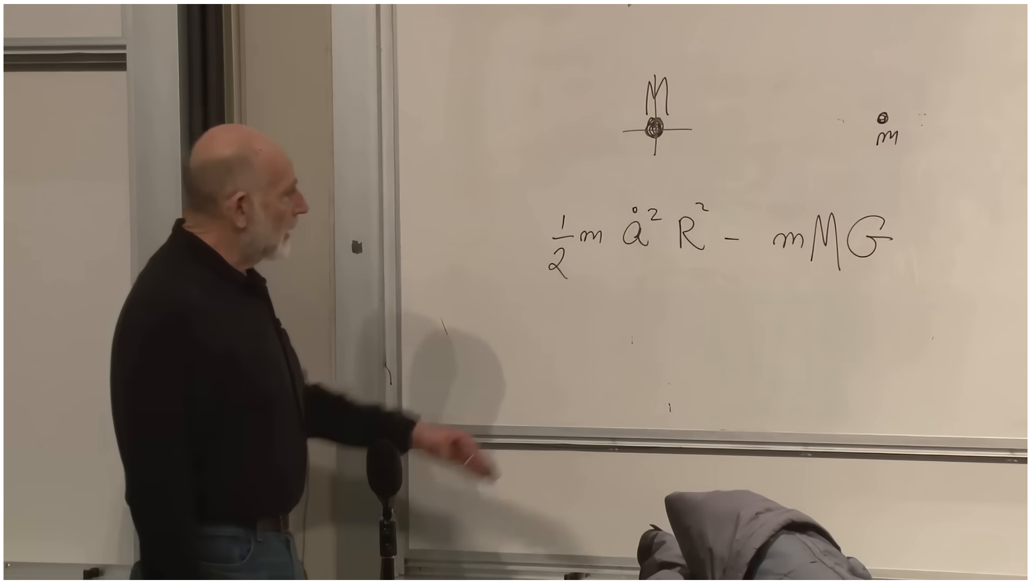

Figure 1.6: The cosmological kinetic and potential terms on the board. The complete denominator aR in equation (1.28) follows from the physical radius and the accompanying derivation.

Lecture frame: 01:06:12.

The factors $4\pi/3$ and $8\pi G/3$ are not arbitrary decoration. The $4\pi/3$ is the volume of a sphere, and the factor of two comes from kinetic energy. In general relativity the same $8\pi G$ appears in Einstein's field equation, and its time-time component leads to the relativistic Friedmann equation. That agreement is not accidental, but the present derivation does not yet include pressure, spatial curvature as geometry, or a cosmological constant.

1.8 The Critical Matter-Dominated Solution

Use the density law $\rho = \nu/a^3$ in equation (1.32):

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\nu}{3a^3}. \tag{1.33}$$

The constant ν depends on the arbitrary normalization of comoving coordinates. To expose the time dependence, define

$$K \equiv \frac{8\pi G\nu}{3} > 0. \tag{1.34}$$

Then

$$\dot{a}^2 = \frac{K}{a}. \tag{1.35}$$

The right-hand side of equation (1.35) is positive at every finite positive a. A turnaround would require $\dot{a} = 0$, so the critical universe never turns around. Its expansion grows steadily more tired, but never tired enough to stop.

Now solve the equation. One may separate variables directly,

$$a^{1/2}\, da = \sqrt{K}\, dt, \tag{1.36}$$

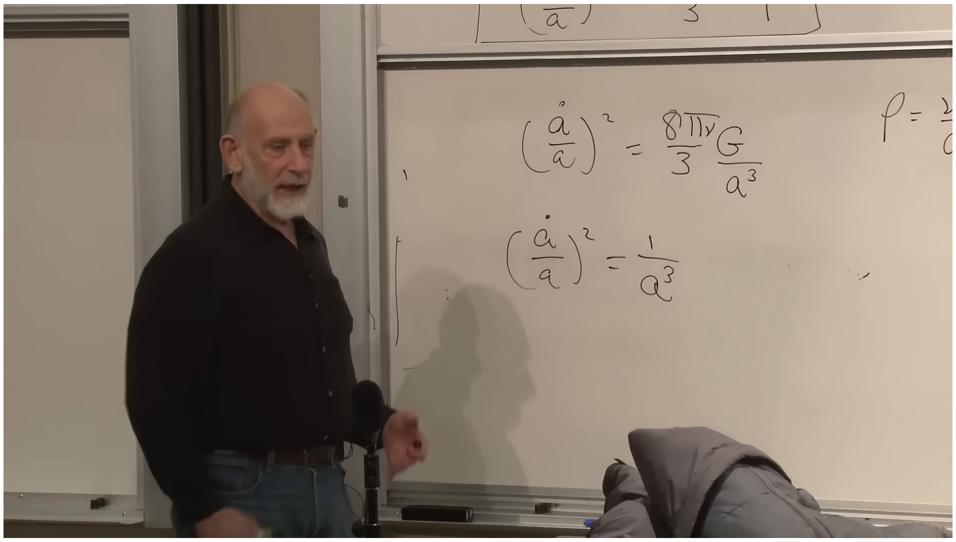

Figure 1.7: The matter-dominated Friedmann scaling on the board, first with its constants and then with the essential inverse power of the scale factor exposed.

Lecture frame: 01:14:44.

or follow the lecture and try a power law,

$$a(t) = Ct^p. \tag{1.37}$$

Then

$$\frac{\dot a}{a} = \frac{p}{t}, \qquad \frac{K}{a^3} = \frac{K}{C^3 t^{3p}}. \tag{1.38}$$

Substitution into equation (1.33) gives

$$\frac{p^2}{t^2} = \frac{K}{C^3 t^{3p}}. \tag{1.39}$$

Matching powers and coefficients yields

$$3p = 2, \qquad C^3 = \frac{K}{p^2} = \frac{9K}{4}. \tag{1.40}$$

Thus

$$\boxed{a(t) = \left(\frac{3\sqrt{K}}{2}t\right)^{2/3} \propto t^{2/3}.} \tag{1.41}$$

The additive origin of time has been chosen so that $a = 0$ at $t = 0$. More generally one may replace t by $t - t_0$.

The scale factor grows forever, while

$$\dot a \propto t^{-1/3} \longrightarrow 0, \qquad H = \frac{2}{3t} \longrightarrow 0. \tag{1.42}$$

Its graph is always increasing and always concave downward. It approaches no horizontal line because $a(t)$ itself is unbounded, but its slope tends to zero.

Question from the audience. Was $t^{2/3}$ found only because that trial form happened to work? Could another trial give another zero-energy solution?

Response. The trial identifies the solution rather than merely one possibility. The first-order expanding-branch equation $\dot{a} = \sqrt{K/a}$ integrates uniquely, up to the normalization and the choice of time origin, to $a \propto (t - t_0)^{2/3}$. Different total energies give different differential equations and therefore different cosmological histories. [11]

Question from the audience. Does the critical curve eventually become horizontal?

Response. Its slope tends to zero: $\dot{a} \propto t^{-1/3}$. It is nevertheless positive at every finite time, so the scale factor continues to grow without bound. The universe asymptotically approaches rest without reaching it. [12]

This simple result is historically striking. Newton possessed the mechanics and shell theorem needed for the derivation, and he considered questions about an infinite gravitating universe, but he did not formulate the expanding model. The classroom discussion suggested that philosophical and theological commitments about cosmic history may have influenced what questions seemed natural to him. That historical suggestion is not part of the derivation; the mathematical point is only that the required Newtonian tools already existed.

Question from the audience. Does this Newtonian construction assume an infinite universe?

Response. It assumes Euclidean, spatially flat Newtonian space. In the usual Newtonian model that space extends indefinitely, so the model is spatially infinite even while the distances between comoving galaxies change with time. Curved finite spatial geometries belong to the relativistic treatment. [13]

1.9 Beyond the Zero-Energy Model

The zero-energy case is a deliberate simplification, not the only Newtonian universe. Keeping $E \neq 0$ in equation (1.28) gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho + \frac{2E}{mR^2a^2}. \tag{1.43}$$

Homogeneity requires the combination $2E/(mR^2)$ to be the same for every representative comoving sphere. Writing it as a constant C_E,

$$H^2 = \frac{8\pi G}{3}\rho + \frac{C_E}{a^2}. \tag{1.44}$$

Positive C_E corresponds to energy above escape; negative C_E permits a turnaround. In relativistic notation this energy term is reinterpreted through spatial curvature, a connection reserved for the later treatment.

[11] Lecture timestamp: 01:27:07–01:27:43.
[12] Lecture timestamp: 01:27:48–01:28:57.
[13] Lecture timestamp: 01:22:12–01:22:55.

All three matter-only histories have negative acceleration while matter dominates:

$$\ddot{a} = -\frac{4\pi G}{3}\rho a < 0. \tag{1.45}$$

The recollapsing curve rises, reaches $\dot{a} = 0$, and falls. The critical curve $t^{2/3}$ rises forever with slope tending to zero. The positive-energy curve also rises forever and approaches linear growth. A provisional board sketch of these cases was corrected during the lecture; the stable conclusion is the shared concave-down behavior, not every detail of the first sketch.

The observed universe does something more. Its early matter-dominated expansion is well approximated by a decelerating curve, but at late times the curve bends upward:

$$\ddot{a} > 0. \tag{1.46}$$

Ordinary pressureless matter cannot produce that sign in equation (1.18). An additional term is required. In the simplest modern model it is a cosmological constant; more generally it is called dark energy. The present chapter intentionally omits it so that the Newtonian matter dynamics can be understood cleanly first.

> **Question from the audience.** Is the $8\pi G$ in the Friedmann equation the same factor that appears in Einstein's field equation, and have we included a cosmological constant?
>
> **Response.** The same gravitational normalization appears in general relativity, and the time-time field equation reproduces the relativistic Friedmann equation. The model here contains no cosmological constant. It is a matter-dominated universe; the missing cosmological-constant term is precisely the kind of term that can account for late-time acceleration. [14]

1.10 The Average Flow and Local Departures

Homogeneous expansion is a coarse-grained description. It does not say that every object expands away from every nearby neighbor. Gravitationally or otherwise bound systems can decouple from the Hubble flow. The Milky Way and Andromeda, for example, approach one another rather than following the large-scale recession law.

The departure of an individual galaxy's velocity from the Hubble flow is called its *peculiar velocity*. The term is technical, not a judgment on the galaxy. Hubble's law becomes a better approximation as one averages over larger distances and larger volumes. On the scale of a bound galaxy, a cluster, or a locally overdense region, local forces can dominate the background expansion.

> **Question from the audience.** Can some regions contract even though the universe expands overall?
>
> **Response.** Certainly. Bound systems and sufficiently dense fluctuations need not follow the mean expansion. The Milky Way–Andromeda system is a familiar example. A cosmological calculation averages over volumes large enough that these peculiar motions and local density fluctuations do not dominate; the resulting $a(t)$ describes the background, not every pair of objects. [15]

[14] Lecture timestamp: 01:30:15–01:31:06.
[15] Lecture timestamp: 01:31:00–01:34:13.

The analogy is an ordinary gas. Calling the air in a room uniform does not deny molecular fluctuations or local variations in density. It says that averages over sufficiently large regions are stable. Cosmic homogeneity has the same logical status, though on vastly larger scales. A region born slightly denser than average can carry enough self-gravity to resist expansion and eventually collapse into bound structure.

Question from the audience. Can observation decide whether galaxies move through space or whether expanding space carries them apart?

Response. Not when the two descriptions assign the same distance $d_{AB}(t)$ to every pair. One description lets galaxies move on a fixed-looking coordinate background; the comoving description freezes the galaxies and lets the scale factor carry the change. They are mathematically equivalent. General relativity makes the second language more natural, but it does not create a separate observable in this comparison. [16]

Question from the audience. Besides Type Ia supernova brightness, is there independent evidence for cosmic acceleration?

Response. Yes. The inference comes from a network of observations, especially the way supernova measurements and the cosmic microwave background fit together. No single datum bears the entire conclusion; consistency among independent cosmological measurements is the important point. [17]

1.11 What Has Been Established

The argument has proceeded in a strict order:

1. Large-scale isotropy, together with the rejection of a privileged center, motivates homogeneity.
2. Homogeneity permits a comoving grid with one scale factor $a(t)$.
3. Differentiating physical distance gives Hubble's law,
$$H = \frac{\dot{a}}{a}, \qquad v = Hd.$$
4. Conservation of mass in a comoving cell gives
$$\rho = \frac{\nu}{a^3}.$$
5. Newton's shell theorem gives the acceleration equation
$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\rho.$$
6. Conserved energy classifies recollapsing, critical, and unbound expansion.

[16] Lecture timestamp: 01:34:14–01:34:59.
[17] Lecture timestamp: 01:35:00–01:35:38.

7. At zero total energy,
$$H^2 = \frac{8\pi G}{3}\rho, \qquad a(t) \propto t^{2/3}.$$

This is a complete Newtonian matter-dominated model, but not yet a complete account of our universe. It captures the early decelerating logic, shows why a filled homogeneous universe cannot remain static, and identifies the critical expansion at the edge of escape. Nonzero energy, relativistic spatial geometry, pressure, radiation, and the term responsible for late acceleration remain to be added.

Chapter 2

Energy, Curvature, and Radiation

Overview. The Newtonian model becomes much more revealing once two restrictions are removed. First, a comoving galaxy need not have exactly zero total energy; keeping its energy arbitrary introduces the a^{-2} term that general relativity will reinterpret as spatial curvature. Second, the cosmic source need not be pressureless matter. Photon wavelengths stretch with the scale factor, so radiation dilutes as a^{-4} and drives the law $a(t) \propto t^{1/2}$. A universe containing both matter and radiation is therefore radiation dominated at small a and matter dominated at larger a. Along the way we shall separate local Newtonian motion from global relativistic geometry and identify the questions that cannot be settled without general relativity.

2.1 What Newton Can See

Before extending the model, we should ask why the previous Newtonian calculation worked at all. The physical universe is described by general relativity, and its spatial geometry need not be globally Euclidean. Nevertheless, every smooth curved space is approximately flat in a sufficiently small neighborhood. A small patch of the surface of the Earth looks planar; in the same way, a region of the universe much smaller than its radius of curvature can be treated as flat.

"Small" here is relative to the curvature scale. The patch may still span billions of light-years. Within it, neighboring galaxies can be treated by Newtonian mechanics when their relative peculiar velocities are nonrelativistic and the gravitational field is weak. The Newtonian calculation is therefore not a rival to Einstein's theory. It is the local, slow-motion limit of the more complete description.

There is an immediate warning. Radiation does not move slowly. Photons, and neutrinos while they are relativistic, cannot be treated as a gas of ordinary Newtonian particles. Newtonian reasoning will continue to organize the expansion equation, but the source on its right-hand side must be enlarged from rest-mass density to energy density. This is where the approximation begins to reveal its own boundary.

2.2 The Scale Factor Recalled

Take a homogeneous distribution of galaxies and attach coordinates to a grid that moves with the average cosmic flow. A galaxy at fixed comoving coordinate x has physical distance

$$D(t) = a(t)x \tag{2.1}$$

from the chosen origin. The scale factor $a(t)$ converts coordinate separation into physical separation.

The absolute value of a is conventional. If all comoving coordinates are rescaled by a constant, the same physical distances can be represented by the inverse rescaling of a. Thus neither a nor the mass assigned to one coordinate cell is separately invariant. Ratios of scale factors and the fractional expansion rate are invariant:

$$\frac{a(t_2)}{a(t_1)}, \qquad H(t) \equiv \frac{\dot{a}}{a}. \tag{2.2}$$

Indeed, if $a' = \alpha a$ for a fixed constant α, then $\dot{a}'/a' = \dot{a}/a$.

Let ν_m be the mass contained in one unit of comoving coordinate volume. Its numerical value changes if the grid is relabeled, but it is constant in time when ordinary matter is conserved. A comoving unit cube has physical volume a^3, so the matter density is

$$\boxed{\rho_m(a) = \frac{\nu_m}{a^3}.} \tag{2.3}$$

This a^{-3} dilution is simply volume growth: the rest energy in the comoving box remains fixed while its volume grows.

Real galaxies are not nailed rigidly to the grid. They orbit in groups, fall into clusters, and carry small velocities relative to the average flow. The scale factor describes the smoothed motion on large scales; such local deviations are called *peculiar velocities*.

> **Question from the audience.** Must every galaxy move exactly radially away from every other galaxy?
>
> **Response.** Only the averaged Hubble flow has that form. A galaxy's actual motion is the sum of the common expansion and a peculiar velocity caused by nearby structure. The distinction is like a river current with eddies superposed on it: the eddies matter locally without erasing the large-scale flow. [1]

For pressureless matter with zero Newtonian total energy, the preceding lecture found

$$H^2 = \frac{8\pi G}{3}\rho_m = \frac{8\pi G\nu_m}{3a^3}. \tag{2.4}$$

Because the overall grid normalization is arbitrary, we may temporarily choose units for which $8\pi G\nu_m/3 = 1$. On the expanding branch,

$$\frac{\dot{a}}{a} = \frac{1}{a^{3/2}}, \qquad \dot{a} = \frac{1}{\sqrt{a}}. \tag{2.5}$$

[1] Lecture timestamp: 00:24:47–00:26:33.

Inverting the derivative and integrating gives

$$\frac{dt}{da} = \sqrt{a}, \qquad t - t_B = \frac{2}{3}a^{3/2}, \tag{2.6}$$

where t_B is the time at which this idealized solution reaches $a = 0$. Hence

$$\boxed{a(t) \propto (t - t_B)^{2/3}.} \tag{2.7}$$

The universe expands forever in this critical matter-only model. Its expansion slows because $\dot{a} \propto a^{-1/2}$, but $\dot{a}$ does not vanish at any finite value of a.

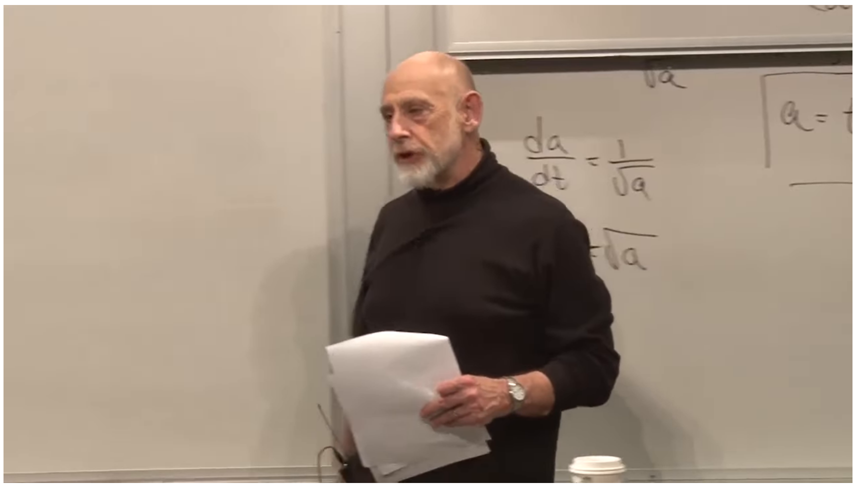

Figure 2.1: The recap board records $\dot{a} = 1/\sqrt{a}$ and the boxed matter-dominated law $a \propto t^{2/3}$. The intermediate algebra is partly obscured and is supplied by equations (2.4)–(2.7).

Lecture frame: 00:29:20.838.

> **Question from the audience.** Why was the negative square root discarded?
>
> **Response.** It was not discarded by the equation. Equation (2.4) fixes H^2, not the sign of H. The positive root describes expansion and the negative root describes contraction. A rock in a planet's gravitational field can be moving outward or inward while obeying the same energy equation; its initial condition selects the branch. We selected the expanding branch because that is the universe under discussion. [2]

The next model will still omit dark energy. That omission is deliberate: we first understand the older matter-and-radiation framework before adding the component that controls the present accelerated era. Dark matter, by contrast, is already included in the matter term because nonrelativistic dark matter also dilutes as a^{-3}.

> **Question from the audience.** Is dark energy not the dominant component today?
>
> **Response.** Yes. The two-component story developed here is the standard pedagogical cosmology from before the discovery of the present acceleration. Dark matter belongs with matter; dark energy is a third component to be introduced separately. Beginning with the older model lets us see what each additional term changes. [3]

[2] Lecture timestamp: 00:23:09–00:24:46.
[3] Lecture timestamp: 00:28:02–00:28:45.

2.3 Restoring the Energy Constant

The zero-energy choice is a special initial condition. To remove it, return to a galaxy of mass m on the boundary of a homogeneous sphere. Let its comoving radius be x, so that

$$D = ax, \qquad \dot{D} = \dot{a}x. \tag{2.8}$$

Newton's shell theorem says that only the mass $M(D)$ inside the sphere affects the radial acceleration; exterior spherical shells exert no net force. The galaxy's conserved Newtonian energy is

$$\frac{1}{2}m\dot{D}^2 - \frac{GM(D)m}{D} = E. \tag{2.9}$$

The factor m belongs in both terms. It is easy to omit while writing rapidly on a board, but retaining it makes clear that the final constant is the energy per unit test mass.

Homogeneity allows us to choose the representative galaxy at $x = 1$. Then $D = a$, $\dot{D} = \dot{a}$, and the sphere has

$$V_{\rm sph} = \frac{4\pi}{3}a^3, \qquad M(a) = \rho_m V_{\rm sph} = \frac{4\pi}{3}\rho_m a^3. \tag{2.10}$$

The board uses V both for volume and, in a side annotation, for radial velocity. We shall keep the unambiguous symbols $V_{\rm sph}$ and $\dot{D}$.

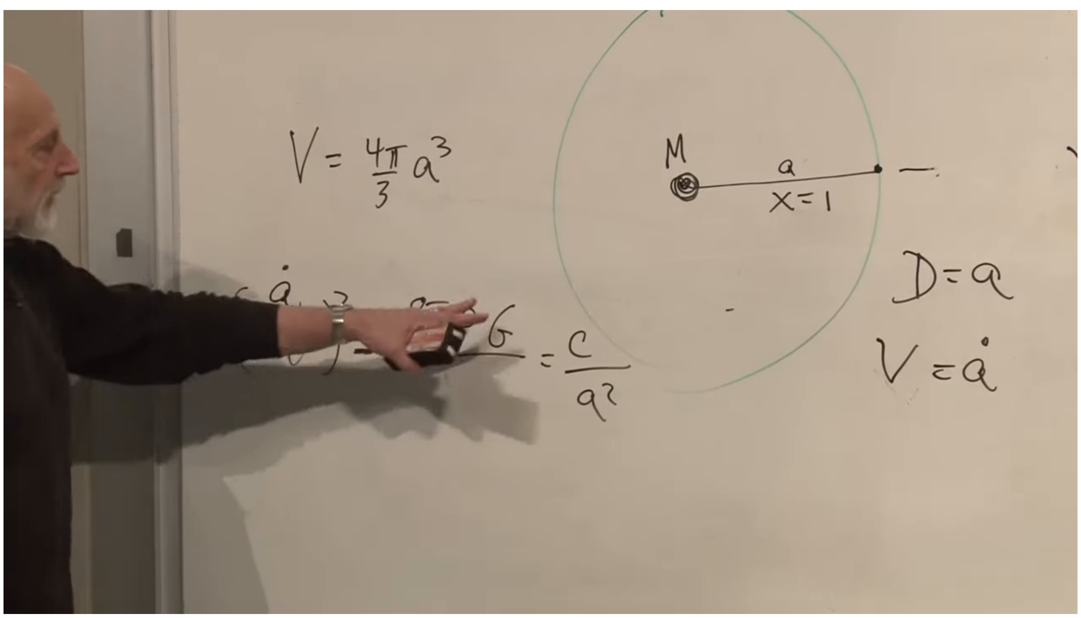

Figure 2.2: The enclosed-mass sphere and the nonzero-energy Friedmann equation. The visible labels include $V_{\rm sph} = 4\pi a^3/3$, the central mass M, $x = 1$, $D = a$, and $\dot{D} = \dot{a}$. Part of the equation is hidden by the lecturer and is reconstructed algebraically below.

Lecture frame: 00:37:59.876.

Insert $D = a$ into equation (2.9), divide by $m/2$, and define the constant

$$C \equiv \frac{2E}{m}. \tag{2.11}$$

We obtain

$$\dot{a}^2 - \frac{2GM(a)}{a} = C. \tag{2.12}$$

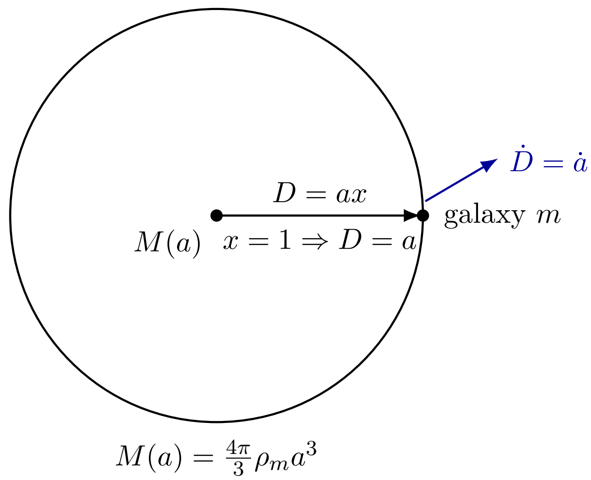

Figure 2.3: A clean reconstruction of the Newtonian sphere at $x = 1$. The exterior shells cancel, so the boundary galaxy responds only to the enclosed mass.

Dividing by a^2 and using equation (2.10) gives

$$\left(\frac{\dot{a}}{a}\right)^2 - \frac{8\pi G}{3}\rho_m = \frac{C}{a^2}, \tag{2.13}$$

or, equivalently,

$$\boxed{H^2 = \frac{8\pi G}{3}\rho_m + \frac{C}{a^2}.} \tag{2.14}$$

This is the first Friedmann equation for pressureless matter, written in the Newtonian energy convention. The new a^{-2} term remembers whether the representative galaxy was launched above, at, or below escape energy.

Substituting $\rho_m = \nu_m/a^3$, it is convenient to define

$$A_m \equiv \frac{8\pi G \nu_m}{3} > 0, \tag{2.15}$$

so that

$$H^2 = \frac{A_m}{a^3} + \frac{C}{a^2}, \qquad \dot{a}^2 = \frac{A_m}{a} + C. \tag{2.16}$$

The equation is more informative when read by limits than when immediately integrated.

2.3.1 Positive Energy

If $C > 0$, the matter term A_m/a^3 dominates at small a. The early expansion therefore retains the matter law $a \propto t^{2/3}$. At large a, however, the C/a^2 term dominates in H^2, or equivalently

$$\dot{a}^2 \longrightarrow C. \tag{2.17}$$

The expansion approaches constant speed,

$$a(t) \sim \sqrt{C}\, t. \tag{2.18}$$

This is the cosmological counterpart of throwing an apple faster than escape velocity. Gravity continually reduces its speed, but far away the apple retains a nonzero asymptotic velocity.

2.3.2 Zero Energy

For $C = 0$, equation (2.16) returns the critical solution $a \propto t^{2/3}$. It expands without bound while $\dot{a}$ tends to zero. This is the exact escape-energy case.

2.3.3 Negative Energy

If $C = -|C| < 0$, then

$$H^2 = \frac{A_m}{a^3} - \frac{|C|}{a^2} = \frac{A_m - |C|a}{a^3}. \tag{2.19}$$

The scale factor cannot exceed

$$a_{\max} = \frac{A_m}{|C|}, \tag{2.20}$$

because the right-hand side reaches zero there. The expanding branch arrives at $a_{\max}$ with $\dot{a} = 0$; the continuation is the contracting branch. In the Newtonian picture this is an apple launched below escape velocity: it rises, turns, and falls.

Question from the audience. How can the model recollapse when the right-hand side equals H^2 and therefore cannot be negative?

Response. It never becomes negative. During expansion the right-hand side decreases to zero at $a_{\max}$. There $H = 0$. The solution then changes from the positive square root to the negative square root and retraces the allowed region with $\dot{a} < 0$. A tempting but incorrect continuation would let the expression pass through zero. It cannot: zero is the boundary, not a value through which a real H can continue. [4]

Thus the three Newtonian energies give three qualitative histories: recollapse for $C < 0$, critical escape for $C = 0$, and eternal expansion with asymptotically constant $\dot{a}$ for $C > 0$. The surprising next step is that general relativity assigns these same cases a geometric meaning.

2.4 From Energy to Curvature

Move the energy term in equation (2.14) to the geometric side and define

$$k \equiv -C. \tag{2.21}$$

Then

$$\boxed{H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho.} \tag{2.22}$$

The board alternates among C, K, and κ, and a sign is absorbed when moving the term across the equality. Equation (2.21) fixes our convention: positive Newtonian energy corresponds to $k < 0$, zero energy to $k = 0$, and negative Newtonian energy to $k > 0$.

This arrangement is a miniature version of Einstein's equation. The left side describes geometry: expansion through H, and spatial curvature through k/a^2. The right side describes the material

[4] Lecture timestamp: 00:48:43–00:53:03.

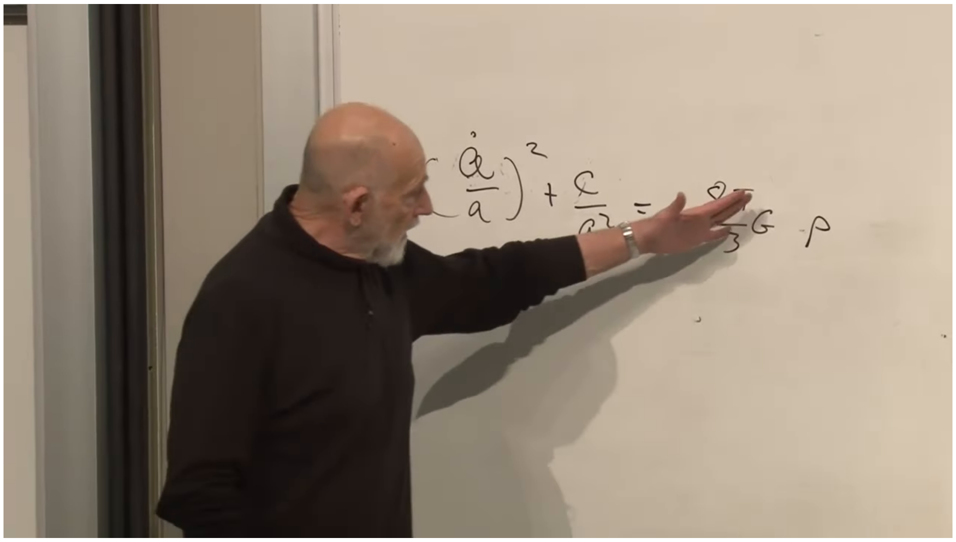

Figure 2.4: The Friedmann equation arranged as geometry on the left and source on the right. The frame supports this conceptual division; it does not display the full tensor Einstein equation.

Lecture frame: 00:55:22.810.

source. In Newtonian mechanics that source was mass density. In relativity it is energy and momentum, represented fully by the stress-energy tensor. For a homogeneous and isotropic universe the first Friedmann equation depends on the total energy density ρ.

If ordinary units are retained, a mass density contributes through its rest energy $\rho_{\text{mass}}c^2$. From here onward we set $c = 1$, so mass and energy use the same units. The symbol ρ may then include rest mass, kinetic energy, radiation, and any other homogeneous energy component.

The geometric interpretation is global. In the relativistic model, $k = 0$ describes flat spatial slices, $k > 0$ positively curved spherical slices, and $k < 0$ negatively curved hyperbolic slices. The Newtonian derivation reproduced the same evolution equation without possessing these global geometries. Its nonzero energy constant is therefore a remarkably accurate mimic of curvature, but not a derivation of curved space itself.

> **Question from the audience.** How can Newtonian mechanics and general relativity produce the same expansion equation if their geometries are different?
>
> **Response.** General relativity includes the flat case, and locally every smooth geometry is approximately flat. More remarkably, the Newtonian energy constant enters the evolution equation in exactly the place occupied by relativistic spatial curvature. The dynamical formula agrees, while the global interpretation does not: Euclidean Newtonian space has not secretly become a sphere or a hyperbolic space. The three geometries will have to be constructed within general relativity. [5]

[5] Lecture timestamp: 01:20:59–01:24:20.

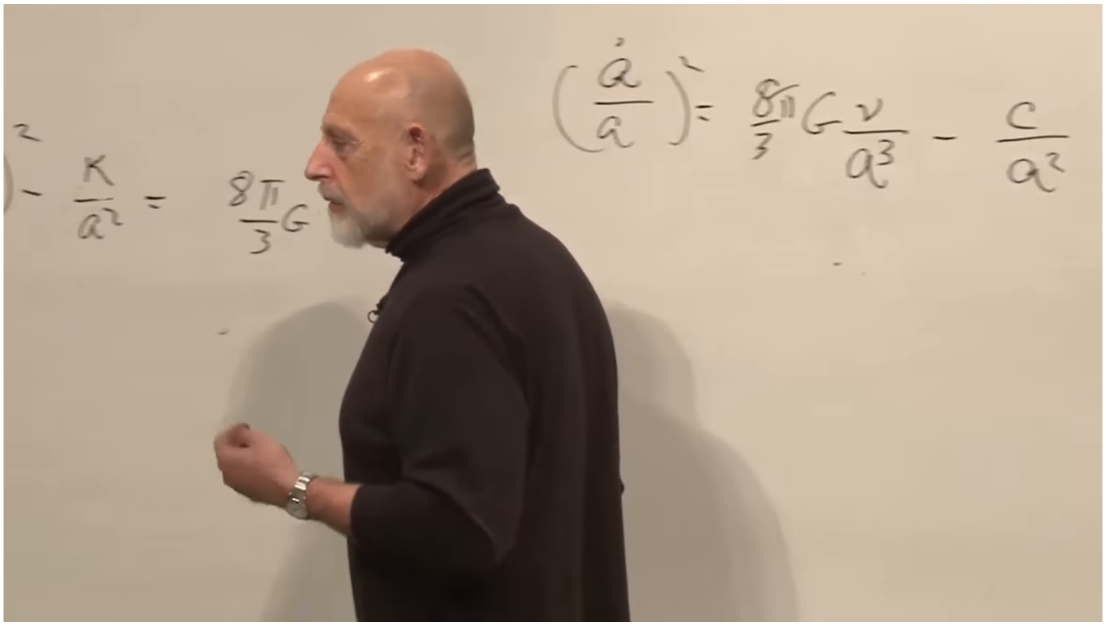

Figure 2.5: Side-by-side board forms of the Friedmann equation: a curvature form using ρ, and a matter form obtained after $\rho_m = \nu_m/a^3$. The notation changes on the board; equation (2.21) keeps the sign convention fixed here.

Lecture frame: 00:57:52.659.

2.5 Radiation in an Expanding Box

We now change the contents of the universe. Consider one comoving unit cube. Its physical volume is a^3. For nonrelativistic matter, the number of particles in that box is fixed and each particle's rest energy is approximately fixed, giving $\rho_m \propto a^{-3}$. For radiation there is another effect: each photon loses energy as the universe expands.

A photon of wavelength λ has energy

$$E_\gamma = \frac{hc}{\lambda}, \qquad E_\gamma \propto \lambda^{-1}. \tag{2.23}$$

Under homogeneous expansion, a freely propagating wavelength stretches with the scale factor:

$$\frac{\lambda(t_2)}{\lambda(t_1)} = \frac{a(t_2)}{a(t_1)}, \qquad \lambda \propto a. \tag{2.24}$$

This is cosmological redshift. It implies

$$E_\gamma \propto a^{-1}. \tag{2.25}$$

If photon number in the comoving box is conserved, its total radiation energy falls as a^{-1}. Dividing by its physical volume a^3 gives

$$\boxed{\rho_r(a) = \frac{\nu_r}{a^4}.} \tag{2.26}$$

Here ν_r is the constant amplitude of the freely streaming radiation density. Radiation has one more inverse power of a than matter: three powers from spatial dilution and one from the redshift of every photon.

The guitar-string analogy gives useful intuition. A mode fitted to a string has an integer number of half-wavelengths. If the string is lengthened slowly, that integer cannot change continuously; instead the wavelength adjusts to the new length. Likewise, for a wave on a slowly expanding ring, the number of nodes is an adiabatic invariant, so its wavelength follows the circumference. A general wave can be decomposed into such modes.

This is not a substitute for solving Maxwell's equations in an expanding spacetime. It is the controlled intuition behind equation (2.24): slow geometric expansion changes the physical wavelength while preserving the mode label.

> **Question from the audience.** Would this argument require a conducting or reflecting box? What happens in empty expanding space with no physical walls?
>
> **Response.** The walls are a bookkeeping device, not a claim that the universe is enclosed by a conductor. A full proof follows from wave propagation in the expanding metric. For the present argument we use the observed and relativistically derived fact that freely propagating wavelengths scale with a. The box helps count photons and physical volume; it does not cause the cosmological redshift. [6]

> **Question from the audience.** Why should a wavelength track the grid rather than keep a fixed physical length?
>
> **Response.** Consider a mode on a ring whose circumference changes adiabatically. Its integer node count cannot drift continuously, so the allowed wavelength must change with the ring. Expanding a general wave into Fourier modes gives the same conclusion mode by mode. This supplies the needed adiabatic intuition; the exact statement belongs to field theory in curved spacetime. [7]

2.5.1 The Radiation-Dominated Solution

With curvature omitted for the moment, insert equation (2.26) into Friedmann's equation:

$$H^2 = \frac{8\pi G \nu_r}{3a^4} \equiv \frac{A_r}{a^4}, \qquad A_r > 0. \tag{2.27}$$

On the expanding branch,

$$\frac{\dot a}{a} = \frac{\sqrt{A_r}}{a^2}, \qquad \dot a = \frac{\sqrt{A_r}}{a}. \tag{2.28}$$

Therefore

$$a\,da = \sqrt{A_r}\,dt, \qquad \frac{a^2}{2} = \sqrt{A_r}(t - t_B), \tag{2.29}$$

and

$$\boxed{a(t) \propto (t - t_B)^{1/2}.} \tag{2.30}$$

The distinction between the exponents 1/2 and 2/3 is visually modest but physically decisive. Expansion rate, temperature, reaction freeze-out, and horizon size in the early universe all depend on which component dominates.

[6] Lecture timestamp: 01:02:21–01:04:37.
[7] Lecture timestamp: 01:34:18–01:38:33.

2.6 A Universe of Matter and Radiation

For separately conserved matter and radiation, their densities add:

$$\rho(a) = \rho_m(a) + \rho_r(a). \tag{2.31}$$

Ignoring curvature for this comparison,

$$\boxed{H^2 = \frac{C_m}{a^3} + \frac{C_r}{a^4},} \tag{2.32}$$

where $C_m, C_r > 0$. The board first labels the two coefficients C_1 and C; the second symbol is corrected verbally to a distinct constant. The notation in equation (2.32) makes that correction explicit.

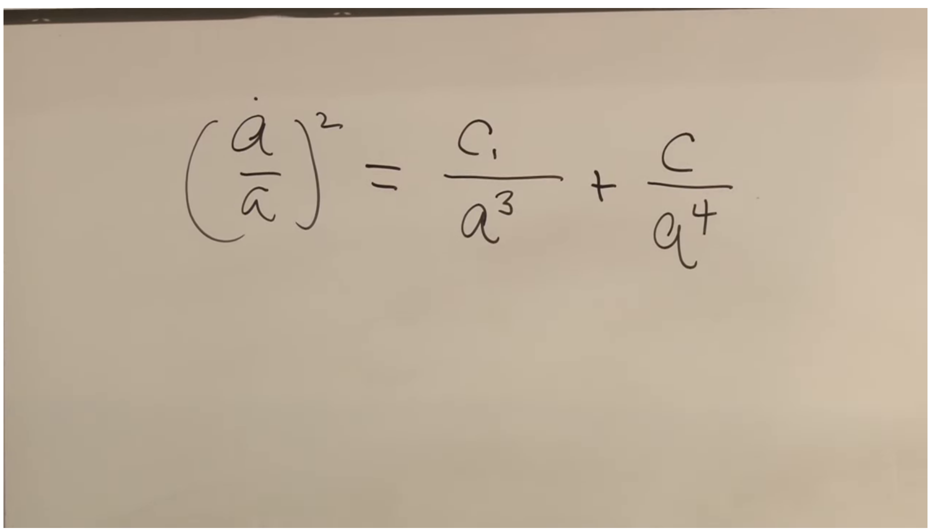

Figure 2.6: The mixed matter-and-radiation equation on the board. The written second coefficient is subsequently corrected to a radiation-specific constant, denoted C_r in equation (2.32).
Lecture frame: 01:11:37.997.

At small a, the radiation term wins because a^{-4} grows faster than a^{-3}:

$$a \ll a_{\mathrm{eq}} \implies H^2 \simeq \frac{C_r}{a^4}, \qquad a(t) \propto t^{1/2}. \tag{2.33}$$

At larger a, matter wins:

$$a \gg a_{\mathrm{eq}} \implies H^2 \simeq \frac{C_m}{a^3}, \qquad a(t) \propto t^{2/3}. \tag{2.34}$$

The equality scale follows directly by setting the two terms equal:

$$a_{\mathrm{eq}} = \frac{C_r}{C_m}. \tag{2.35}$$

This equality is not the same event as photon decoupling. Equality asks when the two energy densities match; decoupling asks when interactions cease to keep matter and radiation in thermal contact.

The radiation sector includes photons and any species moving ultrarelativistically. Very light neutrinos behave this way at sufficiently high temperature, and a background of gravitons would scale in the same manner. Once a massive species becomes nonrelativistic, its dominant rest-mass contribution begins to scale like matter. Cold dark matter belongs in C_m/a^3, not in the radiation term.

Question for the class. Which term in equation (2.32) dominates first, and which dominates later?

Response. For small a, a^{-4} outruns a^{-3}, so radiation controls the early expansion and gives $a \propto t^{1/2}$. As a grows, the radiation-to-matter ratio falls as $1/a$; matter then controls the expansion and gives $a \propto t^{2/3}$. [8]

Question from the audience. Are the relative amounts of matter and radiation being assumed constant?

Response. Their densities are not constant, but the coefficients C_m and C_r can be treated as constant after the components cease exchanging energy efficiently and particle creation or annihilation is negligible. Matter then carries a conserved comoving rest mass and free radiation a conserved comoving photon number, with redshift supplying its extra factor of a^{-1}. During epochs of strong interaction or species annihilation, the bookkeeping must be refined. [9]

Question from the audience. When did matter and radiation stop interacting strongly?

Response. A rough order-of-magnitude answer is 10^5 years for photons to decouple from ordinary matter. The precise recombination history is a later calculation, and it should not be confused with matter–radiation equality. The point needed here is only that, after decoupling, the two components may be followed approximately separately. [10]

Question from the audience. How can we know that an early $a \propto t^{1/2}$ era occurred before there were galaxies to trace the expansion?

Response. We infer the expansion history from the physical records it leaves. Primordial nucleosynthesis depends on the early competition between reaction rates and $H(t)$; the microwave background records the later thermal state and perturbations. Galaxy observations make the matter-era behavior more direct, while the radiation era is reconstructed from this early-universe evidence. [11]

The matter-plus-radiation model is deliberately incomplete. It is the classic framework in which the universe passes from radiation domination to matter domination. Present accelerated expansion requires another source, dark energy, which will add a qualitatively different term.

[8] Lecture timestamp: 01:11:46–01:13:38.
[9] Lecture timestamp: 01:13:39–01:14:53.
[10] Lecture timestamp: 01:27:29–01:27:46.
[11] Lecture timestamp: 01:27:46–01:30:10.

2.7 Where the Redshifted Energy Goes

The decrease $E_\gamma \propto a^{-1}$ raises an unavoidable question: where has the photon energy gone? In a literal expanding box with reflecting walls, radiation exerts pressure and performs work on the walls. The thermodynamic statement is

$$dE = -p\,dV. \tag{2.36}$$

As V increases, the radiation energy decreases. The lost energy may appear as kinetic energy, strain, or heat in whatever moves the walls.

The comoving box has no material walls. It is an imaginary boundary moving with the average expansion. Photons cross it, but in a homogeneous radiation bath the statistical flux entering balances the flux leaving. The box remains a valid bookkeeping volume even though it is not a container. The pressure-work law still describes how the energy in a comoving volume evolves.

For radiation $p = \rho/3$, and $E = \rho V$. Equation (2.36) then gives

$$d(\rho V) = -\frac{\rho}{3}\,dV, \tag{2.37}$$

$$V\,d\rho = -\frac{4}{3}\rho\,dV, \tag{2.38}$$

$$\frac{d\rho}{\rho} = -\frac{4}{3}\frac{dV}{V}. \tag{2.39}$$

Since $V \propto a^3$, integration yields $\rho \propto a^{-4}$, agreeing with the wavelength argument. This thermodynamic derivation also makes clear that the energy of each photon, not merely the number density, decreases.

Question from the audience. Where does the energy of the redshifted photons go? Is total cosmic energy conserved?

Response. With physical walls the radiation does work on them, so the transfer is ordinary mechanical bookkeeping. For the universe as a whole the question is more subtle. Global energy conservation normally follows from time-translation symmetry, but an expanding spacetime is not generally invariant under time translation and need not possess a unique conserved total energy. One can build useful Newtonian analogies involving the energy of expansion, but a definitive global statement requires general relativity. Both descriptions are useful here, but no final global verdict follows from the Newtonian model. [12]

Question from the audience. If the comoving box is imaginary and photons cross its boundary, how can pressure work still be a useful picture?

Response. Homogeneity supplies the missing bookkeeping. On average, every photon leaving one comoving volume is replaced statistically by one entering from its neighbor. The local fluxes cancel in the mean, while the physical volume and wavelengths change. A reflecting box makes the work mechanism concrete; the comoving box carries the same averaged conservation law without literal walls. [13]

[12] Lecture timestamp: 01:16:51–01:20:58.
[13] Lecture timestamp: 01:24:22–01:27:28.

Question from the audience. Does expansion merely reduce photon density, or does it reduce the energy of every photon?

Response. Both occur. The number per physical volume falls as a^{-3}, and cosmological redshift lowers each photon energy by another factor a^{-1}. This is the radiative analogue of adiabatic cooling. An ordinary gas also cools while doing work during slow expansion, although a sudden free expansion is a different thermodynamic process. [14]

2.8 Recession, Horizons, and the Limit of the Model

Hubble's law assigns the recession rate

$$v_{\text{rec}} = HD \tag{2.40}$$

to an object at proper distance D on a chosen cosmic-time slice. For sufficiently large D, equation (2.40) can exceed c. This does not violate special relativity, because v_{rec} is not the velocity of a nearby object measured in one local inertial frame. It compares widely separated comoving worldlines in a changing geometry. Every local observer still measures light at speed c, and no local material object passes another faster than light.

Question from the audience. Can a sufficiently distant galaxy recede faster than light?

Response. Yes, its cosmological recession rate HD can exceed c. No local object then overtakes a neighboring observer faster than light, and the result cannot be used for superluminal signaling. It is a statement about the changing proper distance between distant comoving worldlines, not an ordinary special-relativistic velocity measured at one event. [15]

Superluminal recession by itself does not decide whether light emitted now will ever reach us. That is a question about the entire future expansion history. In a decelerating matter-dominated model, regions currently receding faster than light can later become observable because the relevant horizon evolves. If accelerated expansion persists, an event horizon may permanently hide sufficiently distant events.

Question from the audience. Do galaxies disappear once their recession rate exceeds the speed of light?

Response. Not simply. A Hubble-radius crossing and a true event horizon are different notions. In a decelerating model, light from a currently superluminally receding region may still reach us later. Persistent dark-energy domination changes the global causal structure and can create an ultimate horizon. Deciding which signals arrive requires the spacetime geometry, not a snapshot of HD. [16]

Special-relativistic language can also mislead when applied globally. A particle's invariant rest mass does not increase with speed. Its energy and momentum depend on the observer's local frame. Massive particles cannot be accelerated through the local speed of light, but the recession of a distant comoving galaxy is not its passage through a single local frame.

[14] Lecture timestamp: 01:42:30–01:45:55.
[15] Lecture timestamp: 01:30:17–01:32:13.
[16] Lecture timestamp: 01:32:14–01:34:18.

Question from the audience. Does a rapidly receding galaxy acquire a larger relativistic mass?

Response. It is cleaner not to use "relativistic mass." The galaxy has an invariant rest mass, while its energy and momentum are frame dependent. Moreover, a recession rate assigned across cosmological distance is not the local velocity appearing in the special-relativistic formula. These concepts agree only after the distant events are compared using the spacetime metric. [17]

Question from the audience. What happens to the time and space axes when a distant recession rate is superluminal?

Response. There is no single global Lorentz frame whose tilted axes settle the question. Special relativity compares inertial frames in flat spacetime; cosmology requires a metric that relates local frames over an extended curved spacetime. Newtonian reasoning remains reliable in a small patch, but the global construction must wait for general relativity. [18]

Newtonian energy conservation has produced the correct scale-factor equation for matter and has even anticipated the curvature term. Wave mechanics and thermodynamics supply the radiation scaling. Spatial geometry, global energy, and horizons, however, are not local Newtonian questions; they require the metric description that follows.

2.9 Matter, Radiation, and Curvature

The matter-only critical solution obeys

$$\rho_m \propto a^{-3}, \qquad a(t) \propto t^{2/3}. \tag{2.41}$$

Restoring the Newtonian energy per unit mass gives

$$H^2 = \frac{8\pi G}{3}\rho_m + \frac{C}{a^2}. \tag{2.42}$$

Positive C yields eternal expansion with $a \sim t$ at late times, $C = 0$ gives critical escape, and negative C yields a maximum scale factor followed by recollapse. Defining $k = -C$ places the same term on the geometric side of the relativistic Friedmann equation.

Radiation differs from matter because expansion changes both number density and energy per quantum:

$$\begin{aligned} \lambda &\propto a, & E_\gamma &\propto a^{-1}, \\ \rho_r &\propto a^{-4}, & a(t) &\propto t^{1/2}. \end{aligned} \tag{2.43}$$

The two-component equation

$$H^2 = \frac{C_m}{a^3} + \frac{C_r}{a^4} \tag{2.44}$$

therefore describes a radiation-dominated early universe followed by a matter-dominated era. Dark energy, spatial geometry, and the causal structure of horizons remain outside this model, exactly where the Newtonian approximation tells us to hand the problem over to general relativity.

[17] Lecture timestamp: 01:38:34–01:40:05.
[18] Lecture timestamp: 01:40:07–01:42:30.

Chapter 3

The Geometry of Homogeneous Space

Overview. Flatness is an excellent approximation to the observed universe, but it is not a principle. If we retain homogeneity and isotropy while allowing curvature, three simply connected spatial geometries emerge: Euclidean, spherical, and hyperbolic space. Their metrics can be built from the way concentric shells grow with distance, and their curvature can be tested through angular sizes and object counts. Flat space may also be finite when its topology is toroidal. Finally, attaching a time-dependent scale factor to any of these homogeneous spaces gives the cosmological spacetime metrics and the Hubble law; dynamics for that scale factor must come from general relativity.

3.1 Flatness Is an Observation, Not a Principle

Until now we have treated space as flat and have said almost nothing about spacetime geometry. Observation justifies flat space as a very good approximation, but it does not elevate flatness into a law. What the measurements really tell us is that any radius of curvature must be large compared with the region we have surveyed. Space might be exactly flat, positively curved, or negatively curved. On scales far beyond the observable region, we do not yet know which description is correct.

We therefore loosen one assumption while keeping another. We shall not assume flatness, but we shall continue to assume that space is homogeneous and isotropic. This is a working hypothesis, not an article of faith. To learn its consequences, we assume it and calculate; afterward we may abandon it and ask what an inhomogeneous universe would do differently.

Homogeneity means that every point has the same geometric character. An observer who moves elsewhere and repeats local geometric experiments obtains the same results. Isotropy means that no direction is preferred. These conditions exclude almost every curved surface one draws first. A paraboloid is more sharply curved near its tip than far from it. A long ellipsoid differs at its poles and waist. A bumped sphere, or the Earth's surface with mountains and valleys resolved, is curved but not homogeneous. The cosmological question is narrower:

> Which spaces can be curved while still looking geometrically the same from every point and in every direction?

Ignoring global identifications for the moment, there are three constant-curvature answers: flat space, positively curved spherical space, and negatively curved hyperbolic space. The qualification about global identifications matters; later we shall construct a flat finite torus and correct the oversimplified statement that these three exhaust every possibility.

3.2 The Metric and the Observer's Coordinates

We first discuss space alone, not spacetime. Its geometry is encoded in a metric: a rule for the squared distance ds^2 between neighboring points. Once that local rule is known everywhere, lengths, angles, areas, volumes, geodesics, and curvature can all be recovered from it.

For a Euclidean plane and Euclidean three-space, Cartesian coordinates give

$$ds_2^2 = dx^2 + dy^2, \qquad ds_3^2 = dx^2 + dy^2 + dz^2. \tag{3.1}$$

Adding the third dimension simply adds the orthogonal term dz^2, but the important point is not the simplicity of this formula. It is that equation (3.1) completely specifies the local geometry.

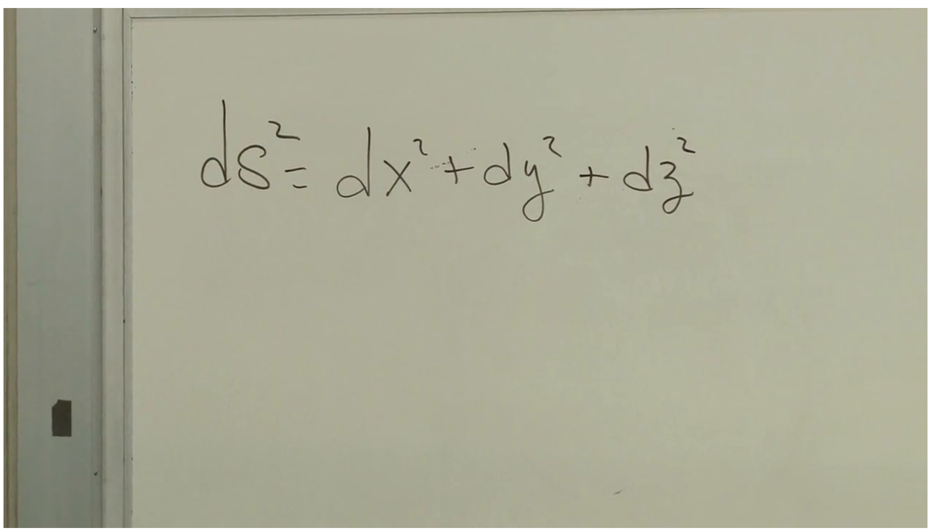

Figure 3.1: The Cartesian metric of flat three-dimensional space, as written on the board.
Lecture frame: 00:05:13.138.

Cartesian coordinates conceal the way an astronomer encounters the universe. We occupy one point, look outward by direction, and sort what we see by angle and distance. In a two-dimensional universe the visual field would literally be a circle of directions around us. Polar coordinates are therefore not merely elegant; they expose the angular geometry that cosmology uses.

Let r be radial distance and θ the direction. The same Euclidean plane now has metric

$$ds^2 = dr^2 + r^2 d\theta^2. \tag{3.2}$$

For a fixed small angle $d\theta$, the transverse separation is $r\,d\theta$. It grows linearly as we move outward, and its squared contribution to the metric is consequently $r^2 d\theta^2$.

The angular term is itself the metric of a one-dimensional space. A circle is a one-sphere, denoted S^1, and the metric of the unit circle is conventionally written

$$d\Omega_1^2 = d\theta^2, \qquad \text{so that} \qquad ds^2 = dr^2 + r^2 d\Omega_1^2. \tag{3.3}$$

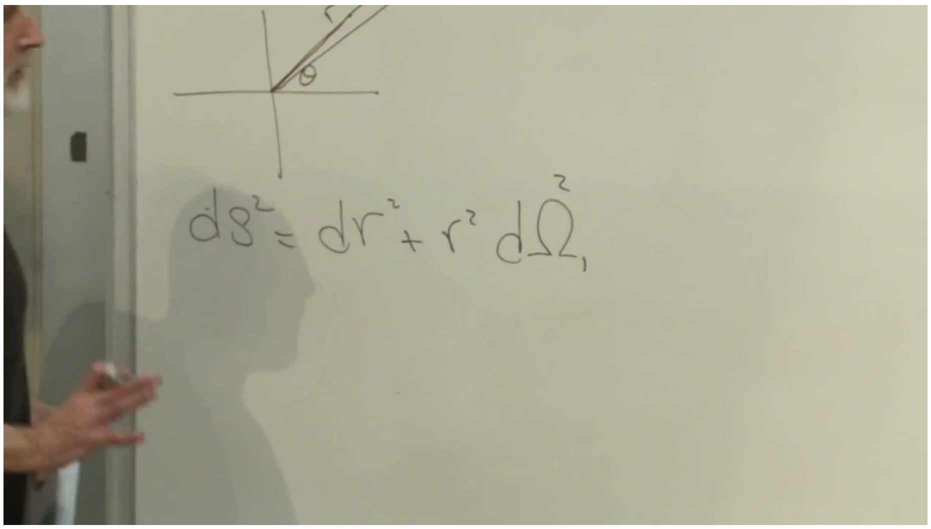

Figure 3.2: The polar-coordinate sketch and metric of flat space. The board shows $d\Omega^2$; the explanation identifies it as the unit-circle metric $d\Omega_1^2 = d\theta^2$.

Lecture frame: 00:08:37.254.

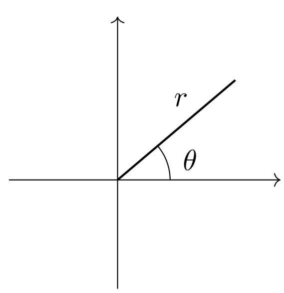

Figure 3.3: A clean reconstruction of the observer-centered polar coordinates used in equation (3.3).

The board uses $d\Omega^2$ without a legible subscript; the spoken definition fixes the intended object as $d\Omega_1^2$.

> **Question from the audience.** Does $d\theta^2$ really measure distance along the circumference?
>
> **Response.** Yes. On a unit circle, arc length equals angle in radians, so neighboring points separated by $d\theta$ have squared intrinsic separation $d\theta^2$. The circle has circumference 2π, and integrating $ds = d\theta$ once around it gives precisely 2π. Multiplying by r^2 enlarges the unit circle to the shell of radius r in the plane. [1]

The word *intrinsic* is essential. If we embed the unit circle in a Euclidean plane, we may describe it by

$$x^2 + y^2 = 1. \tag{3.4}$$

That equation describes how the circle sits in a larger space. It is not needed by an inhabitant who can measure only distances along the circle. We could bend a flexible optical fiber into a lopsided closed loop without stretching it. Its appearance in the plane would change, but its intrinsic circumference and its metric would not. Equal increments of θ would still mark equal distances along the fiber. Geometry must therefore be separated from the accidental shape supplied by an embedding.

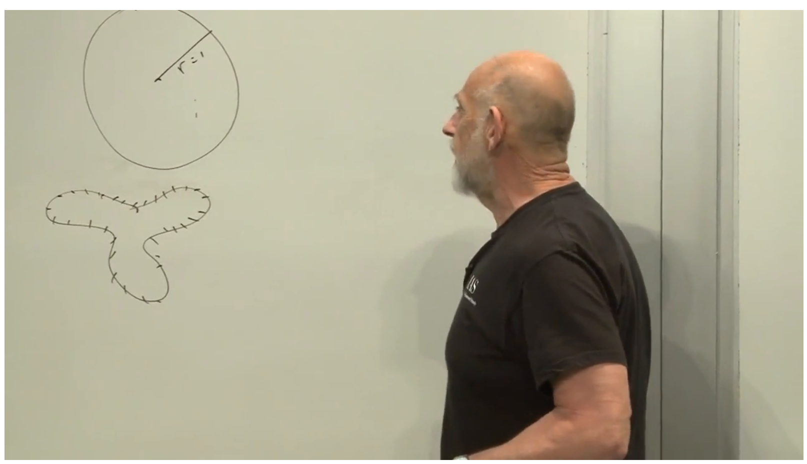

Figure 3.4: The board compares a round unit circle with a deformed closed loop. If the loop is bent without stretching, the two have the same intrinsic one-dimensional metric.

Lecture frame: 00:10:34.205.

Flat space can now be read as a nested family of circles around the observer. The shell at distance r is a copy of the unit circle enlarged by r, which is exactly what the factor $r^2 d\Omega_1^2$ records.

3.3 Positive Curvature: Shells That Close

A sphere is curved but homogeneous: no point on a perfectly round sphere is geometrically distinguished from another. We can build its metric without appealing to an external three-dimensional picture. Imagine two-dimensional astronomers who live on a unit 2-sphere and place the origin of their coordinates at their own location.

[1] Lecture timestamp: 00:09:11–00:09:26.

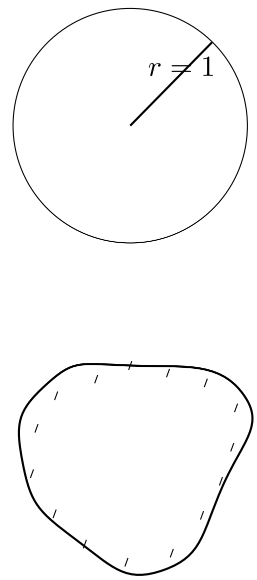

Figure 3.5: Intrinsic distance survives an extrinsic deformation: a one-dimensional inhabitant measures the same circumference in either embedding.

The circle at a small geodesic distance r around them is small. As r increases, the circles grow, reach their greatest circumference halfway around the sphere, and then shrink until they collapse at the antipodal point. This growth and contraction occur with *distance*, not with time. The sphere is static throughout this construction.

Here r is angular distance along the unit sphere rather than Euclidean distance from a center:

$$0 \leq r \leq \pi.$$

At $r = 0$ lies the chosen origin, at $r = \pi/2$ the great circle halfway around, and at $r = \pi$ the antipode. The radius of the shell is now $\sin r$, so the flat factor r^2 is replaced by $\sin^2 r$:

$$d\Omega_2^2 = dr^2 + \sin^2 r \, d\Omega_1^2 = dr^2 + \sin^2 r \, d\theta^2. \tag{3.5}$$

The endpoints and midpoint behave exactly as required:

$$\sin 0 = 0, \qquad \sin \frac{\pi}{2} = 1, \qquad \sin \pi = 0. \tag{3.6}$$

Thus the shells begin at zero size, attain their maximum size, and close again.

The notation $d\Omega_2^2$ means the metric of the unit 2-sphere, just as $d\Omega_1^2$ means the metric of its circular shells. The same recursive construction works one dimension higher. At fixed radius in flat three-space, the shell around us is a 2-sphere:

$$ds^2_{\text{flat},3} = dr^2 + r^2 d\Omega_2^2. \tag{3.7}$$

If those 2-sphere shells grow and then close with the factor $\sin r$, the resulting space is a unit 3-sphere:

$$d\Omega_3^2 = dr^2 + \sin^2 r \, d\Omega_2^2. \tag{3.8}$$

The 3-sphere is difficult to picture because its familiar embedding would require four Euclidean dimensions. The metric, however, is complete and needs no picture. Around any chosen observer sit nested 2-spheres; they grow to a maximum and then shrink to the antipodal point.

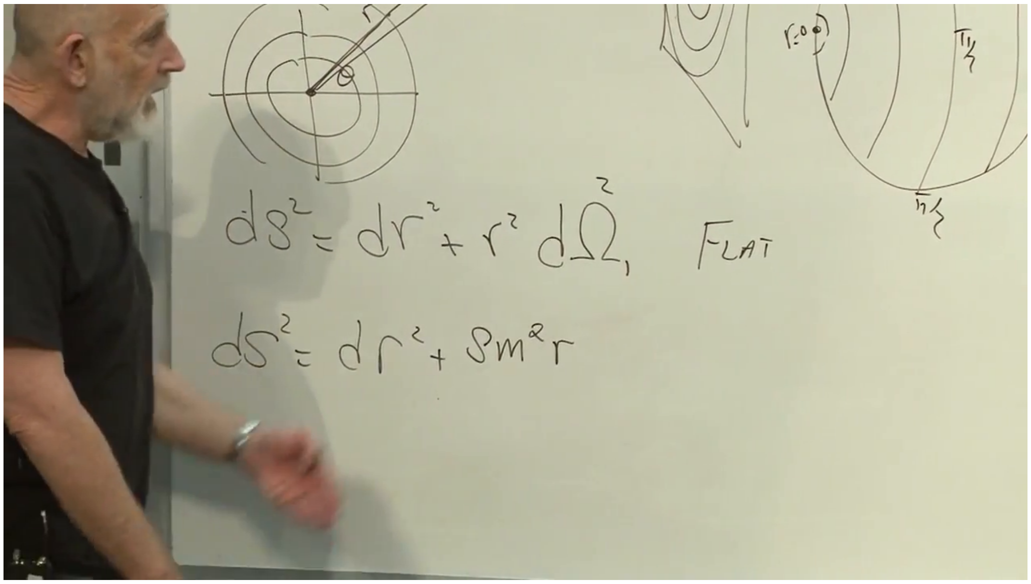

Figure 3.6: The transition from nested circles in flat space to shells on the sphere. The lower formula is partly obscured in the frame; its spoken derivation gives the replacement $r^2 \mapsto \sin^2 r$ in equation (3.5).

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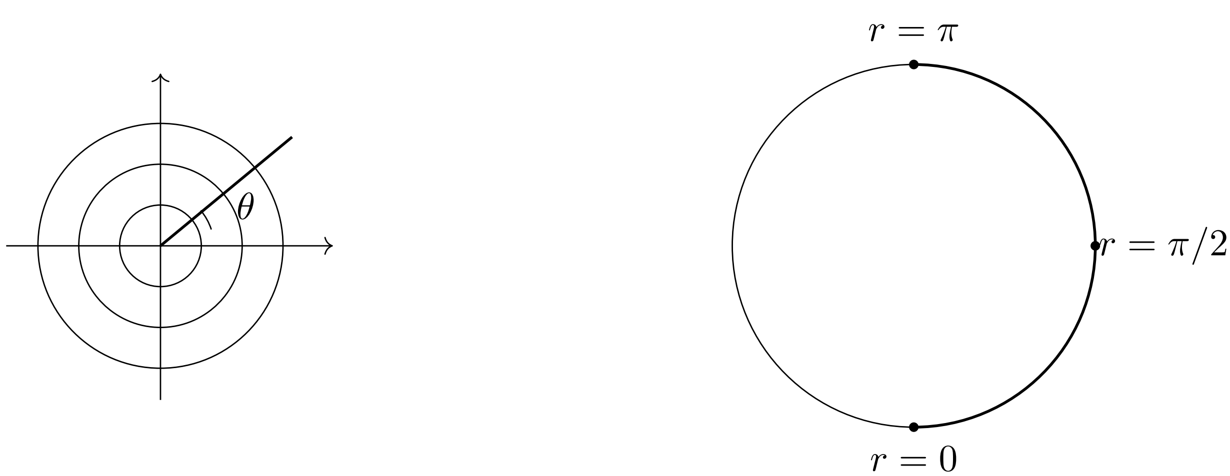

Figure 3.7: Flat shells grow forever, whereas spherical shells grow to a maximum at $r = \pi/2$ and close at $r = \pi$.

Question from the audience. Is the observer at the center of the sphere?

Response. The observer is at the center of the *coordinate shells*, not at a privileged geometric center. The origin is a point on the space, and any other observer may choose their own point as $r = 0$ and obtain the same metric. Homogeneity is precisely the statement that no such choice is preferred. [2]

For visualization alone, the unit circle and the unit 2-sphere can be embedded in Euclidean spaces:

$$x^2 + y^2 = 1, \qquad x^2 + y^2 + z^2 = 1. \tag{3.9}$$

These equations describe embeddings, not extra physics. An ant confined to the circle or sphere can determine the intrinsic metric without detecting the surrounding Euclidean space. Likewise, S^3 is not an ordinary ball with a mysterious boundary. It is the complete three-dimensional space described by equation (3.8).

3.4 How Curvature Changes What We See

Geometry becomes physics only when it changes an observation. Suppose an object has known transverse diameter d, its radial distance r has been inferred independently, and its angular diameter $d\theta$ is measured on the sky. Galaxies provide a rough intuitive example; more precise cosmology uses calibrated standard rulers and standard candles. Distance information can be assembled from spectral redshift, the Hubble relation, luminosity, and cross-calibration among different classes of objects.

In flat space, hold r fixed across the transverse width of the object. Equation (3.2) gives

$$d^2 = r^2 d\theta^2. \tag{3.10}$$

Therefore

$$d\theta = \frac{d}{r}. \tag{3.11}$$

Angular size falls inversely with distance.

On a unit sphere, the transverse shell size is $\sin r$ rather than r:

$$d^2 = \sin^2 r \, d\theta^2, \qquad \text{hence} \qquad d\theta = \frac{d}{\sin r}. \tag{3.12}$$

For $0 < r < \pi$, $\sin r < r$, so the same object at the same radial distance looks larger than it would in flat space. Its angular size first decreases, reaches a minimum at the equator, and then increases as the shell contracts. In the limiting idealization at the antipode, all directions meet and the image spreads around the sky.

Question from the audience. Is this the same idea used to infer curvature from the cosmic microwave background?

Response. Yes. The early plasma supplies a characteristic physical scale through its acoustic oscillations. We measure the angle subtended by that standard ruler and compare it with the angle predicted by each geometry. The objects need not literally be galaxies; the essential comparison is a known transverse scale against its observed angular size. [3]

[2] Lecture timestamp: 00:19:21–00:20:27.
[3] Lecture timestamp: 00:37:37–00:38:12.

Shell volumes provide a second test. In three flat dimensions, the area of a shell is proportional to r^2; on a unit 3-sphere it is proportional to $\sin^2 r$. A homogeneous population therefore first becomes more numerous with distance on a spherical shell and then less numerous after the equator. Positive curvature predicts a paired signature relative to flat space: sufficiently distant standard rulers look too large, and sufficiently distant shells contain too few objects.

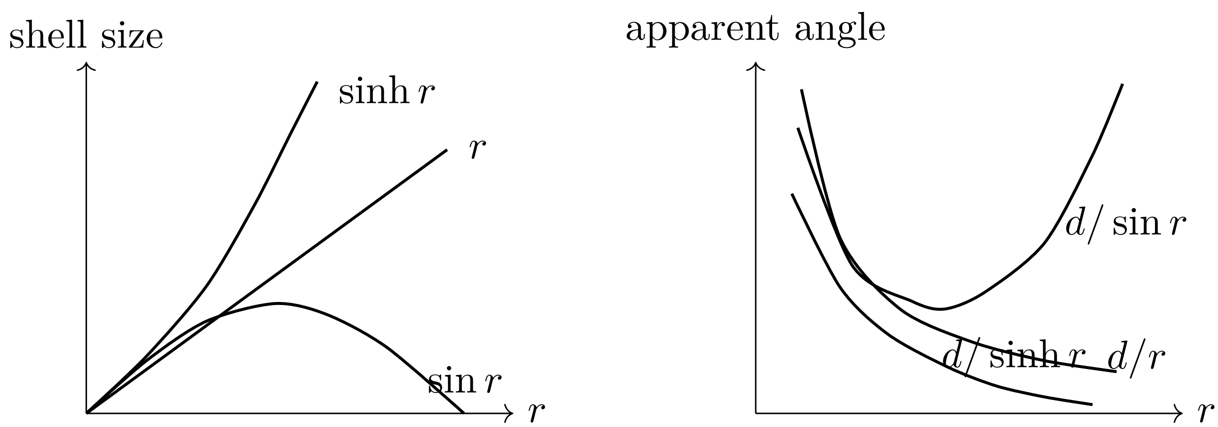

Figure 3.8: The radial shell factors and corresponding angular-size laws. Positive curvature makes the shell close; negative curvature makes it grow exponentially.

Question from the audience. Does the radial coordinate become infinite as we continue around the sphere?

Response. No. On the unit sphere, r is geodesic angular distance and ranges only from 0 to π. At $r = \pi$ the shell has collapsed at the antipode. We are examining a fixed spatial geometry here, not following an object outward through time. Introducing a curvature radius later multiplies all physical distances by that radius but does not change this dimensionless range. [4]

3.5 Projection Is Not Geometry

Before turning to negative curvature, it is useful to see how badly a faithful geometry can be distorted by a picture. Rest a sphere on a plane tangent at its south pole. Draw a line from the north pole through each point P on the sphere and continue it to the plane at P'. This is stereographic projection. The north pole itself is sent to infinity. A small region near the south pole is represented with little distortion, whereas equal intrinsic regions near the north pole become enormous on the plane.

The distortion belongs to the map, not to the sphere. The source sphere remains homogeneous even though its planar image assigns wildly different coordinate sizes to equivalent regions. This discipline will be needed again when an infinite hyperbolic space is compressed into a finite disk.

[4] Lecture timestamp: 00:39:30–00:40:35.

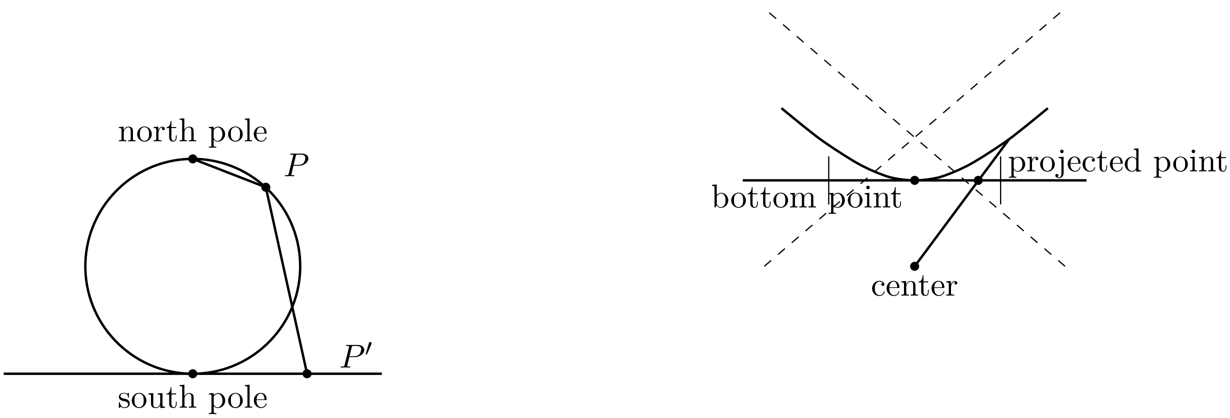

Figure 3.9: Left: stereographic projection sends the north pole of a sphere to infinity. Right: a hyperboloid can be projected into a finite disk, represented here by a cross-sectional interval. Coordinate distortion does not destroy intrinsic homogeneity.

3.6 Negative Curvature: Shells That Outgrow Flat Space

The third simply connected homogeneous geometry replaces $\sin r$ by the hyperbolic sine. Its unit metrics in two and three dimensions are

$$dh_2^2 = dr^2 + \sinh^2 r \, d\Omega_1^2, \tag{3.13}$$

$$dh_3^2 = dr^2 + \sinh^2 r \, d\Omega_2^2. \tag{3.14}$$

The coordinate r now ranges over

$$0 \le r < \infty. \tag{3.15}$$

Near the origin, $\sinh r \approx r$, so the geometry is locally Euclidean. Far away, its shells grow much faster than Euclidean shells and never turn around.

The growth law follows from

$$\sin r = \frac{e^{ir} - e^{-ir}}{2i}, \qquad \sinh r = \frac{e^{r} - e^{-r}}{2}. \tag{3.16}$$

At large r, $\sinh r \sim e^r/2$. In two dimensions, circumference therefore grows exponentially with radius; in three dimensions, shell area grows as $\sinh^2 r \sim e^{2r}/4$. A homogeneous distribution contains correspondingly many objects on a distant shell.

For a transverse object of size d,

$$d^2 = \sinh^2 r \, d\theta^2, \qquad \text{so} \qquad d\theta = \frac{d}{\sinh r}. \tag{3.17}$$

At large r,

$$d\theta \sim 2d \, e^{-r}. \tag{3.18}$$

Negative curvature therefore gives the opposite signature from positive curvature: distant standard rulers look too small, while distant shells contain too many objects relative to flat space.

Question from the audience. How do the radial ranges distinguish spherical and hyperbolic space?

Response. On the unit sphere, r stops at the antipode, $r = \pi$. In hyperbolic space it continues without bound, $0 \le r < \infty$. The shell factor reveals the same contrast: $\sin r$ returns to zero, whereas $\sinh r$ increases forever. [5]

There is an embedding that makes homogeneity less mysterious. Introduce ambient coordinates T, x, y, where T is only an embedding coordinate and is *not* cosmological time, and take the upper sheet

$$T^2 - x^2 - y^2 = 1. \tag{3.19}$$

Equip the ambient space with the Lorentz-signature metric

$$d\ell^2_{\text{emb}} = -dT^2 + dx^2 + dy^2. \tag{3.20}$$

The metric induced on the hyperboloid is positive definite. Lorentz transformations preserve both equations (3.19) and (3.20), and they can move any point of the upper sheet to any other. In that sense, changing the origin on hyperbolic space is like tilting the time axis in the auxiliary ambient space. Every point is geometrically equivalent.

Project the hyperboloid into a disk and infinitely distant points accumulate at the boundary circle. Nearby regions are drawn nearly faithfully; regions farther away are compressed ever more strongly. This is the Poincaré disk representation familiar from repeating figures of angels and devils. A coordinate transformation can move an apparently tiny figure near the edge to the center, where it becomes large, while preserving all intrinsic lengths and angles. In three dimensions the corresponding model is a ball rather than a disk. The names *Poincaré space*, *Lobachevsky space*, and *space of uniform negative curvature* refer to this same local geometry.

Question from the audience. Is the finite disk like a Penrose diagram?

Response. It is similar in the limited sense that an infinite domain is represented inside a finite picture. Here, however, the disk represents space alone, not spacetime or causal infinity. Its boundary is infinitely far away in the intrinsic hyperbolic metric even though the drawing places it at a finite coordinate radius. [6]

3.7 The Radius of Curvature

The metrics written so far describe unit geometries. To give a sphere or hyperbolic space an arbitrary curvature radius, multiply every length by a constant. The board denotes that constant by a; to avoid confusing it with the time-dependent cosmological scale factor used later, we shall call the static curvature radius R_c. Since the metric measures squared length,

$$ds^2_{\text{sph}} = R_c^2 d\Omega_n^2 = R_c^2 \left(dr^2 + \sin^2 r\, d\Omega^2_{n-1}\right), \tag{3.21}$$

$$ds^2_{\text{hyp}} = R_c^2 dh_n^2 = R_c^2 \left(dr^2 + \sinh^2 r\, d\Omega^2_{n-1}\right). \tag{3.22}$$

[5] Lecture timestamp: 00:52:46–00:53:25.
[6] Lecture timestamp: 01:02:55–01:03:05.

The physical radial distance is $R_c r$. Enlarging R_c makes the space less curved; the magnitude of constant sectional curvature scales as $1/R_c^2$. In the limit $R_c \to \infty$, both curved metrics approach Euclidean geometry on every fixed finite region.

Question from the audience. If the sphere is made twice as large, does it become more curved or less curved?

Response. Less curved. The first answer offered in class was corrected by comparing a basketball with Earth: the smaller surface bends more rapidly over a given distance. Doubling R_c halves the inverse-radius curvature scale and divides sectional curvature $1/R_c^2$ by four. [7]

This explains why all three geometries look alike nearby. For $r \ll 1$,

$$\sin r = r - \frac{r^3}{6} + \cdots , \qquad \sinh r = r + \frac{r^3}{6} + \cdots . \tag{3.23}$$

The leading term in either curved metric is the flat polar metric; curvature first appears through corrections of relative order $(L/R_c)^2$ across a physical scale L.

Question from the audience. Do observations tell us whether space is flat, spherical, or hyperbolic?

Response. They tell us that the observable region is very close to flat. The rough classroom estimate was that a one-percent bound on curvature effects implies R_c at least about ten times the size of the visible region, because $(L/R_c)^2 \lesssim 10^{-2}$. A corresponding curvature volume would then be at least roughly 10^3 visible volumes. This is an order-of-magnitude lecture estimate, not a current precision-data quotation, and exact flatness remains the $R_c = \infty$ possibility. [8]

3.8 Flat but Finite: The Torus

We must now correct the statement that flat, spherical, and hyperbolic spaces are the only homogeneous possibilities. They are the three simply connected constant-curvature geometries. A flat space can also have nontrivial global topology.

Begin with a Euclidean rectangle. Identify its left edge point by point with its right edge, and identify its bottom edge with its top. Crossing any edge returns through the opposite one. The local metric remains $dx^2 + dy^2$, but the global space is periodic and finite. This quotient is a flat two-torus T^2. One periodic dimension gives a circle T^1, identifying opposite faces of a rectangular box gives a flat three-torus T^3, and the construction extends to any number of dimensions.

Question from the audience. Can a universe therefore be both flat and finite?

Response. Yes. Flatness is a local statement about the metric; finiteness can be a global statement about topology. A sufficiently large three-torus would look Euclidean throughout the region we can observe even though its total volume is finite. [9]

[7] Lecture timestamp: 01:10:14–01:10:56.
[8] Lecture timestamp: 01:11:00–01:13:37.
[9] Lecture timestamp: 01:13:37–01:17:59.

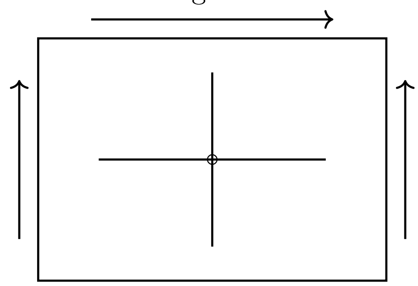

Figure 3.10: A flat torus constructed by identifying opposite sides of a rectangle. The edges are not boundaries; paired points are the same points of the quotient space.

The universal-cover picture makes the observable consequence transparent. Instead of identifying the edges, tile an infinite Euclidean plane with copies of the fundamental rectangle. A single object in the torus appears once in every copy. In three dimensions the copies form a periodic crystal-like array.

> **Question from the audience.** Could we live in a toroidal universe without noticing it?
>
> **Response.** Yes, if the fundamental cell is larger than the region from which light has reached us. If it were small enough, light could traverse the identifications and we would see repeated images of the same galaxies, perhaps even repeated views of our own neighborhood, in different directions. The local geometry would remain flat in either case. [10]

> **Question from the audience.** Why does the sphere stop at its antipode while the torus repeats?
>
> **Response.** On the sphere, the surrounding shell genuinely shrinks to zero at $r = \pi$; all radial directions meet at the antipode. On a torus, the edge of a chosen rectangular cell never shrinks. It is identified with an equally large opposite edge, so crossing it continues the same space in a periodic coordinate description. [11]

3.9 From Space to Spacetime

We can now restore time. In units with the speed of light $c = 1$, the Minkowski metric is

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2. \tag{3.24}$$

The sign convention is a choice; what matters is that the temporal term has the opposite sign from the spatial terms. A light ray follows a null trajectory,

$$ds^2 = 0. \tag{3.25}$$

[10] Lecture timestamp: 01:18:16–01:22:04.
[11] Lecture timestamp: 01:22:06–01:23:02.

For motion only along x,

$$-dt^2 + dx^2 = 0 \qquad \Rightarrow \qquad dx = \pm dt. \tag{3.26}$$

The two signs describe light moving left or right at unit speed.

For cosmology, keep the proper-time term $-dt^2$, replace the Euclidean spatial part by a homogeneous spatial metric, and let its overall scale vary with time. We denote this scale factor by $a(t)$. It is now a dynamical function rather than the fixed radius R_c used in the static comparison.

As a pictureable $2+1$-dimensional example, take

$$ds^2 = -dt^2 + a(t)^2 d\Omega_2^2. \tag{3.27}$$

At each fixed t, space is a 2-sphere of radius $a(t)$. Stacking these spheres in time gives the spacetime. The familiar $3+1$-dimensional construction is

$$ds^2 = -dt^2 + a(t)^2 d\Sigma^2, \tag{3.28}$$

where $d\Sigma^2$ is a time-independent unit spatial metric. The three simply connected cases are

$$\text{flat:} \quad ds^2 = -dt^2 + a(t)^2 \left(dx^2 + dy^2 + dz^2\right), \tag{3.29}$$

$$\text{spherical:} \quad ds^2 = -dt^2 + a(t)^2 d\Omega_3^2, \tag{3.30}$$

$$\text{hyperbolic:} \quad ds^2 = -dt^2 + a(t)^2 dh_3^2. \tag{3.31}$$

The relative scale of comoving coordinates is fixed while $a(t)$ changes. In the spherical and hyperbolic normalizations, $a(t)$ is also the instantaneous curvature radius. In the flat case its absolute normalization can be absorbed into the comoving coordinates.

The Hubble law follows before any gravitational field equation is imposed. Consider two nearby comoving points with fixed coordinate separation χ. Their physical separation is

$$D(t) = a(t)\chi. \tag{3.32}$$

Differentiating at fixed χ gives

$$V = \dot{D} = \dot{a}(t)\chi. \tag{3.33}$$

Eliminating χ,

$$V = \frac{\dot{a}}{a} D. \tag{3.34}$$

Thus

$$H = \frac{\dot{a}}{a}. \tag{3.35}$$

This is a kinematic identity for homogeneous expansion or contraction. It holds for all three spatial geometries: $\dot{a} > 0$ gives recession, and $\dot{a} < 0$ gives approach.

Geometry alone does not determine $a(t)$. Functions such as

$$a(t) \propto t^{2/3}, \qquad a(t) \propto t^{1/2}, \qquad a(t) \propto e^{Ht} \tag{3.36}$$

all fit the homogeneous metric ansatz but describe different cosmic matter and energy contents. General relativity supplies the dynamical equation that selects among them. The resulting Friedmann equation will also connect the flat, spherical, and hyperbolic cases with the zero-, negative-, and positive-energy alternatives of the Newtonian cosmology developed earlier.

Question from the audience. Could the geometry and the cosmic evolution be much more complicated than these models?

Response. Certainly. Homogeneity and isotropy are restrictive assumptions, chosen because they are natural, observationally successful on large scales, and mathematically tractable. Many more complicated geometries and evolutions exist. An important dynamical question is whether broad classes of initially irregular models evolve toward one of these simple homogeneous forms. [12]

Question from the audience. Is the 3-sphere the boundary of a four-dimensional object?

Response. It can be represented that way as an embedding, but the cosmological space is the 3-sphere itself. It has three intrinsic dimensions and no boundary. The four-dimensional Euclidean picture is optional scaffolding; equation (3.8) already defines the geometry completely. [13]

Question from the audience. Is luminosity the only way to obtain the distance needed for the curvature test?

Response. No. Spectral lines identify redshift, nearby distance ladders calibrate the Hubble relation, and supernovae provide approximately standard luminosities. These methods are cross-checked rather than trusted in isolation. Even the inverse-square luminosity law depends on geometry because the area of a light front is set by r^2, $\sin^2 r$, or $\sinh^2 r$. The agreement of several observables is what supports near-flatness. [14]

3.10 What to Carry Forward

The geometry of a homogeneous universe can be read from the growth of shells around any observer:

$$f(r) = \begin{cases} r, & \text{flat space,} \\ \sin r, & \text{positive curvature,} \\ \sinh r, & \text{negative curvature.} \end{cases} \tag{3.37}$$

The spatial metric is $dr^2 + f(r)^2 d\Omega_{n-1}^2$. The same function controls angular sizes and the number of objects available on a shell, so curvature has direct observational meaning. A large curvature radius hides these differences locally, and a toroidal identification shows that local curvature does not by itself determine global size or topology.

Adding $-dt^2$ and the common factor $a(t)^2$ turns any of the three homogeneous spatial metrics into a cosmological spacetime. It immediately yields $V = HD$ with $H = \dot{a}/a$, but it leaves $a(t)$ undetermined. The next step is therefore not another exercise in geometry. It is the dynamical problem: use Einstein's equations to determine how matter, radiation, curvature, and other forms of energy make the scale factor evolve.

[12] Lecture timestamp: 01:36:00–01:36:50.
[13] Lecture timestamp: 01:36:55–01:37:59.
[14] Lecture timestamp: 01:38:00–01:41:07.

Chapter 4

Friedmann Dynamics and the Equation of State

Overview. Homogeneous expansion must be separated from local bound physics, and intrinsic geometry must be separated from the shape of an embedding. Once those distinctions are secure, the cosmological principle reduces the spacetime metric to one scale factor $a(t)$ and one curvature label k. The time-time component of Einstein's equation then becomes the Friedmann energy equation. To solve it we still need the material relation between pressure and energy density. Adiabatic thermodynamics and the equation of state $P = w\rho$ give $\rho \propto a^{-3(1+w)}$, unifying matter, radiation, and vacuum energy in one framework.

4.1 Questions That Fix the Physical Picture

4.1.1 Could Space Repeat Before the Horizon?

A finite flat universe can be represented by a periodic cell. If that cell were smaller than the observable region, light could traverse it and the sky would contain repeated views of the same structures. Detecting those repetitions is harder than simply recognizing a familiar galaxy: the images arrive from different directions and epochs and are altered by cosmic evolution. Nevertheless, periodic topology has observable consequences.

> **Question from the audience.** Could the whole universe be smaller than the observable universe?
>
> **Response.** A periodically identified torus is the natural proposal. We would then be observing repeated copies of the same physical region, figuratively seeing ourselves through the back door. Searches for the corresponding cosmic periodicity have not produced supporting evidence and do place constraints on small fundamental cells. The possibility is geometrically consistent, but a universe smaller than the region already observed is not favored by this evidence. [1]

[1] Lecture timestamp: 00:01:17–00:02:33.

4.1.2 Expansion Does Not Pull Apart Bound Systems

Cosmic expansion describes the large-scale motion of freely moving matter in the average cosmological geometry. It is not a universal command that every proper length must increase. Electromagnetic forces bind atoms, gravity binds planetary systems, and ordinary molecular forces bind solid objects. These interactions are enormously stronger than the cosmological correction on such scales.

> **Question from the audience.** Is the expansion caused by dark energy, and why do atoms or planetary orbits not expand with space?
>
> **Response.** The universe was known to expand before dark energy was discovered. Dark energy changes the expansion by accelerating it; it is not the definition or original cause of expansion. A bound system does not follow the Hubble flow because its local forces dominate. An accelerated background can produce an exceedingly small perturbation of an atom or an orbit, but ordinary binding restores an almost unchanged equilibrium rather than allowing the system to fly apart. [2]

4.1.3 Energy in a Time-Dependent Geometry

The remaining objection is not about force but bookkeeping. If a bound orbit continually resists a small cosmological perturbation, where does the required energy come from? The premise imports a conservation law from a static background into a spacetime that has no global time-translation symmetry.

> **Question from the audience.** Where does the energy come from when local binding resists expansion?
>
> **Response.** Energy conservation in a static world follows from time-translation invariance: the background and its laws look the same when an experiment is shifted in time. An expanding spacetime is not time-translation invariant in that global sense, so there need not be a single conserved total energy for the entire universe. Local stress-energy conservation still holds and will appear below as the continuity equation. What fails is the naive claim that a globally defined cosmic energy must remain fixed while the geometry changes. [3]

4.1.4 Homogeneity Is a Coarse-Grained Statement

The cosmological principle is scale dependent. Air in a closed room may be nearly uniform on the scale of the room but is granular on molecular scales. The universe likewise contains planets, galaxies, clusters, and density fluctuations. Homogeneity and isotropy refer to its averaged behavior over distances large enough that those structures cease to select a location or direction.

> **Question from the audience.** Do small-scale bound structures contradict homogeneity and isotropy?

[2] Lecture timestamp: 00:02:36–00:06:50.
[3] Lecture timestamp: 00:06:52–00:09:24.

Response. No. Exact homogeneity fails on small scales, just as it fails for air when individual molecules are resolved. On scales of hundreds of millions of light-years and larger, the observed distribution is approximately uniform and direction independent. Beyond the region we can observe, homogeneity remains an assumption whose consequences must be tested rather than an established fact. [4]

4.2 Homogeneity and Isotropy as Metric Statements

Homogeneity and isotropy are geometrical statements before they become dynamical ones. Einstein's equation will tell us how geometry changes with time. First we must say what kind of spatial geometry is allowed at any one time.

Question from the audience. What do homogeneity and isotropy mean mathematically?

Response. Describe the space by a metric and choose coordinates x^i centered on one point. Now choose coordinates y^i centered on another point. Homogeneity means that a transformation can move the origin while leaving the metric with the same functional form. Every point is then geometrically equivalent to every other. Isotropy adds rotational symmetry about each point, so no direction is preferred. [5]

Definition 4.1. A space is *homogeneous* if an isometry can carry any point into any other point. It is *isotropic* about a point if rotations through that point leave its geometry unchanged. A space that is homogeneous and isotropic has neither a preferred position nor a preferred direction.

The Euclidean plane makes the definition transparent. In Cartesian coordinates,

$$ds^2 = dx_1^2 + dx_2^2. \tag{4.1}$$

Move the origin by constants a_1 and a_2:

$$y_1 = x_1 + a_1, \qquad y_2 = x_2 + a_2. \tag{4.2}$$

Because $dy_i = dx_i$, the translated metric is

$$ds^2 = dy_1^2 + dy_2^2. \tag{4.3}$$

The symbols used to label the point have changed, but the distance rule has not. This is the simplest realization of homogeneity. Rotations likewise preserve the metric, so the plane is isotropic as well.

The sphere is less trivial only because its coordinates are less convenient. With the south pole chosen as origin, the unit two-sphere can be written

$$d\Omega_2^2 = dr^2 + \sin^2 r\, d\theta^2. \tag{4.4}$$

Rotate the sphere so that an east pole becomes the new origin. The coordinate transformation is cumbersome, but the result is again equation (4.4). No point on a round sphere is geometrically distinguished.

[4] Lecture timestamp: 00:09:27–00:10:55.
[5] Lecture timestamp: 00:11:13–00:14:26.

Question from the audience. Is a periodically identified flat torus both homogeneous and isotropic?

Response. It is homogeneous under translations: every point in the periodic cell has the same local geometry. It is not fully isotropic, because the identifications pick out preferred periodic directions and lengths. Looking along one of those axes is globally different from looking in a generic oblique direction. The torus therefore supplies a clean example of homogeneity without complete isotropy. [6]

Observed large-scale uniformity motivates the cosmological principle within our horizon. Extending it beyond the observable region is a postulate, not a direct observation. We accept that postulate because it gives a sharply defined geometry whose consequences can be confronted with data.

4.3 Intrinsic Geometry and Embedding

When space is called spherical, it is tempting to imagine the rubber surface of a balloon placed inside a larger room. That picture may help us draw, but the surrounding room is not part of the cosmological assertion. The geometry that matters is *intrinsic*: it consists of distances and angles measurable by observers who remain within the space.

4.3.1 Closed One-Dimensional Worlds

Consider a closed one-dimensional world. We may draw it as a circle, a square with rounded corners, or a peanut-shaped loop. An inhabitant confined to the line measures only distance along it. If two loops have the same total circumference, then those inhabitants encounter the same intrinsic geometry even when the drawings look entirely different.

Every one-dimensional metric is locally flat. A short piece of string can be straightened without stretching it, so visible bending in the surrounding plane is extrinsic rather than intrinsic curvature. A closed loop has a global modulus, its circumference, but it has no intrinsic local bending that its one-dimensional inhabitants can measure.

Question from the audience. Would a finite line segment be the same kind of one-dimensional space?

Response. No. A segment has two endpoints, whereas a closed loop has none. That topological distinction is intrinsic and cannot be removed by bending either object in a larger space. Once the topology is fixed, the segment is characterized by its length and the closed loop by its circumference. [7]

4.3.2 Why Embeddings Are Useful but Optional

We visualize a one-dimensional circle by placing it in two dimensions and a two-dimensional sphere by placing it in three. The pattern tempts us to demand a fourth Euclidean dimension before

[6] Lecture timestamp: 00:20:18–00:21:30.
[7] Lecture timestamp: 00:26:00–00:26:39.

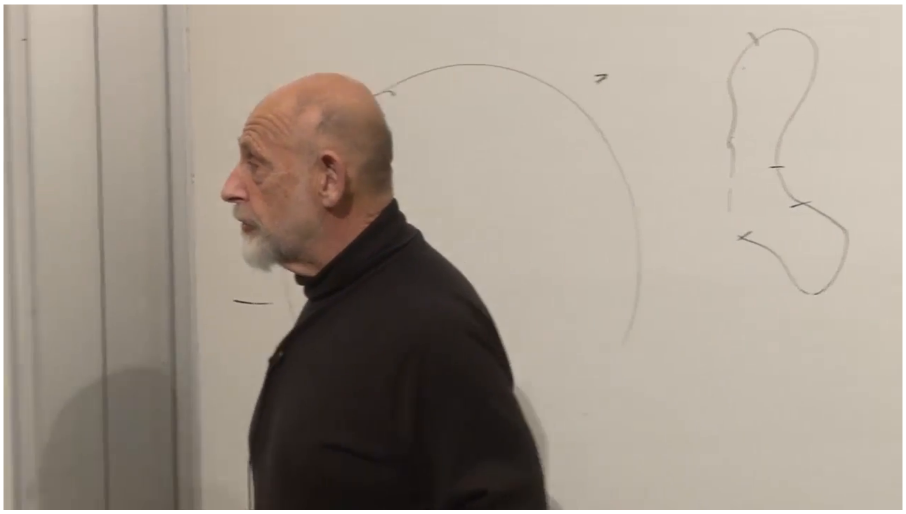

Figure 4.1: Round and deformed closed loops used to distinguish intrinsic geometry from embedding shape. This later frame preserves drawings made during the earlier discussion; the relevant content occurs at 00:23:21–00:25:59.

Lecture frame: 00:59:54.899.

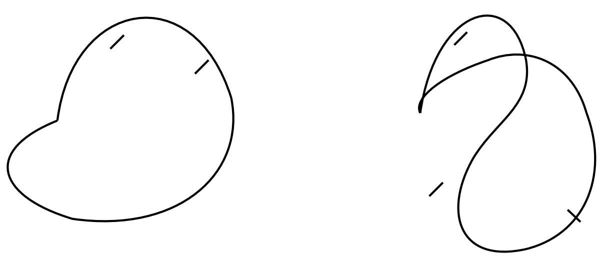

Figure 4.2: A clean reconstruction of the circle and deformed loop. Equal circumference means equal intrinsic one-dimensional geometry.

discussing a three-sphere. That demand reflects human visualization, not a mathematical necessity. Our visual machinery evolved to navigate three dimensions; coordinates and metrics let us reason beyond that limit without constructing a literal external viewpoint.

Question from the audience. Could additional spatial dimensions exist but be too small to see?

Response. Yes. Geometry does not require all dimensions to have the same observable extent. A compact dimension could be much smaller than the scales probed by ordinary experience. The immediate point, however, is simpler: whether or not extra dimensions exist physically, an embedding dimension is unnecessary for defining the intrinsic metric of our three-dimensional space. [8]

Question from the audience. How would an inhabitant confined to a curved line know whether the line is bent?

Response. It could not know from intrinsic measurements alone. Imagine light and matter confined to an optical fiber. Bending the fiber without stretching it changes the embedding but not the distances or signal travel times measured along the fiber. Only a signal that escaped into the surrounding space could reveal the external bend. [9]

The thought experiment can be made operational. A creature restricted to the fiber could launch a signal, such as a sufficiently energetic photon, and discover an external direction only if the signal escaped the fiber and later returned by a route unavailable within the one-dimensional metric. In the same spirit, the statement that ordinary space has three large dimensions is ultimately experimental, not a deduction from what the human mind can picture.

Question from the audience. If every one-dimensional space is locally flat, must every two-dimensional space also be locally flat?

Response. No. A sheet rolled into a cylinder stays intrinsically flat: it can be unrolled without stretching. A hemisphere cannot be flattened onto a plane without stretching or tearing, so it has intrinsic curvature. Every sufficiently small patch looks approximately Euclidean, but measurable deviations appear on a finite scale. The contrast between a cylinder and a sphere is precisely the contrast between extrinsic bending and intrinsic curvature. [10]

Question from the audience. Does intrinsic curvature necessarily mean that a gravitational field is present?

Response. In general relativity, spacetime curvature is the invariant manifestation of tidal gravity. The cosmological spatial curvature k/a^2 is therefore part of the gravitational geometry, not merely a bend in an imagined external space. Matter influences curvature through Einstein's equation, but one should not sharpen the statement into "curvature exists only where matter sits": vacuum spacetime can carry Weyl curvature, as outside a star or in a gravitational wave. [11]

[8] Lecture timestamp: 00:29:55–00:30:12.
[9] Lecture timestamp: 00:30:15–00:31:14.
[10] Lecture timestamp: 00:31:14–00:35:00.
[11] Lecture timestamp: 00:35:01–00:35:51.

We now impose the large-scale idealization. Average over stars, galaxies, clusters, and local irregularities, just as one ignores mountains when describing Earth as round. The resulting spatial geometry is taken to be homogeneous and isotropic. Under these assumptions there are three constant-curvature possibilities: flat, spherical, and hyperbolic.

4.4 The FRW Ansatz and the Three Curvature Cases

The symmetry assumptions collapse the geometry to one time-dependent scale and one discrete curvature sign. In comoving coordinates the Friedmann-Robertson-Walker ansatz is

$$ds^2 = -dt^2 + a(t)^2\, d\Sigma_k^2, \qquad k \in \{-1, 0, +1\}. \tag{4.5}$$

The time coordinate is the proper time of comoving observers. Homogeneity permits a to depend on time but not position, while isotropy forbids a preferred spatial direction.

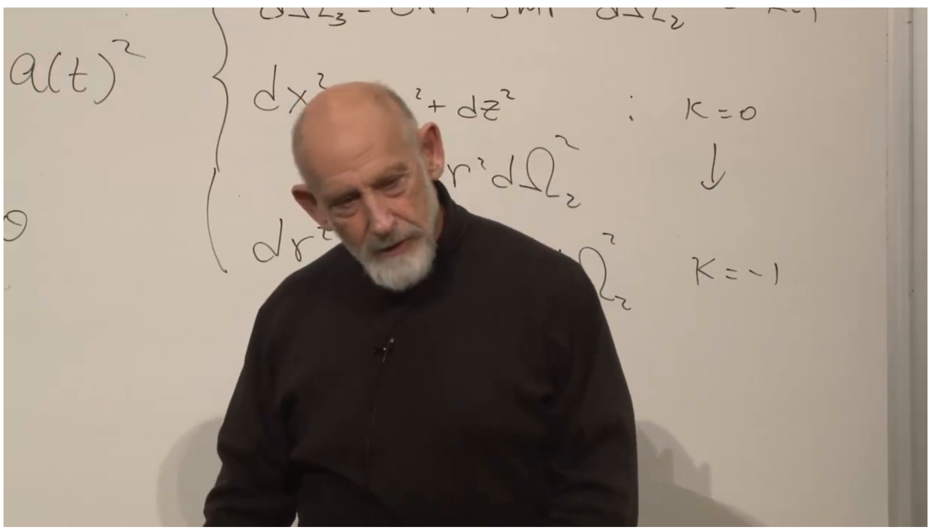

Figure 4.3: The three constant-curvature spatial metrics grouped beneath the common scale factor. The frame was captured later, after the board had accumulated other material, but it preserves the metric discussion from 00:40:23–00:42:42.

Lecture frame: 00:45:48.386.

Using a radial coordinate r and the unit two-sphere metric $d\Omega_2^2$, the three normalized spatial metrics are

$$d\Sigma_0^2 = dr^2 + r^2 d\Omega_2^2, \qquad k = 0, \tag{4.6}$$
$$d\Sigma_{+1}^2 = dr^2 + \sin^2 r\, d\Omega_2^2, \qquad k = +1, \tag{4.7}$$
$$d\Sigma_{-1}^2 = dr^2 + \sinh^2 r\, d\Omega_2^2, \qquad k = -1. \tag{4.8}$$

At a fixed time, space is therefore flat, positively curved, or negatively curved. Multiplication by $a(t)^2$ changes every spatial distance by the same factor while preserving the normalized shape.

Take two galaxies at fixed comoving separation $\Delta\chi$. Their physical distance is

$$D(t) = a(t)\Delta\chi. \tag{4.9}$$

Differentiating and eliminating $\Delta\chi$ gives

$$\dot{D} = \dot{a}\,\Delta\chi = \frac{\dot{a}}{a}D = H(t)D, \qquad H(t) \equiv \frac{\dot{a}}{a}. \tag{4.10}$$

Thus Hubble's law follows kinematically from homogeneous scaling. The Hubble parameter can change with time, but at any fixed cosmic time it is the same everywhere.

4.5 From Einstein's Equation to Friedmann's Equation

The metric tells us how distances are measured, but not how $a(t)$ evolves. Curved spacetime requires Einstein's equation. Computing every Christoffel symbol and curvature component would obscure the physical argument, so we use the form forced by symmetry and isolate its time-time component.

In units with $c = 1$ and with a cosmological constant temporarily omitted, the full tensor equation is

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G\,T_{\mu\nu}. \tag{4.11}$$

Geometry occupies the left-hand side; energy density, momentum density, pressure, and stress occupy the right. The factor 1/3 does not belong in the full tensor equation. It appears only after the time-time component has been reduced to Friedmann form.

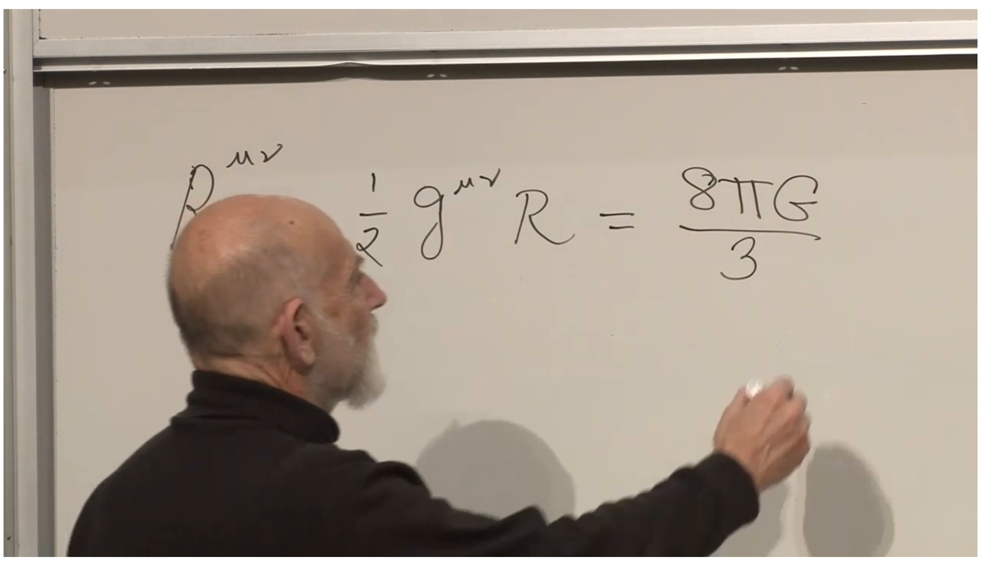

Figure 4.4: An intermediate board state of Einstein's equation. The geometric left-hand side is visible, while the matter tensor is not yet legible and the displayed normalization is incomplete. Equation (4.11) records the standard completed tensor equation used in the derivation.

Lecture frame: 00:48:19.394.

For a homogeneous fluid at rest in comoving coordinates,

$$T_{00} = \rho. \tag{4.12}$$

The time-time Einstein tensor has a useful structural property. Its second derivatives cancel, leaving one term quadratic in the first time derivative and one term from spatial curvature:

$$G_{00} = 3\left[\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2}\right]. \tag{4.13}$$

Substituting equations (4.12) and (4.13) into equation (4.11) gives the first Friedmann equation,

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho}. \tag{4.14}$$

Equivalently,

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{4.15}$$

Equation (4.15) has exactly the form found in the earlier Newtonian argument. There the integration constant was interpreted as the total mechanical energy or escape condition of a sphere of matter. General relativity places intrinsic spatial curvature in the same mathematical slot. The agreement is not an accident: in a sufficiently small region, curvature effects become negligible and relativistic cosmology must reproduce local Newtonian behavior. Relativity also clarifies that ρ means total energy density, so relativistic particles and photons gravitate even though a nonrelativistic mass-density argument would miss them.

Question from the audience. Why are there no terms mixing space and time in the metric?

Response. A mixed term would behave like a preferred spatial vector or flow relative to the comoving observers. Exact homogeneity and isotropy leave no such direction available. In coordinates adapted to the cosmic matter, the shift therefore vanishes and the metric takes the block-diagonal form in equation (4.5). If the assumed symmetries are preserved, this is not an extra dynamical guess but their geometrical consequence. [12]

Question from the audience. Why is the time-time component enough, and where did the second derivatives go?

Response. The homogeneous and isotropic ansatz contains only one unknown function, $a(t)$. The mixed components reduce to identities, while the spatial components are related to the time-time equation and stress-energy conservation. One may combine them into an acceleration equation containing $\ddot{a}$, but the time-time Einstein tensor is special: its second-derivative terms cancel, leaving the first-order energy equation (4.14). One nontrivial equation plus the material evolution law is enough to determine the single geometric degree of freedom. [13]

4.6 How the Contents of the Universe Dilute

Friedmann's equation is not closed until ρ is known as a function of a. Geometry has supplied one equation; the material in the universe must supply the second relation.

For nonrelativistic matter, the number of particles in a comoving volume is fixed while the physical volume grows as a^3. Neglecting the particles' small kinetic energies,

$$\rho_{\mathrm{m}}(a) = \frac{\rho_{\mathrm{m},0}}{a^3}. \tag{4.16}$$

Here the normalization $a = 1$ may be assigned at any convenient reference epoch; only ratios of scale factors are directly meaningful.

[12] Lecture timestamp: 00:58:27–00:59:08.
[13] Lecture timestamp: 00:59:09–01:00:56.

Question from the audience. If a flat universe is infinite, what does it mean for a to shrink toward the big bang?

Response. An infinite flat space remains infinite for every nonzero value of a; all finite comoving separations are multiplied by the same shrinking factor. The absolute normalization of a is arbitrary in the flat case, so only $a(t_2)/a(t_1)$ has invariant meaning. For a closed spherical universe a is naturally its curvature radius, and for a hyperbolic universe it sets the curvature radius. The limit $a \to 0$ signals the cosmological singularity, not an ordinary finite ball sitting in external space. [14]

Radiation loses energy in two ways. Its number density falls as the inverse volume, and every photon's wavelength stretches in proportion to a, reducing its energy by another factor of a^{-1}. Therefore

$$\rho_{\mathrm{r}}(a) = \frac{\rho_{\mathrm{r},0}}{a^4}. \tag{4.17}$$

Because a^{-4} grows faster toward the past than a^{-3}, radiation dominates sufficiently early even if matter dominates much later.

If matter and radiation were the only components, the sign of k would strongly track the future: $k = +1$ permits recollapse, $k = 0$ is the marginal case that expands forever while slowing, and $k = -1$ expands forever with curvature eventually important. Dark energy breaks that simple connection. A closed universe can expand forever, and spatial flatness by itself does not determine the ultimate fate.

4.7 Equation of State and the Continuity Law

Rather than memorize a separate dilution rule for every substance, characterize the material by an equation of state. The program has two parts. First derive $\rho(a)$ for a given relation between pressure and energy density. Then determine that relation from the microscopic physics of matter, radiation, or vacuum. The first part is enough for the present lecture.

Pressure is momentum flux. Molecules striking the wall of a box reverse their momentum and push on the wall; faster particles transfer more momentum and produce greater pressure. Energy density and pressure therefore cannot be chosen independently once the kind of material is specified. For a simple cosmological component we write

$$P = w\rho, \tag{4.18}$$

where w is dimensionless in units with $c = 1$. Cold matter has negligible pressure and hence $w = 0$. Radiation has $w = 1/3$, with the factor $1/3$ reflecting the average over three spatial directions.

4.7.1 Thermodynamics in a Comoving Box

Take a comoving box with physical volume V, energy density ρ, and total energy

$$E = \rho V. \tag{4.19}$$

The first law is

$$dE = T\,dS - P\,dV. \tag{4.20}$$

[14] Lecture timestamp: 01:05:06–01:05:53.

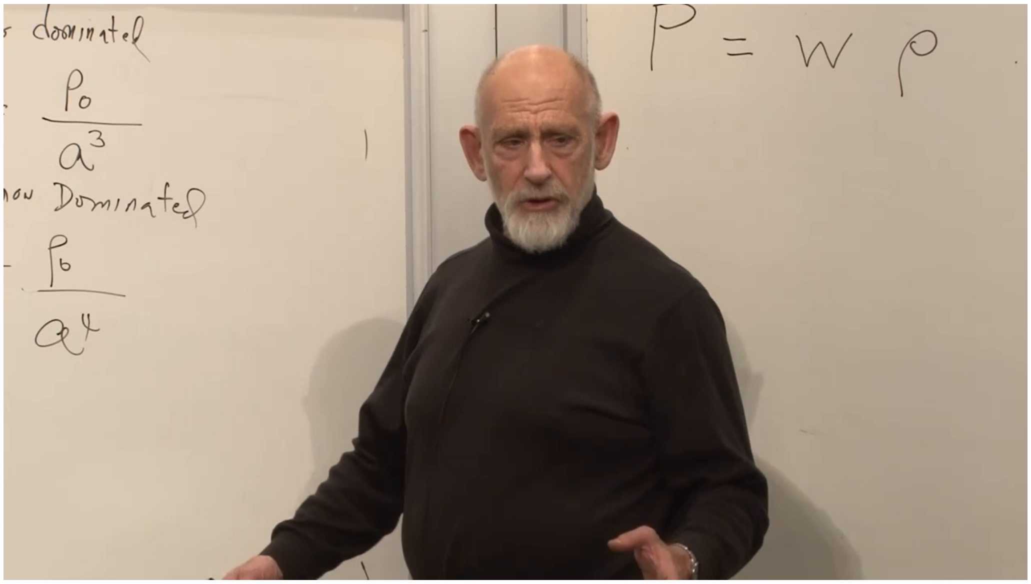

Figure 4.5: The board collects the matter and radiation scalings and writes the equation of state as $P = W\rho$. We use the conventional lowercase w.

Lecture frame: 01:15:08.166.

Cosmological expansion is treated as adiabatic, so $dS = 0$ for the comoving box and

$$dE = -P\,dV. \tag{4.21}$$

The minus sign says that an expanding system does work and loses internal energy. Differentiating $E = \rho V$ independently gives

$$dE = \rho\,dV + V\,d\rho. \tag{4.22}$$

Equating the two expressions,

$$\rho\,dV + V\,d\rho = -P\,dV, \tag{4.23}$$

$$V\,d\rho = -(\rho + P)\,dV. \tag{4.24}$$

With $P = w\rho$,

$$\frac{d\rho}{\rho} = -(1+w)\frac{dV}{V}. \tag{4.25}$$

Integration gives

$$\rho \propto V^{-(1+w)}. \tag{4.26}$$

Since a comoving volume scales as $V \propto a^3$,

$$\boxed{\rho(a) \propto a^{-3(1+w)}}. \tag{4.27}$$

In differential form, the same statement is the local energy-conservation equation

$$\dot{\rho} + 3H(\rho + P) = 0. \tag{4.28}$$

This local conservation law remains exact even though a changing cosmological geometry need not admit a globally conserved total energy.

The two familiar cases follow immediately:

$$w = 0 \quad \Longrightarrow \quad \rho_{\mathrm{m}} \propto a^{-3}, \tag{4.29}$$

$$w = \frac{1}{3} \quad \Longrightarrow \quad \rho_{\mathrm{r}} \propto a^{-4}. \tag{4.30}$$

The extra inverse power for radiation is the thermodynamic expression of cosmological redshift.

Question from the audience. What radiation remains in the universe today?

Response. The classroom discussion points chiefly to the cosmic microwave background, the relic photon bath from the hot early universe. Radiation is now a small part of the total cosmic energy density, but its a^{-4} scaling means that it overtakes matter when the equations are run far enough backward. The precise present fractions are observational inputs and are not needed for this scaling argument. [15]

4.8 Vacuum Energy and the Failure of the Curvature-Fate Rule

Dark energy introduces the equation of state

$$P = -\rho, \qquad w = -1. \tag{4.31}$$

Negative pressure is tension: instead of pushing outward on the wall of a box, the contents pull inward on it. A stretched spring is a familiar object with tension, though it is only an analogy and not a microscopic model of vacuum energy.

Question from the audience. Is negative pressure connected with absolute zero?

Response. Not in any simple general way. Cooling an ordinary material to zero temperature does not by itself produce the cosmological equation of state $P = -\rho$. The exchange is useful because it distinguishes thermal pressure from vacuum stress: vacuum energy is characterized by Lorentz-invariant stress-energy, whose spatial components act as negative pressure. Its microscopic origin is a separate question. [16]

Substituting $w = -1$ into equation (4.27) gives

$$\rho_\Lambda \propto a^0 = \text{constant}. \tag{4.32}$$

As a comoving region expands, its volume and total vacuum energy increase together, leaving the energy per unit volume unchanged. The work term in equation (4.21) accounts for that behavior because P is negative. Once such a component is present, positive spatial curvature no longer guarantees recollapse.

[15] Lecture timestamp: 01:26:53–01:28:34.
[16] Lecture timestamp: 01:31:01–01:31:46.

Question from the audience. Where do dark matter, black holes, and other compact objects enter?

Response. On scales large compared with their separations, any nonrelativistic component whose random velocities are small is part of the pressureless matter density. That includes ordinary matter, cold dark matter, and a population of compact objects such as black holes. Their local structure is complicated, but after coarse-graining their average cosmological equation of state is approximately $w = 0$. [17]

Question from the audience. Do the relativistic equations reproduce the matter- and radiation-dominated solutions found earlier?

Response. Yes. For $k = 0$, inserting $\rho_{\mathrm{m}} \propto a^{-3}$ into Friedmann's equation gives $\dot{a}^2 \propto a^{-1}$ and therefore $a(t) \propto t^{2/3}$. Inserting $\rho_{\mathrm{r}} \propto a^{-4}$ gives $\dot{a}^2 \propto a^{-2}$ and $a(t) \propto t^{1/2}$. General relativity changes the interpretation and enlarges the possible sources, but it preserves those solutions. [18]

Question from the audience. What sort of force would dark energy resemble in a Newtonian picture?

Response. A positive cosmological constant produces a repulsive relative acceleration proportional to separation,

$$\ddot{D} = \frac{\Lambda}{3} D$$

in units with $c = 1$. A negative cosmological constant reverses the sign and is attractive. Because the positive-Λ effect grows with distance, it can eventually dominate matter and curvature; even a closed $k = +1$ universe can then continue expanding. This Newtonian form is only a useful shadow of the relativistic dynamics. [19]

4.9 What to Carry Forward

The cosmological principle is an averaged geometrical assertion: every point and direction is equivalent after local structure is smoothed away. It restricts the spatial metric to the three constant-curvature cases and packages all time dependence into $a(t)$:

$$ds^2 = -dt^2 + a(t)^2 d\Sigma_k^2. \tag{4.33}$$

Comoving separations then obey $\dot{D} = HD$, while bound systems need not join that flow.

Einstein's time-time equation turns the ansatz into dynamics:

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho. \tag{4.34}$$

Its resemblance to the Newtonian energy equation is real, but general relativity identifies the integration slot with intrinsic spatial curvature and allows every form of energy to contribute.

[17] Lecture timestamp: 01:33:34–01:34:10.
[18] Lecture timestamp: 01:34:14–01:35:12.
[19] Lecture timestamp: 01:35:13–01:36:55.

The remaining input is thermodynamic. For an adiabatically expanding component with $P = w\rho$,

$$\dot{\rho} + 3H(\rho + P) = 0, \qquad \rho \propto a^{-3(1+w)}. \tag{4.35}$$

Matter, radiation, and vacuum energy correspond respectively to

$$w = 0, \qquad w = \frac{1}{3}, \qquad w = -1. \tag{4.36}$$

Their different dilution rates determine which component controls the expansion at each epoch. The next task is to understand the physics behind these equations of state and the acceleration produced by vacuum energy.

Chapter 5

Equation of State, Radiation, and Vacuum Energy

Overview. The history of a homogeneous universe is largely the history of its scale factor $a(t)$. Friedmann's equation supplies the geometry, but the material equation of state tells us how the energy density changes while space expands. We first derive $\rho \propto a^{-3(1+w)}$ from work done by a fluid, then obtain $P = \rho/3$ for radiation by counting photon momentum transfer, and finally identify vacuum energy by $P = -\rho$. Positive vacuum energy produces de Sitter expansion; curvature and the sign of the cosmological constant produce bounce and recollapse solutions that can be read as simple mechanical systems.

5.1 The Scale Factor and the Missing Material Law

Let us first recover where we are going. Within the homogeneous and isotropic approximation, the main cosmological history is encoded in one function, $a(t)$. Knowing that function does not answer every question about the universe, but it determines distances, expansion rates, and the sequence of large-scale eras. The purpose of this lecture is not yet to establish those eras observationally. It is to derive, from basic physics, the equations of state that make the models possible.

Two benchmark models have already appeared:

$$a(t) \propto t^{2/3} \qquad \text{(matter domination)}, \tag{5.1}$$

and

$$a(t) \propto t^{1/2} \qquad \text{(radiation domination)}. \tag{5.2}$$

Neither law is exact for all times. Radiation gives the useful early-time approximation and matter the later intermediate one; vacuum energy eventually changes the late-time behavior again. The immediate question is not how observations reveal these eras, but what distinguishes the corresponding right-hand sides of Friedmann's equation.

The missing bridge is the equation of state. It determines how the energy density on the right-hand side of Friedmann's equation changes with the scale factor. The two benchmark laws are

$$\rho_{\text{matter}} = \frac{\rho_0}{a^3}, \qquad \rho_{\text{rad}} = \frac{\rho_0}{a^4}. \tag{5.3}$$

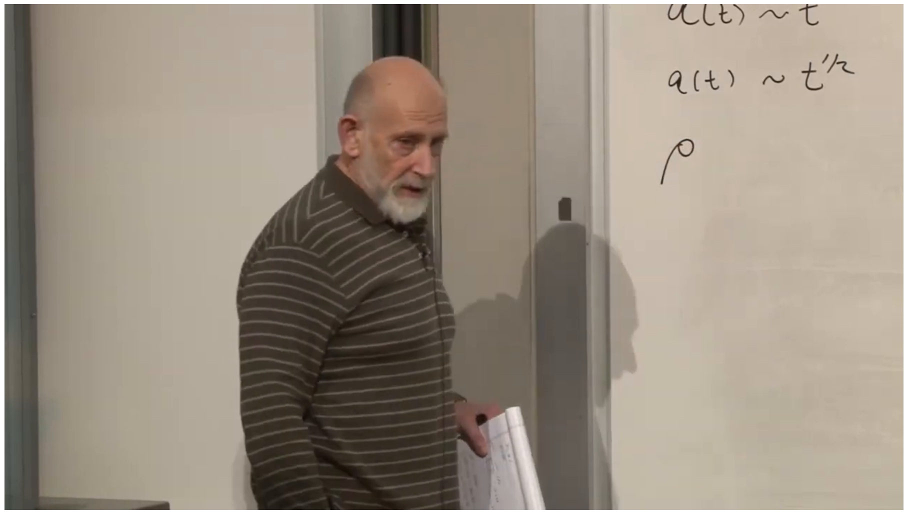

Figure 5.1: The reviewed scale-factor power laws and the newly introduced density symbol. This is a transitional board state rather than a completed derivation.

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Matter dilutes with volume, and volume scales as a^3. Radiation falls with one additional power of a. We want to understand both rules as consequences of the relation between pressure and energy density.

5.2 From Pressure to the Density-Scaling Law

An equation of state may generally involve density, pressure, temperature, and other variables. For the cosmological components considered here, adopt the useful constant-w form

$$P = w\rho, \tag{5.4}$$

where w characterizes the fluid. The board uses an uppercase W; we use the conventional lowercase w. This linear form covers the important cases below, but it is a simplifying ansatz rather than a law obeyed by every possible fluid.

For nonrelativistic matter, energy density is dominated by rest energy. If n is the number density and m the particle mass, then schematically

$$\rho \simeq nmc^2. \tag{5.5}$$

The factor c^2 makes even a small rest mass an enormous reservoir of energy; complete annihilation of a dust grain would release a macroscopic amount. Pressure, by contrast, is produced by momentum transfer when particles strike a wall and is of order nmv^2. A bowling ball transfers more momentum than a ping-pong ball, but for a cold gas every relevant speed satisfies $v \ll c$. Hence

$$\frac{P}{\rho} \sim \frac{v^2}{c^2} \ll 1, \qquad w \simeq 0. \tag{5.6}$$

Radiation moves at c, so it will not share this approximation. We shall derive its value of w rather than assume it.

We now need a general machine that turns equation (5.4) into a scaling law for ρ. A laboratory box is enough. Galaxies cannot literally be placed in a box, but after coarse-graining they behave as particles, and the relation between energy and pressure can be derived locally.

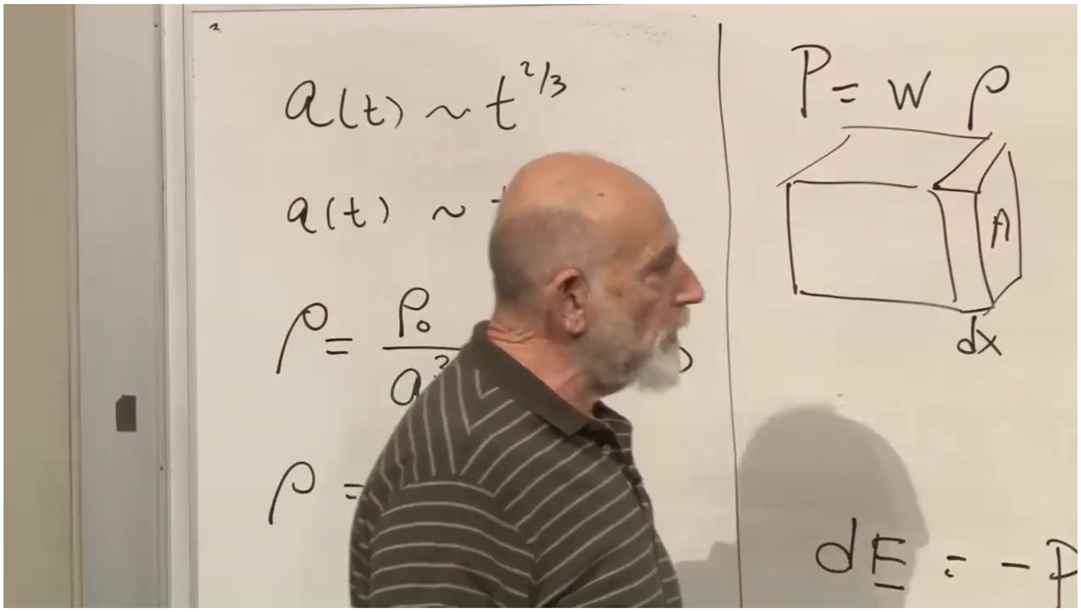

Figure 5.2: The equation of state, reviewed density scalings, and the thermodynamic box on one board. The lecturer obscures part of the earlier writing, so the image is used for layout and visible equations rather than as a complete transcription.

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The left side of the board carries the reviewed cosmological scalings; the right starts the box argument. Read together, the visible content connects

$$\rho_{\text{matter}} = \frac{\rho_0}{a^3}, \qquad \rho_{\text{rad}} = \frac{\rho_0}{a^4}, \qquad P = w\rho. \tag{5.7}$$

Our task is to derive the two density laws from the equation of state.

One wall has area A. Move it outward by dx, so

$$dV = A\,dx. \tag{5.8}$$

Pressure is force per area, so the force on the wall is PA. The gas does work $PA\,dx = P\,dV$, and the energy remaining inside decreases.

$$dE = -P\,dV. \tag{5.9}$$

The total energy is

$$E = \rho V. \tag{5.10}$$

Differentiating gives

$$d(\rho V) = \rho\,dV + V\,d\rho. \tag{5.11}$$

Substitution into equation (5.9) gives

$$\rho\,dV + V\,d\rho = -P\,dV. \tag{5.12}$$

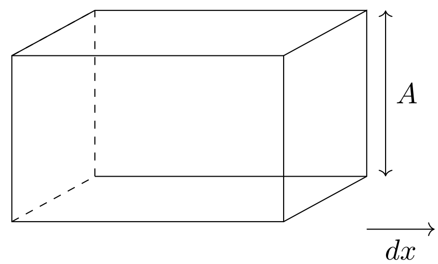

Figure 5.3: A clean reconstruction of the thermodynamic box, with face area A and displacement dx.

Insert $P = w\rho$:

$$V\,d\rho = -(1+w)\rho\,dV. \tag{5.13}$$

The result is preliminary to the useful equation, but not quite there. Divide by ρV.

$$\frac{d\rho}{\rho} = -(1+w)\frac{dV}{V}. \tag{5.14}$$

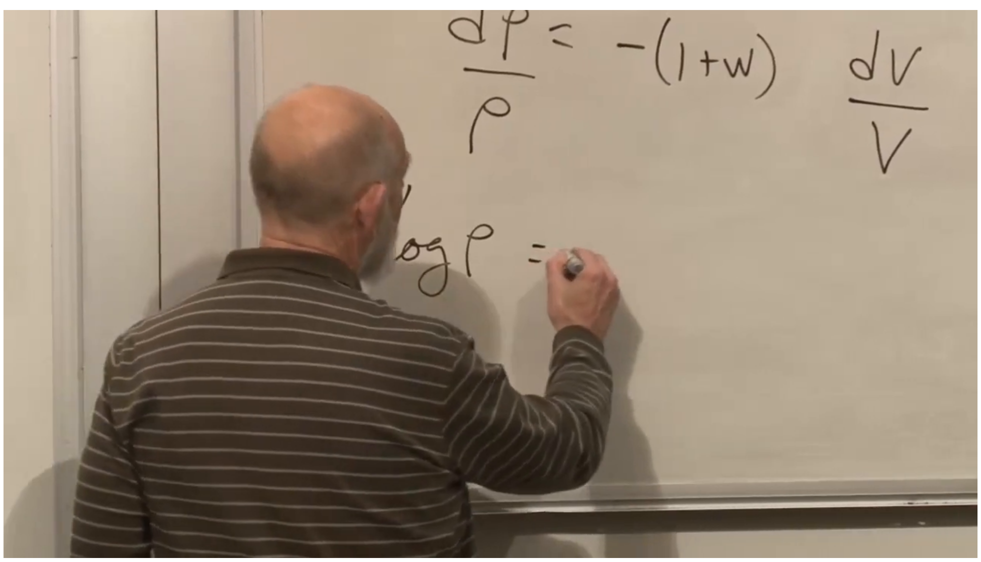

Figure 5.4: The separated density-volume equation and the beginning of the logarithmic integration. The complete right-hand side is reconstructed from the preceding algebra, not from hidden pixels.
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The frame shows the separated equation and the beginning $\log\rho =$. Complete the integration directly:

$$d(\log\rho) = -(1+w)\,d(\log V), \tag{5.15}$$

so that

$$\log\rho = -(1+w)\log V + \text{const.} \tag{5.16}$$

Exponentiating,

$$\rho \propto V^{-(1+w)}. \tag{5.17}$$

This is the density-volume law.

Return now from the box to cosmology. Moving only one wall was a convenient derivation, not an anisotropic assumption. Expanding all three directions uniformly gives the same work relation, and a comoving volume scales as

$$V \propto a^3. \tag{5.18}$$

Therefore

$$\boxed{\rho \propto a^{-3(1+w)}}. \tag{5.19}$$

In d spatial dimensions the exponent 3 would become d; the rest of the argument is unchanged.

5.3 First Applications: Matter and Radiation

We can now recover the review formulas as consequences rather than assumptions.

For nonrelativistic matter we take

$$w = 0. \tag{5.20}$$

Then

$$\rho \propto a^{-3}, \tag{5.21}$$

which is exactly the matter-dilution law.

For radiation, provisionally take

$$w = \frac{1}{3}. \tag{5.22}$$

If this is correct, then

$$\rho \propto a^{-3(1+1/3)} = a^{-4}. \tag{5.23}$$

The extra inverse power is therefore equivalent to $w = 1/3$. The unresolved step is to derive that value from photon physics.

5.4 Why Radiation Pressure Is One Third of Its Energy Density

Radiation may be described as electromagnetic waves or as massless particles. Use photons and compute the pressure of a photon gas in a box. The box is drawn in two dimensions, but the gas and its angular distribution are three-dimensional. The photons move at the speed of light and bounce elastically from the provisional walls.

For the moment, pretend all photons have the same energy ϵ. This simplification will disappear when the contributions from different energies are added. Let $\vec{\pi}$ be a photon momentum vector; the usual p is avoided because P already denotes pressure. A massless particle obeys the following relations.

$$\epsilon = c\,\pi, \tag{5.24}$$

with $\pi = |\vec{\pi}|$. In the units used in the lecture, $c = 1$, so

$$\epsilon = \pi. \tag{5.25}$$

If ν is the photon number density, then the energy density is

$$\rho = \epsilon\,\nu. \tag{5.26}$$

Let the x-axis be normal to one wall. During a short interval Δt, a photon moving at angle θ to that normal reaches the wall only if it starts within a slab of thickness

$$\Delta x = \Delta t \cos\theta. \tag{5.27}$$

If it reflects, its x-component of momentum reverses sign, so the momentum transferred to the wall is

$$\Delta \pi_x = 2\epsilon \cos\theta. \tag{5.28}$$

Force is momentum transfer per unit time, so one hit contributes $\Delta\pi_x/\Delta t$. The number of photons in the slab is $\nu A \Delta x$, but only half of them move toward the wall, so the relevant number is $\frac{1}{2}\nu A\Delta x$. Dividing the total force by the wall area A, we find

$$\begin{aligned} P(\theta) &= \frac{1}{A}\left(\frac{2\epsilon\cos\theta}{\Delta t}\right)\left(\frac{1}{2}\nu A\Delta x\right) && (5.29)\\ &= \epsilon\nu\cos\theta\frac{\Delta x}{\Delta t} && (5.30)\\ &= \epsilon\nu\cos^2\theta && (5.31)\\ &= \rho\cos^2\theta. && (5.32)\end{aligned}$$

The result depends on angle because it is only the contribution from photons traveling in one direction.

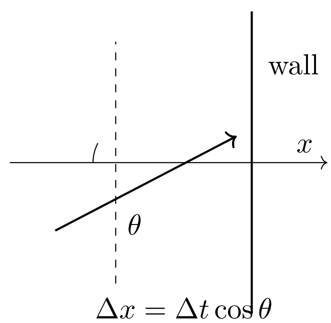

Figure 5.5: Photon momentum incident on a wall at angle θ, the geometry used to derive radiation pressure.

Average over the isotropic distribution. Let $\mathbf{n} = (n_x, n_y, n_z)$ be the unit vector along a photon's direction. Then

$$n_x = \cos\theta, \qquad n_x^2 + n_y^2 + n_z^2 = 1. \tag{5.33}$$

In an isotropic gas, the averages of n_x^2, n_y^2, and n_z^2 are equal. Averaging the identity above,

$$\langle n_x^2\rangle + \langle n_y^2\rangle + \langle n_z^2\rangle = 1, \tag{5.34}$$

and by symmetry each term is the same, so

$$\langle n_x^2\rangle = \langle\cos^2\theta\rangle = \frac{1}{3}. \tag{5.35}$$

Therefore

$$P = \rho\langle\cos^2\theta\rangle = \frac{\rho}{3}. \tag{5.36}$$

So the radiation equation of state is

$$P = \frac{1}{3}\rho, \qquad w = \frac{1}{3}. \tag{5.37}$$

The factor 1/3 is the number of spatial directions sharing the squared unit vector. In d isotropic spatial dimensions, the same argument would give $\langle n_x^2 \rangle = 1/d$. This completes the microscopic derivation promised at the start.

5.4.1 Questions That Test the Radiation Argument

Question from the audience. Do all photons need to have the same energy?

Response. No. Perform the calculation separately in each energy interval. Every interval contributes pressure equal to one third of its own energy density, so summing over the spectrum preserves $P = \rho/3$. The equal-energy assumption only kept the notation short. [1]

Question from the audience. Would absorbing black walls give the same answer as perfect mirrors, and what is the wall in empty space?

Response. A black wall in thermal equilibrium absorbs and re-emits radiation, producing the same average momentum balance. Cosmological space has no literal wall; the surface is an imagined boundary. For every photon that crosses outward, isotropy supplies an average compensating flux inward, so the local momentum-flow calculation is unchanged. The classroom radiation is chiefly the thermal cosmic microwave background, for which this equilibrium picture is especially natural. [2]

Question from the audience. Do quantum statistics change $P = \rho/3$?

Response. No. Bose statistics determine how photons populate the available modes and therefore determine the spectrum and total ρ, but the isotropic stress of massless particles remains one third of that energy density. [3]

Question from the audience. What if photons scatter from one another?

Response. Ordinary photon-photon scattering is extremely weak, so its correction here is negligible. At sufficiently high temperature and density, collisions can create electron-positron pairs and change the effective particle content and equation of state. That is a real effect, but it belongs to a much hotter regime than the one used in this argument. [4]

Question from the audience. What happened to the wall area A?

Response. The intermediate product is the force from one photon times the number striking the wall, and that number is proportional to A. Pressure is force divided by area, so the final division cancels A. Calling the intermediate expression pressure too early was the board omission; restoring the force-to-pressure step repairs it. [5]

[1] Lecture timestamp: 00:39:08–00:39:36.
[2] Lecture timestamp: 00:39:37–00:41:19.
[3] Lecture timestamp: 00:41:51–00:42:02.
[4] Lecture timestamp: 00:42:02–00:43:08.
[5] Lecture timestamp: 00:43:11–00:44:01.

5.5 Negative Pressure and the Meaning of Tension

Up to now, particles have struck the walls and pushed them outward. That picture naturally produces positive pressure, but it does not exhaust the possibilities.

Question from the audience. Can energy density itself be negative?

Response. Quantum theory permits negative energy densities in restricted circumstances, and a cosmological constant can formally have either sign. Such cases require care and are not the immediate starting point. The more familiar doorway is negative *pressure*, which can occur even while the energy density is positive. [6]

Question from the audience. How can pressure be negative?

Response. In a one-dimensional world, a box is a line interval. Freely moving particles bounce from its ends and push outward, giving positive pressure. Connect the ends instead by a stretched string or spring. Pulling the ends apart makes the spring pull them inward. That inward force is tension, the one-dimensional model of negative pressure. [7]

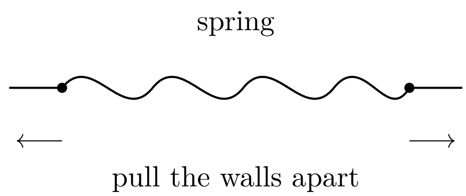

Figure 5.6: A one-dimensional spring as an analogy for negative pressure.

The work bookkeeping reverses accordingly. Expansion lets a positive-pressure gas do work and lose internal energy. Pulling outward against tension stores energy in the spring. More generally, attractive constituents can pull one another and the boundary inward. Thus a system may have $\rho > 0$ and $P < 0$, giving $w < 0$, without any contradiction. This possibility is central because vacuum energy has precisely that sign pattern.

5.6 Vacuum Energy and the Cosmological Constant

Vacuum energy is assigned to empty space itself. Quantum field theory motivates it through the zero-point motion of the harmonic oscillators associated with field modes, but its macroscopic description does not require us to solve that microscopic problem. The defining property is that empty space has the same energy density wherever and however large a region is chosen.

$$\rho = \rho_0 = \text{const.} \tag{5.38}$$

If a region of volume V contains only vacuum energy, then the total energy in that region is

$$E = \rho_0 V. \tag{5.39}$$

[6]Lecture timestamp: 00:44:25–00:44:59.
[7]Lecture timestamp: 00:45:04–00:46:31.

Question from the audience. Does a box change the vacuum density by restricting the allowed field modes?

Response. The total energy in a finite region is smaller than the energy in all of space because the region has smaller volume; the bulk density ρ_0 remains the same. Real boundaries can change mode energies and produce the Casimir effect, but that correction matters when separations are comparable to the relevant wavelengths. It is not the large-scale vacuum density used in cosmology. [8]

Introduce

$$\Lambda = \frac{8\pi G}{3}\rho_0. \tag{5.40}$$

This definition absorbs the combination that appears in Friedmann's equation. The inverse relation is

$$\rho_0 = \frac{3\Lambda}{8\pi G}. \tag{5.41}$$

The parameter Λ is the cosmological constant. A positive nearly constant component driving late acceleration is commonly called dark energy.

The scale-factor dependence is already settled, so run the box argument backward to discover its pressure. Since $d\rho_0 = 0$,

$$dE = d(\rho_0 V) = \rho_0\, dV. \tag{5.42}$$

On the other hand the thermodynamic work relation is still

$$dE = -P\, dV. \tag{5.43}$$

If we write $P = w\rho_0$, then

$$dE = -w\rho_0\, dV. \tag{5.44}$$

Comparison gives

$$w = -1. \tag{5.45}$$

and therefore

$$P = -\rho. \tag{5.46}$$

The spring analogy was preparing this sign. Positive vacuum energy has negative pressure; negative vacuum energy has positive pressure. Astronomical measurements of w ask whether the late-time component really behaves as a nondiluting vacuum: the closer w lies to -1, the closer the behavior is to equation (5.38).

Question from the audience. Is $P = -\rho$ the equation of state of an empty universe, and can we compute ρ_0?

Response. It is the equation of state of empty space when vacuum energy is present. Its magnitude is much harder. Every quantum field and every energy scale can contribute, including fields not yet discovered, so no accepted calculation explains the observed value. The main puzzle is not how to write $w = -1$, but why the net ρ_0 is so extraordinarily small. [9]

[8] Lecture timestamp: 00:50:06–00:51:25.
[9] Lecture timestamp: 00:57:27–00:58:29.

5.7 Vacuum Energy in Friedmann Dynamics

Isolate vacuum energy just as we previously isolated pure matter and pure radiation. With $\rho = \rho_0$, Friedmann's equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \Lambda - \frac{k}{a^2}, \tag{5.47}$$

with $k = +1, 0, -1$ for spherical, flat, and hyperbolic space. Positive and negative Λ, combined with three choices of k, give six sign cases, though not every case admits a real solution.

Remark 5.1. The lecture defines $\Lambda = (8\pi G/3)\rho_0$, so Λ here has dimensions of H^2. In the common convention $G_{\mu\nu} + \Lambda_{\mathrm{E}} g_{\mu\nu} = 8\pi G T_{\mu\nu}$, the corresponding parameter is $\Lambda_{\mathrm{E}} = 3\Lambda$.

5.7.1 Flat de Sitter Expansion

Set $k = 0$. Then

$$\left(\frac{\dot{a}}{a}\right)^2 = \Lambda. \tag{5.48}$$

For $\Lambda > 0$, choose the expanding branch:

$$\dot{a} = \sqrt{\Lambda}\, a, \tag{5.49}$$

and therefore

$$a(t) = C\, e^{\sqrt{\Lambda}\, t}. \tag{5.50}$$

The Hubble parameter is then constant:

$$H = \frac{\dot{a}}{a} = \sqrt{\Lambda}. \tag{5.51}$$

This is de Sitter space in flat slicing, named for Willem de Sitter, who found the cosmological-constant solution shortly after Einstein introduced that term in 1917. If $\Lambda < 0$ while $k = 0$, equation (5.47) has no real solution because H^2 cannot be negative.

Question from the audience. Is the Hubble parameter really constant here, and does this spacetime have a big bang?

Response. For the pure flat-slicing solution, $H = \sqrt{\Lambda}$ is exactly constant. Although $a(t) \to 0$ only as $t \to -\infty$, this coordinate patch is not geodesically complete: some past-directed trajectories reach its boundary in finite proper time. The formula therefore does not by itself settle the global big-bang question. The complete de Sitter geometry contains more than this flat patch, a point postponed until the geometry is studied in detail. [10]

A schematic Friedmann equation for a universe containing matter, radiation, curvature, and vacuum energy is

$$\left(\frac{\dot{a}}{a}\right)^2 = \Lambda + \frac{C_m}{a^3} + \frac{C_r}{a^4} - \frac{k}{a^2}. \tag{5.52}$$

At early times, when a is small, the inverse powers dominate and vacuum energy is negligible. At late times, when a is large, the constant term Λ dominates. So the universe makes a transition: early on it looks matter- or radiation-dominated, while late on it approaches de Sitter-like behavior.

[10] Lecture timestamp: 01:06:54–01:08:07.

An exponential has $\ddot{a} > 0$, so vacuum domination is accelerated expansion. Acceleration need not be exactly exponential during the transition because matter still contributes to equation (5.52). The lecture-era observations are described as catching the universe during this competition rather than in an already pure de Sitter phase.

5.8 Questions About Vacuum Energy

Question from the audience. If vacuum energy is ground-state energy, how can it be negative?

Response. A bosonic oscillator mode contributes $+\frac{1}{2}\hbar\omega$, whereas a fermionic mode contributes with the opposite sign. The net vacuum energy is a sum over many fields and scales, so either sign is possible before the total is known. The existence of contributions is natural; explaining their extraordinarily small remainder is the difficult part. [11]

Question from the audience. Does constant $w = -1$ imply a future big rip?

Response. No. A cosmological constant with $w = -1$ gives de Sitter-like exponential expansion but does not produce the finite-time divergence called a big rip. That scenario belongs to a hypothetical phantom component with $w < -1$, for which the density grows as the universe expands. No such component is assumed here. [12]

Question from the audience. Would exact supersymmetry make the vacuum energy vanish?

Response. It can cancel the zero-point contributions because every bosonic degree of freedom is paired with a fermionic one. Nature, however, does not display exact low-energy supersymmetry: observed bosons and fermions are not mass-degenerate partners. Exact cancellation therefore does not explain the vacuum energy in the world as presently observed. [13]

Question from the audience. What is the enormous discrepancy between a natural vacuum-energy estimate and the observed value?

Response. The universal constants c, $\hbar$, and G define Planck units. The speed c limits every signal, $\hbar$ fixes the universal quantum uncertainty scale, and G couples all energy to gravity. From them one constructs a Planck mass of order 10^{-5} grams and a Planck length of order 10^{-33} centimeters, hence a Planck energy density of roughly one Planck energy per cubic Planck length. In those units the observed vacuum density is quoted in the lecture as about 10^{-123}, with the exact exponent depending on convention. The mystery is why contributions that naturally suggest a vastly larger scale leave such a tiny nonzero result. [14]

[11] Lecture timestamp: 01:12:00–01:14:12.
[12] Lecture timestamp: 01:14:13–01:15:50.
[13] Lecture timestamp: 01:15:51–01:16:20.
[14] Lecture timestamp: 01:16:21–01:23:12.

Question from the audience. Why were physicists surprised to discover a nonzero cosmological constant if vacuum contributions are expected?

Response. Repeated observational improvements had continued to find a result consistent with zero. That history encouraged the psychological expectation that an unknown exact principle, perhaps in a future quantum-gravity theory, forced complete cancellation. The eventual detection overturned that expectation. The surprise reflected decades of increasingly stringent null results, not a derivation that vacuum energy had to vanish. [15]

Question from the audience. Can observations determine whether dark energy changes with time?

Response. They cannot rule out an abrupt change in the remote future, but they can compare the recent expansion history across billions of years. Measuring w, or more generally $w(a)$, tests whether the density has evolved during that interval. The lecture reports the then-current constraint only at roughly the ten-percent level; that number is historical, while the enduring point is that $w = -1$ corresponds to an unchanging density. [16]

Question from the audience. Would the result look different in another number of spatial dimensions?

Response. The overall structure remains. In d isotropic spatial dimensions, radiation has $w = 1/d$ and dilutes as $a^{-(d+1)}$, while Lorentz-invariant vacuum energy still has $w = -1$. Numerical coefficients in Friedmann's equation change with dimension, but the distinction between diluting matter, radiation, and nondiluting vacuum survives. [17]

5.9 Curvature and the Mechanical Analogy

5.9.1 Positive Curvature with Positive Vacuum Energy

Take $k = +1$ and $\Lambda > 0$:

$$\left(\frac{\dot{a}}{a}\right)^2 = \Lambda - \frac{1}{a^2}. \tag{5.53}$$

Multiplying by a^2,

$$\dot{a}^2 - \Lambda a^2 = -1. \tag{5.54}$$

Regard a as the coordinate of a fictitious particle. Then $\dot{a}^2$ is its kinetic term, $-\Lambda a^2$ is an upside-down quadratic potential, and the total energy is -1. The physically allowed branch has $a \geq \Lambda^{-1/2}$: the universe contracts from large size, reaches rest at the minimum, and expands again.

The exact solution is

$$a(t) = \Lambda^{-1/2} \cosh(\sqrt{\Lambda}\, t). \tag{5.55}$$

[15] Lecture timestamp: 01:23:14–01:26:05.
[16] Lecture timestamp: 01:26:06–01:27:16.
[17] Lecture timestamp: 01:27:16–01:27:57.

Using

$$\cosh(\sqrt{\Lambda}\,t) = \frac{1}{2}\left(e^{\sqrt{\Lambda}\,t} + e^{-\sqrt{\Lambda}\,t}\right), \tag{5.56}$$

The function is symmetric in time. At $t = 0$, $a = \Lambda^{-1/2}$; toward either temporal direction it grows exponentially. The closed slicing therefore displays a contraction, a nonsingular minimum, and re-expansion. It looks globally different from the flat exponential slicing, yet both coordinate systems cover portions of the same de Sitter geometry.

5.9.2 Negative Cosmological Constant

Write $\Lambda = -|\Lambda|$. If $k = +1$, then equation (5.47) has a negative right-hand side for every a, while $H^2 \geq 0$. No real FRW solution of this pure-vacuum type exists.

For $k = -1$,

$$\left(\frac{\dot{a}}{a}\right)^2 = -|\Lambda| + \frac{1}{a^2}. \tag{5.57}$$

Multiplying through by a^2,

$$\dot{a}^2 + |\Lambda|a^2 = 1. \tag{5.58}$$

This is the energy equation of a harmonic oscillator. Since $a \geq 0$, one physical half-cycle is

$$a(t) = \frac{1}{\sqrt{|\Lambda|}}\sin\left(\sqrt{|\Lambda|}\,t\right), \qquad 0 < t < \frac{\pi}{\sqrt{|\Lambda|}}. \tag{5.59}$$

The open universe begins at $a = 0$, reaches $a_{\max} = |\Lambda|^{-1/2}$, and returns to a big crunch. Open spatial geometry therefore does not guarantee eternal expansion; a negative cosmological constant can reverse it. A very large positive Λ would tear bound structures apart, while a large negative one would cause rapid recollapse.

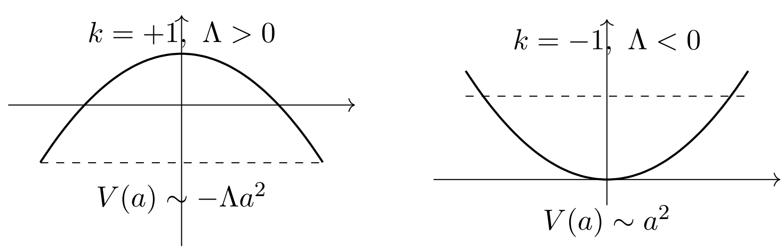

Figure 5.7: Schematic effective potentials for the closed de Sitter bounce and the open negative-Λ recollapse.

Question from the audience. Are these Friedmann equations really Einstein's equations?

Response. Yes. They are Einstein's field equation after homogeneity and isotropy reduce the metric to the single function $a(t)$. The symmetry reduction removes most tensor components, leaving this ordinary differential equation for cosmic expansion. [18]

[18] Lecture timestamp: 01:43:52–01:44:05.

Question from the audience. Could the scale factor a be complex?

Response. Not in the real classical FRW cosmologies being analyzed here. The scale factor measures spatial size and is taken to be real and nonnegative. Complexified geometries can appear in other mathematical approaches, but they have no role in this lecture's solutions. [19]

Question from the audience. Is there a physical reason pressure must be a linear function of density?

Response. No. Constant w is convenient and happens to describe the principal components considered here. A mixture of matter, radiation, and vacuum does not have one constant w over all epochs, and a general material can have nonlinear $P(\rho)$. For a barotropic fluid, $dP/d\rho$ controls the squared sound speed in units with $c = 1$; constant w makes that derivative constant, but nature is not obliged to do so. [20]

5.10 What to Carry Forward

The equation of state connects microscopic physics with the time history of the scale factor. Adiabatic work in a comoving box gives

$$\rho \propto a^{-3(1+w)}.$$

Cold matter has $w = 0$ and $\rho \propto a^{-3}$. Isotropic photon momentum flow gives

$$\left\langle \cos^2\theta \right\rangle = \frac{1}{3}, \qquad P_{\mathrm{r}} = \frac{\rho_{\mathrm{r}}}{3}, \qquad \rho_{\mathrm{r}} \propto a^{-4}.$$

Tension makes negative pressure physically intelligible. A constant vacuum density obeys

$$P_\Lambda = -\rho_\Lambda, \qquad w_\Lambda = -1.$$

Positive vacuum energy gives exponential de Sitter expansion in flat slicing and a cosh bounce in closed slicing. Negative vacuum energy can make even an open universe recollapse. In every sign case, the Friedmann equation can be read as an effective mechanical energy equation. The unresolved question is not why quantum fields contribute vacuum energy, but why their observed net contribution is almost zero without being exactly zero.

[19] Lecture timestamp: 01:44:06–01:44:21.

[20] Lecture timestamp: 01:44:21–01:45:34.

Chapter 6

Observations, Dark Matter, and the Cosmological Fit

Overview. Observation acquires meaning only after it is connected to a cosmological model. We begin with the Friedmann equation and the three homogeneous spatial geometries, organize the present cosmic budget into radiation, matter, vacuum energy, and curvature, and reconstruct why an open universe once seemed unavoidable. Galaxy rotation curves then reveal the missing gravitating matter and lead to the dynamics of a cold, weakly interacting halo. Finally, redshift, brightness, and source counts turn a trial expansion history into observable curves. The old matter-plus-curvature model fails; a matter fraction near 0.3, a vacuum fraction near 0.7, and very small curvature give the lecture's successful fit.

6.1 Put the Observations Back Into the Equations

Astronomical facts cannot be interpreted in isolation. We first need the equations that tell us which combinations of observations distinguish one universe from another. For a homogeneous and isotropic universe the starting point is Friedmann's equation,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{6.1}$$

The quantity

$$H(t) = \frac{\dot{a}(t)}{a(t)} \tag{6.2}$$

is the Hubble *function*. Its present value H_0 is conventionally called the Hubble constant, but $H(t)$ is not constant over cosmic history. When astronomers quote a value in kilometers per second per megaparsec, they are quoting H_0.

The normalized curvature label takes only three values,

$$k \in \{+1, 0, -1\}, \tag{6.3}$$

corresponding respectively to spherical, flat, and hyperbolic spatial geometry. The common FRW metric may be written

$$ds^2 = -dt^2 + a(t)^2 \left[dr^2 + \xi(r)^2 d\Omega_2^2\right], \tag{6.4}$$

where

$$\xi(r) = \begin{cases} \sin r, & k = +1, \\ r, & k = 0, \\ \sinh r, & k = -1. \end{cases} \tag{6.5}$$

This notation will do two jobs. It packages the three geometries into one metric, and later it gives the area of a sphere and therefore the number of sources in a radial shell. Thus a choice of k and an equation for $a(t)$ is already a model with observational consequences.

6.2 The Present Cosmic Budget

The energy density has three principal components in the model under discussion: radiation, nonrelativistic matter, and vacuum energy. Including curvature on the same bookkeeping line gives

$$\frac{C_r}{a^4} + \frac{C_m}{a^3} + \Lambda - \frac{k}{a^2} = H^2. \tag{6.6}$$

Here Λ uses the normalization of the preceding lecture, $\Lambda = (8\pi G/3)\rho_{\text{vac}}$. Radiation dilutes as a^{-4}, matter as a^{-3}, vacuum energy does not dilute, and curvature scales as a^{-2}. Curvature is not a material energy density, but placing it beside the other terms makes the present constraint transparent: the four contributions must add to H_0^2.

Define their present fractions by

$$\begin{aligned} \Omega_r &= \frac{C_r/a_0^4}{H_0^2}, & \Omega_m &= \frac{C_m/a_0^3}{H_0^2}, \\ \Omega_\Lambda &= \frac{\Lambda}{H_0^2}, & \Omega_k &= \frac{-k/a_0^2}{H_0^2}. \end{aligned} \tag{6.7}$$

Then

$$\boxed{\Omega_r + \Omega_m + \Omega_\Lambda + \Omega_k = 1.} \tag{6.8}$$

The Ω's are ratios evaluated today, not time-independent constants. Also note the sign convention: $k = -1$ gives a positive Ω_k.

6.2.1 How the First Terms Are Measured

Hubble obtained the local expansion rate by comparing recession redshift with distance. The redshift supplies a velocity at low z; a calibrated brightness supplies a distance. His first numerical value was badly wrong, but the method was right. Because the galaxies were comparatively nearby, the measurement did not probe a large fraction of cosmic history and could treat H as its present value.

Question from the audience. How can brightness determine distance unless the intrinsic brightness is already known?

Response. It cannot do so in one step. The cosmic distance ladder begins with parallax for nearby stars. Cepheid variables in that calibrated region reveal a period-luminosity relation and become standard candles for greater distances. Overlapping calibrations carry the ladder outward, and supernovae provide the best candles at the cosmological redshifts used later. Distance is therefore inferred from a chain of standards, not from an arbitrary assumption that every object shines equally. [1]

Question from the audience. Are C_r and C_m constants even though their densities change?

Response. Yes. The coefficients remain fixed while the factors a^{-4} and a^{-3} carry the time dependence. Vacuum energy is constant, and curvature scales as a^{-2}. These unequal powers are precisely why the fractional contributions change from one era to another. [2]

The radiation term can be measured locally from the photon background. Most of it is the cosmic microwave background, and its present energy density is tiny compared with the other important components. A sufficiently large averaging volume is required, but no long excursion down the past light cone is needed to establish that Ω_r is negligible today.

Visible matter is estimated from galaxies, stellar populations, hydrogen, and interstellar gas. The methods are technical rather than conceptually obscure: determine the typical stellar and gas content, sum it over galaxies, and average over a large volume. The lecture quotes the old rough scale of one proton mass per cubic meter. This is much larger than the radiation density but still remarkably dilute.

Question from the audience. What about dark matter in this accounting?

Response. It is coming, but first we deliberately reconstruct the older inference based on luminous matter alone. The historical order matters: it shows exactly which apparent conclusion dark matter corrected and which gap it did not close. [3]

6.3 Why an Open Universe Once Seemed Necessary

In the older picture, radiation was correctly judged negligible, vacuum energy was set to zero, and matter meant only the material visible through electromagnetic radiation. The resulting estimate was roughly

$$\Omega_r \simeq 0, \qquad \Omega_\Lambda = 0, \qquad \Omega_m^{\text{luminous}} \sim \frac{1}{20}\text{–}\frac{1}{30}. \tag{6.9}$$

The sum rule then appeared to force

$$\Omega_k \simeq 1. \tag{6.10}$$

Since a positive curvature contribution means $-k/a_0^2 > 0$, this implied $k = -1$: an open, negatively curved universe. These numbers are a reconstruction of the older argument, not a modern parameter estimate.

To interpret the curvature scale, return to the matter-dominated solution

$$a(t) \propto t^{2/3}. \tag{6.11}$$

[1] Lecture timestamp: 00:10:00–00:11:32.
[2] Lecture timestamp: 00:11:35–00:12:02.
[3] Lecture timestamp: 00:15:08–00:15:40.

Differentiating and dividing by a gives

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{6.12}$$

At the present age T,

$$H_0 \simeq \frac{2}{3T}. \tag{6.13}$$

Thus the Hubble time is of the same order as the age of the universe. The radiation-dominated comparison,

$$a(t) \propto t^{1/2}, \qquad H(t) = \frac{1}{2t}, \tag{6.14}$$

shows that this inverse-age relation is not peculiar to dust.

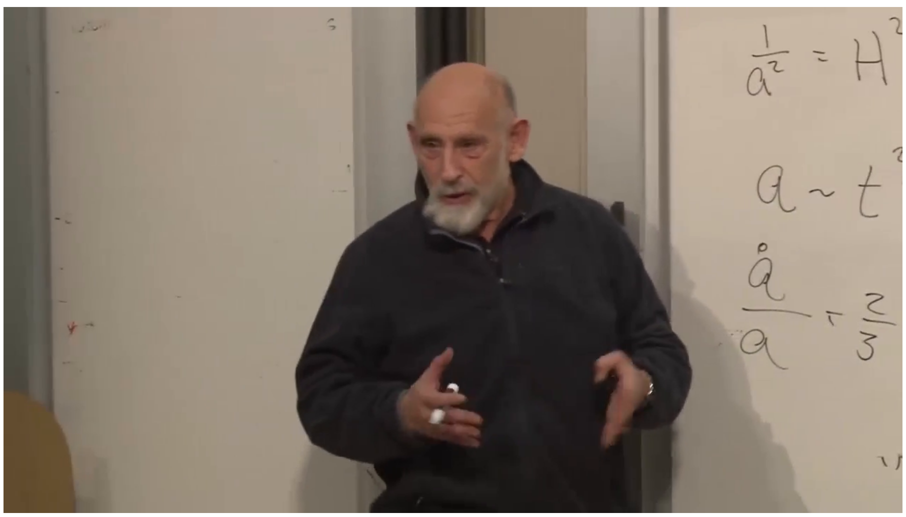

Figure 6.1: The matter-era board review. The visible writing contains $a \sim t^{2/3}$ and $\dot{a}/a = 2/(3t)$; the lecturer obscures the remaining board, so no hidden equation is attributed to the frame.

Lecture frame: 00:24:41.615.

Question from the audience. What units does the Hubble constant have, and where is the speed of light?

Response. Since H relates velocity to distance, it has units of inverse time. It may be quoted in inverse seconds, inverse years, or the astronomers' equivalent kilometers per second per megaparsec. The relation $H \sim 1/T$ needs no extra c. A factor of c appears only when the time T is translated into a length such as cT. [4]

Hubble's original value of H_0 was too large by roughly an order of magnitude, so the corresponding Hubble age was too short, of order a billion rather than ten billion years in the lecture's rounded comparison. Better calibration repaired the number without changing the logic that H_0^{-1} is a cosmic timescale.

The old curvature inference now reads

$$\frac{1}{a_0^2} \sim H_0^2 \sim \frac{1}{T^2}. \tag{6.15}$$

[4] Lecture timestamp: 00:22:50–00:24:25.

In units $c = 1$, the radius of curvature a_0 was therefore thought to be of order the age T; with ordinary units restored it was of order cT. The lecture immediately withdraws this as a statement about the real universe. It is the conclusion one obtained several decades earlier after discarding the terms then believed negligible.

> **Question from the audience.** If curvature was supposed to dominate today, why use the matter-dominated law $a \propto t^{2/3}$?
>
> **Response.** Because matter scales as a^{-3} while curvature scales only as a^{-2}. At smaller a, matter wins. Curvature could become important only late, so matter domination still described most of the elapsed history and kept H_0 of order $1/T$. A complete mixed solution would correct the numerical factor, not erase the timescale argument. [5]

6.4 Dark Matter From Galactic Dynamics

There are two operational ways to inventory matter. One is to count material that emits or absorbs light. The other is to infer mass from gravity. Oort found anomalous galactic dynamics in the early 1930s, and Zwicky found the analogous mass discrepancy in galaxy clusters. The evidence said that gravitating matter exceeded luminous matter. Ordinary charged matter radiates and collides; the missing component was known first by its gravitational action.

> **Question from the audience.** Does luminous matter include diffuse gas and central black holes?
>
> **Response.** Diffuse gas is included insofar as its amount can be inferred from its electromagnetic behavior. A central supermassive black hole may be included in the known mass budget, but it is only a small fraction of a galaxy's total mass and is concentrated at the center. It cannot explain a discrepancy that persists far outside the luminous center. [6]

Consider a star moving approximately in a circular orbit of radius r. If nearly all the mass M were concentrated near the galactic center, Newton's law would give

$$\frac{GM}{r^2} = \frac{v^2}{r}, \tag{6.16}$$

and hence

$$v^2 = \frac{GM}{r}, \qquad v(r) \propto r^{-1/2}. \tag{6.17}$$

This is the Kepler falloff expected outside a central mass distribution.

Observed galactic rotation curves do not fall in this way; over a substantial radial range they are approximately flat. Replace the fixed central mass by the enclosed mass $M(r)$:

$$\frac{GM(r)}{r^2} = \frac{v(r)^2}{r}. \tag{6.18}$$

[5] Lecture timestamp: 00:49:20–00:50:17.
[6] Lecture timestamp: 00:29:50–00:31:14.

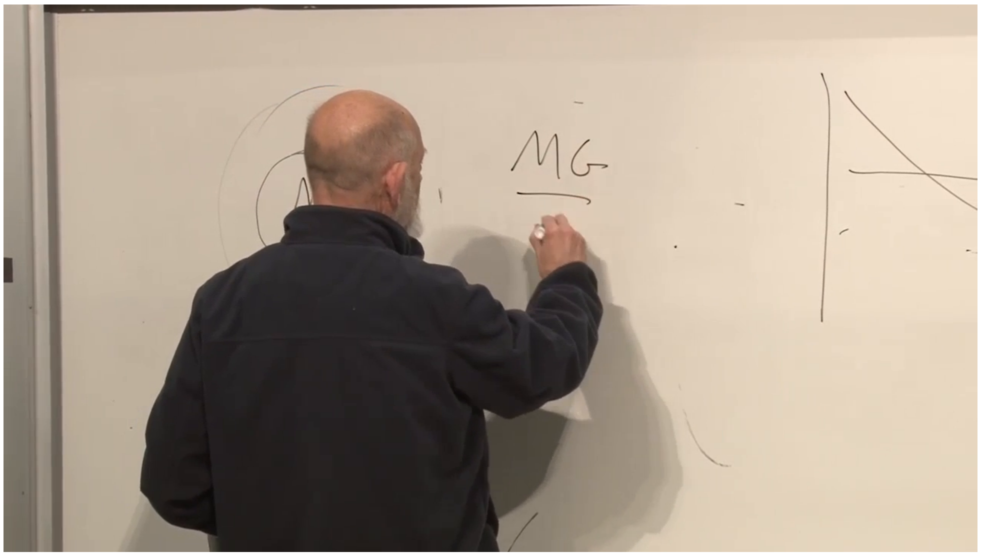

Figure 6.2: The rotation-curve board at the transition from a central mass to an enclosed mass $M(r)$. Only MG, the fraction bar, the orbit sketch, and portions of the curve sketches are visible; equation (6.17) comes from the spoken derivation.

Lecture frame: 00:34:53.642.

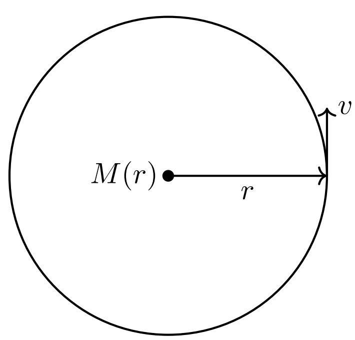

Figure 6.3: The circular-orbit construction used to infer enclosed mass from a measured rotation speed.

If $v(r)$ is nearly constant, then

$$\boxed{M(r) \propto r.} \tag{6.19}$$

This refers to total mass inside radius r, not to local density. The gravitating mass continues to grow well beyond the region where most starlight is concentrated. Motions perpendicular to galactic disks further indicate an approximately spherical distribution: a dark halo rather than a second thin disk.

Question from the audience. Is there a simple theory explaining why the rotation curve is so flat?

Response. Not in this argument. Simulations of galaxy formation may make such profiles natural, but the lecture does not claim a short derivation. The secure statement is observational: nearly constant v implies approximately linear growth of enclosed mass. The neatness of that relation should not be mistaken for an elementary explanation of galaxy formation. [7]

Question from the audience. Are spiral arms simply material pinwheels produced by the varying angular speed v/r?

Response. No simple pinwheel picture works. The visible pattern need not rotate with individual stars; density-wave or star-formation-wave descriptions are among the possibilities. The lecture uses this uncertainty to separate the robust halo inference from the more complicated problem of spiral structure. [8]

The inferred dark mass is several times the luminous mass, quoted in the lecture only as an order-of-magnitude factor between roughly five and ten. The halo also extends beyond the visible galaxy. These are deliberately rough historical scales, not precise universal ratios for every system.

6.5 What the Dark Matter Must Be Like

Why does ordinary matter form bright compact disks while dark matter remains in a broad halo? Luminous matter can lose mechanical energy through electromagnetic collisions and radiation. Once it dissipates energy, gravity can pull it into a denser configuration. Dark matter appears electrically neutral and much more weakly interacting. Its particles orbit through the halo with little collision or radiation, so even over billions of years they do not lose enough energy to settle into the luminous disk. Gravitational radiation from galactic motion is far too weak to provide an alternative cooling mechanism.

Question from the audience. Does weakly interacting mean that dark matter obeys a different law of gravity?

Response. No. It is assumed to gravitate in the ordinary way. Weak interaction here means the absence, or great suppression, of nongravitational collisions and electromagnetic radiation. Two dark particles still attract gravitationally; that attraction organizes a halo but does not efficiently dissipate orbital energy. [9]

[7] Lecture timestamp: 00:35:58–00:37:33.
[8] Lecture timestamp: 00:37:35–00:38:34; 00:47:03–00:47:47.
[9] Lecture timestamp: 00:42:36–00:43:35.

Question from the audience. What observations show that this is not only a peculiarity of one galactic rotation curve?

Response. The evidence appears on several scales: stellar motion within galaxies, motion perpendicular to galactic planes, interactions between galaxies, and the dynamics of whole galaxy clusters. Oort's and Zwicky's examples were different manifestations of the same missing-gravity problem. [10]

Question from the audience. Should dark matter congregate around black holes and galactic centers?

Response. It does become denser toward a gravitational center, but the historical sequence is likely the reverse of a halo forming around a pre-existing black hole. Dark halos formed large gravitational wells; baryonic matter then fell into them and formed galaxies, stars, and central black holes. The black hole is a small part of the mass. Searches for annihilation products nevertheless look toward the galactic center because the dark-matter density is expected to be higher there. [11]

Question from the audience. Is dark matter a gas, dust, compact clumps, or isolated particles? Could some be trapped near the Sun?

Response. Its microscopic nature is unknown, but ordinary dust is bound by electromagnetic forces, exactly the sort of interactions dark matter seems to lack. The default particle picture is therefore a dilute population of individual particles. Some may be gravitationally concentrated near the Sun, yet their density is far too small to compete with solar gravity or disturb planetary dynamics. [12]

The second central requirement is that the dark matter cluster. A hot population with relativistic random speeds resists gravitational aggregation; a cold population can fall into potential wells. Ordinary light neutrinos would have behaved as hot dark matter and do not provide the required primary galactic component. In this context *cold* means nonrelativistic in the local cosmological frame, not literally at laboratory zero temperature.

Question from the audience. Does the requirement of cold clustering impose a dark-matter mass range?

Response. It depends on quantum statistics and production history. Very light fermions cannot all occupy one slow state and are strongly constrained. Very light bosons can occupy a coherent Bose-condensed state; axions provide the lecture's example. Alternatively, heavier weakly interacting particles can be cold without condensation. Thus there is no model-independent single mass bound supplied by this argument. [13]

[10] Lecture timestamp: 00:42:00–00:42:36.
[11] Lecture timestamp: 00:43:36–00:45:43.
[12] Lecture timestamp: 00:45:43–00:47:02; 00:58:41–00:59:29.
[13] Lecture timestamp: 00:53:17–00:54:52; 00:56:42–00:58:05.

Question from the audience. Should accelerators already have found the particles?

Response. Only under additional assumptions. If the particles have roughly weak-interaction cross sections and masses within accelerator reach, collider production may be possible. The lecture's expectation of imminent information was explicitly speculative and historically dated. A particle could instead be too heavy or interact too feebly for the available experiments. [14]

Question from the audience. Could dark matter merely live in an unseen extra dimension?

Response. No such mechanism is needed in the discussion. Dark matter is already observed gravitationally and may simply consist of unusual particles with almost no interaction with ordinary material. Calling an interaction weak is not the same as placing the particles outside observable spacetime. [15]

Modifying gravity rather than adding unseen matter is a logical alternative. The lecture judges that no comparably simple modification then explained the evidence across all relevant scales while retaining the successful solar-system laws. The particle interpretation is therefore preferred, but its microphysics remains unknown.

6.6 Dark Matter Does Not Close the Old Budget

Dark matter enlarges the matter contribution by a factor of several, but the old discrepancy between luminous matter and H_0^2 was of order twenty-five. It therefore narrows the gap without making

$$\frac{C_m}{a_0^3} = H_0^2. \tag{6.20}$$

Within the old no-Λ picture, a positive curvature-budget term and hence $k < 0$ were still required. There was nothing logically inconsistent about such an open model; the point is that dark matter alone did not select the modern answer.

Question from the audience. What did cosmologists actually think about curvature before the modern fit?

Response. The evidence was confusing and prior preference for a closed, bounded universe sometimes pulled against the matter budget. Once the old assumptions were combined consistently, negative curvature was the natural inference. That historical consensus was still provisional because neither dark matter nor vacuum energy was securely incorporated in the way they are today. [16]

We must therefore test the candidate model rather than infer its truth from one budget line. A model consists of a curvature choice k and an expansion history $a(t)$, equivalently a set of material equations of state and present parameters from which $a(t)$ is computed. The old candidate is matter plus negative curvature with negligible radiation and no cosmological constant.

[14] Lecture timestamp: 00:54:52–00:55:54.
[15] Lecture timestamp: 00:55:54–00:56:41.
[16] Lecture timestamp: 00:50:17–00:51:50.

6.7 Turn Geometry Into Observables

6.7.1 Source Counts and Brightness in a Frozen Geometry

Temporarily ignore cosmic evolution. In a homogeneous universe the number of sources in a comoving shell is proportional to its area:

$$dN \propto \xi(r)^2\, dr. \tag{6.21}$$

Thus

$$dN \propto \begin{cases} \sin^2 r\, dr, & k=+1, \\ r^2\, dr, & k=0, \\ \sinh^2 r\, dr, & k=-1. \end{cases} \tag{6.22}$$

At large r, the negatively curved case grows as $\sinh^2 r \sim e^{2r}$, far faster than the flat r^2 law. Negative curvature therefore predicts an anomalously rapid increase in the number of sources with distance.

The same area controls received flux. In flat static space a standard source gives

$$F \propto \frac{1}{r^2}. \tag{6.23}$$

In hyperbolic space the sphere over which radiation spreads grows faster, so the flux falls faster. The lecture sometimes calls this observed quantity luminosity; here F denotes brightness or flux, while intrinsic luminosity belongs to the source. Counts and brightness together can therefore distinguish spatial geometries.

6.7.2 Redshift and the Past Light Cone

The real universe evolves, and looking outward means looking backward in time. Light emitted at t_{em} is stretched while the scale factor grows from $a(t_{\mathrm{em}})$ to a_0:

$$1+z = \frac{\lambda_{\mathrm{obs}}}{\lambda_{\mathrm{em}}} = \frac{a_0}{a(t_{\mathrm{em}})}. \tag{6.24}$$

Equivalently,

$$z = \frac{\lambda_{\mathrm{obs}}}{\lambda_{\mathrm{em}}} - 1. \tag{6.25}$$

Different points along our past light cone correspond simultaneously to different comoving radii and different emission times. Expansion changes the wavelength; Doppler and gravitational shifts are other physical mechanisms, but cosmological expansion is the effect at issue here.

> **Question from the audience.** Why does the definition of z contain the minus one?
>
> **Response.** It is a historical convention chosen so that an unshifted source has $z=0$. The physically direct stretch factor is $1+z=\lambda_{\mathrm{obs}}/\lambda_{\mathrm{em}}$. [17]

Radial light has $d\Omega_2 = 0$ and follows $ds^2 = 0$. Equation (6.4) therefore gives

$$dt^2 = a(t)^2 dr^2. \tag{6.26}$$

[17] Lecture timestamp: 01:10:11–01:10:45.

For an incoming ray followed backward from us,

$$dt = -a(t)\,dr, \qquad \frac{dr}{dt} = -\frac{1}{a(t)}. \tag{6.27}$$

The minus sign says that larger comoving radius lies at earlier time along the ray. Once $a(t)$ is known, this equation determines $r(t)$.

6.7.3 Number Per Unit Redshift

An observer can count standard sources in a redshift bin z to $z + dz$. Begin with equation (6.21) and differentiate equation (6.24) along the past light ray:

$$dz = -\frac{a_0}{a^2}\,da = -\frac{a_0}{a^2}\dot{a}\,dt. \tag{6.28}$$

Using $dt = -a\,dr$,

$$dz = \frac{a_0}{a}\dot{a}\,dr = (1+z)\dot{a}\,dr. \tag{6.29}$$

Hence

$$dr = \frac{dz}{(1+z)\dot{a}}, \tag{6.30}$$

and, for a fixed comoving source density,

$$\boxed{\frac{dN}{dz} \propto \frac{\xi(r)^2}{(1+z)\dot{a}}}. \tag{6.31}$$

The live derivation checks signs and factors more than once; equations (6.27) and (6.24) fix the result unambiguously.

There is nothing further to evaluate until a model is chosen. A specified $a(t)$ determines $\dot{a}(t)$, equation (6.27) determines $r(t)$, and equation (6.24) lets every quantity be expressed in terms of z. Different equations of state produce different functions dN/dz and $F(z)$. This is exactly what makes the observations discriminating.

6.8 From Trial Parameters to the Best Fit

The procedure is direct:

$$(\Omega_r, \Omega_m, \Omega_\Lambda, \Omega_k) \longrightarrow a(t) \longrightarrow \left\{F(z), \frac{dN}{dz}\right\}. \tag{6.32}$$

Choose present parameters, solve Friedmann's equation, propagate radial light, compute brightness and counts versus redshift, and compare the curves with data. A failed comparison rejects the parameter set. Together, the two observables constrain both the expansion history and the spatial geometry.

The old model,

$$H^2 = \frac{C_m}{a^3} - \frac{k}{a^2}, \qquad k = -1, \tag{6.33}$$

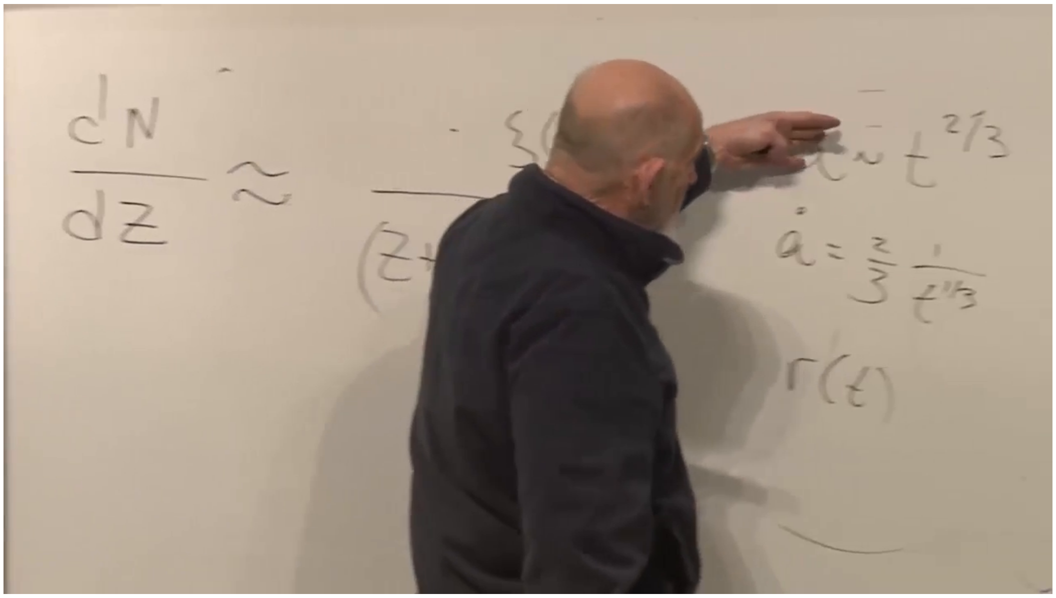

Figure 6.4: The board relation between dN/dz and the model functions. The left side and $a \sim t^{2/3}$, $\dot{a}$, and $r(t)$ on the right are visible, while the lecturer obscures part of the denominator; equation (6.31) is established independently above.

Lecture frame: 01:22:24.114.

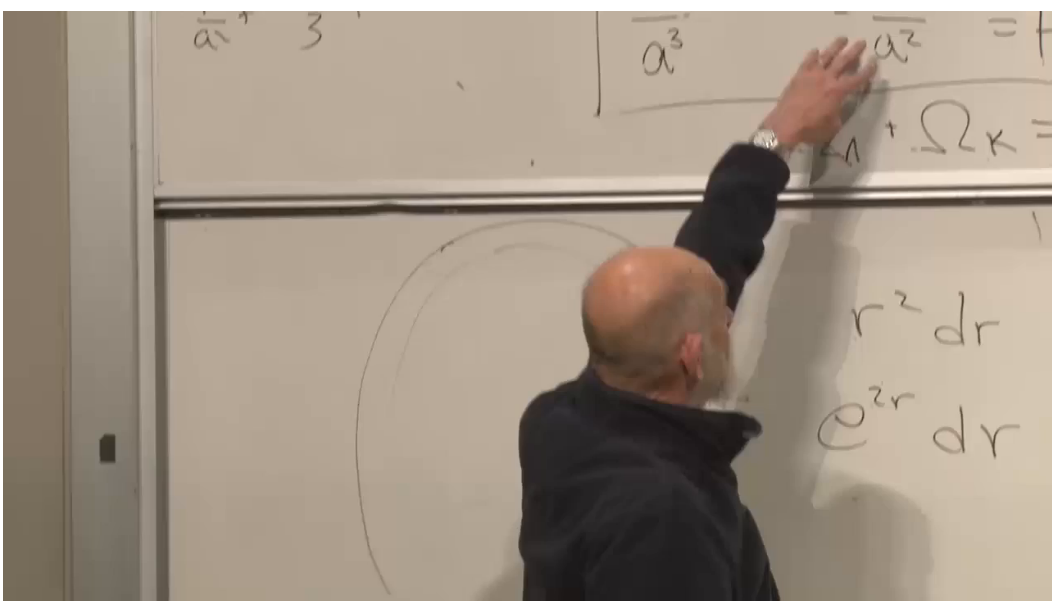

Figure 6.5: The old matter-plus-curvature model on the upper board and the flat and hyperbolic shell factors r^2dr and $e^{2r}dr$ on the lower board. Parts of the Friedmann equation are covered, so the frame is supporting evidence rather than an equation source.

Lecture frame: 01:24:27.863.

fails both the brightness-redshift relation and the source-count test. Supernovae provide the most useful standard candles in the relevant redshift range. The lecture's successful parameter set is

$$\boxed{\Omega_r \simeq 0, \qquad \Omega_m \simeq 0.3, \qquad \Omega_\Lambda \simeq 0.7, \qquad \Omega_k \simeq 0}. \tag{6.34}$$

Matter here includes luminous and dark matter. These values report the lecture-era fit; the durable lesson is the fitting method and the need for a dominant nondiluting component, not the last printed decimal.

6.8.1 Questions About What Is Measured and What Is Fitted

Question from the audience. Was Λ already assumed while the old open-universe model was being discussed?

Response. No. The first part deliberately reconstructed the older model with $\Lambda = 0$. The values in equation (6.34) describe the later, modern fit. The old candidate was not amended silently; it was tested against $F(z)$ and dN/dz and failed. [18]

Question from the audience. How is Ω_Λ obtained if vacuum energy is not measured directly like local matter?

Response. Begin with trial parameter sets. Each set defines a Friedmann solution and therefore predicted functions $F(z)$ and dN/dz. Compare those predictions with observations, discard poor fits, and retain the best region of parameter space. The number 0.7 is an inference from the fit, not a direct reading from a vacuum-energy meter. [19]

Question from the audience. Where does the present scale a_0 come from?

Response. The normalization of a is partly conventional, but a completed model tied to H_0 supplies the present physical curvature scale when $k \neq 0$. Once the model fits the redshift observables, its function $a(t)$ identifies the present point and the corresponding scale. [20]

Question from the audience. Does looking billions of years into the past distort a fit based on today's parameters?

Response. The lookback is not a distortion; it is the signal. Today's parameters are inserted into Friedmann's equation, which evolves them into a complete $a(t)$. The predicted light cone already includes the fact that matter was more important at smaller a, and sufficiently far back radiation dominates. A model that ignored this evolution would fail the data. [21]

[18] Lecture timestamp: 01:26:45–01:27:07.
[19] Lecture timestamp: 01:27:08–01:29:59; 01:31:57–01:33:40.
[20] Lecture timestamp: 01:29:59–01:30:42.
[21] Lecture timestamp: 01:30:43–01:31:55.

Question from the audience. Which present quantities are measured comparatively directly?

Response. The radiation density is measured and negligible today. The Hubble rate is measured from the distance-redshift relation. The total matter density is inferred gravitationally, including luminous and dark matter, and divided by the critical scale $3H_0^2/(8\pi G)$ to obtain Ω_m. The remaining late-time freedom lies mainly in vacuum energy and curvature. [22]

Question from the audience. If both Ω_Λ and Ω_k are unknown, why is there only one independent late-time trial parameter after Ω_m is fixed?

Response. Because equation (6.8) imposes one constraint. With Ω_r negligible and Ω_m measured, choosing a trial Ω_Λ fixes the leftover $\Omega_k = 1 - \Omega_m - \Omega_\Lambda$. One may equally parametrize the same one-dimensional family by curvature. For example, $\Omega_\Lambda = 0.4$ and $\Omega_k = 0.3$ is a valid trial when $\Omega_m = 0.3$; it is rejected only because its predicted curves fit poorly, not because the sum rule forbids it. [23]

Question from the audience. How can k be only $+1, 0, -1$ while Ω_k varies continuously, and are the dimensions correct?

Response. The discrete k records only the sign and normalized geometry. The continuous curvature radius is carried by a_0, so $-k/(a_0^2 H_0^2)$ can take a continuum of values. In units $c = 1$, inverse length and inverse time have the same dimensions; restoring conventional units supplies the required powers of c. [24]

Question from the audience. Does the count method assume homogeneity and isotropy, even though galaxy populations evolve?

Response. It assumes spatial homogeneity and isotropy on sufficiently large slices, and those symmetries can be tested by comparing different directions at the same redshift. Source evolution is a separate issue: populations at different epochs need not be identical, so practical analyses model selection and evolution rather than pretending the number density is eternally fixed. The simple derivation isolates the geometric factor before those refinements are added. [25]

6.9 The Density Fractions Evolve

Equation (6.6) holds at every time, but the Ω's in equation (6.7) were defined at the present time. Their time-dependent analogues change because the four terms have different powers of a. Far in the past,

$$\Omega_r(t) \longrightarrow 1. \tag{6.35}$$

[22] Lecture timestamp: 01:33:40–01:35:32.
[23] Lecture timestamp: 01:35:33–01:39:09.
[24] Lecture timestamp: 01:39:10–01:40:07.
[25] Lecture timestamp: 01:40:08–01:41:08.

As radiation dilutes faster, matter rises toward dominance. Later, matter itself dilutes while Λ remains fixed, so in the far future,

$$\Omega_\Lambda(t) \longrightarrow 1. \tag{6.36}$$

The matter fraction therefore begins small, becomes close to unity during the matter era, and falls again after vacuum energy takes over. Curvature, if nonzero, eventually loses to a positive constant Λ as well.

Question from the audience. Does a small present curvature fraction mean the universe becomes flatter as it expands?

Response. Its physical curvature scale grows with a, and relative to a positive cosmological constant the curvature contribution a^{-2} becomes ever less important. This is the sense of the phrase *big is flat*. It does not change the discrete topology label k; it says local curvature and the fractional dynamical importance of curvature decrease. [26]

Question from the audience. Did the fitted model show unexplained systematic residuals, and how precise was the claim?

Response. The lecture reports no known systematic failure of the framework at that level and describes it as precision cosmology at roughly the percent scale, not at a tenth or hundredth of a percent. That statement is historical and should not be converted into a timeless current error bar. [27]

Question from the audience. Can spatial curvature be tested by direct triangulation rather than only by supernova brightness and counts?

Response. Yes. The angular pattern of the cosmic microwave background supplies a standard physical scale and therefore a large cosmic triangle. Supernova data alone did not provide the lecture's full percent-level curvature statement; combining them with microwave-background geometry sharpened the test. This is the subject of the next lecture. [28]

6.10 What to Carry Forward

The central lesson is not a catalogue of four numbers. It is a method. A cosmological parameter set produces $a(t)$; the metric then predicts the propagation of light, source brightness, and source counts. Agreement tests both the chosen parameters and the Friedmann framework that connects them. There is no reason an arbitrary set must fit.

The old luminous-matter, negative-curvature model failed. Dark matter repaired the gravitating matter inventory but did not by itself close the old Friedmann budget. The successful lecture-era fit required approximately

$$\Omega_m \simeq 0.3, \qquad \Omega_\Lambda \simeq 0.7, \qquad \Omega_k \simeq 0, \tag{6.37}$$

and its curvature precision depended importantly on the microwave background. The model is credible because several independent observations pass through one dynamical theory and meet in the same region of parameter space.

[26] Lecture timestamp: 01:47:13–01:47:42.
[27] Lecture timestamp: 01:45:40–01:46:19.
[28] Lecture timestamp: 01:46:19–01:47:12.

Chapter 7

Thermal History, Last Scattering, and the Primordial Soup

Overview. The expansion history of the universe is also a thermal history. We begin by defining temperature through equilibrium and equilibrium through scattering. A photon gas acquires the Planck spectrum while free charges keep it thermalized; when electrons bind to protons, the radiation decouples but retains its blackbody shape as every wavelength stretches with the scale factor. The enormous photon-to-matter ratio then explains why recombination occurs far below the hydrogen ionization temperature. Continuing backward takes us through last scattering, matter–radiation equality, electron–positron pair equilibrium, and finally the small matter–antimatter excess that survives annihilation.

7.1 From Model Fitting to Thermal History

A cosmological model is specified, at minimum, by a scale-factor history and a spatial geometry,

$$a(t), \qquad k \in \{+1, 0, -1\}. \tag{7.1}$$

The practical method is to choose the model, calculate what it predicts, compare those predictions with data, and adjust the parameters. Running the inference backward directly from the data is possible, but it is often less transparent.

Curvature is difficult to determine from comparatively nearby sources. Just as the curvature of Earth is a small correction over a short terrestrial baseline, spatial curvature is a weak effect when the cosmological baseline is not large enough. This is both a disadvantage and an advantage: ordinary source surveys do not determine k sharply, but moderate uncertainty in k does not prevent a useful reconstruction of the expansion history.

Two families of observables carry much of the load. Standard candles give brightness as a function of redshift, while source surveys give the number of objects in a redshift interval. The resulting fits favor an expansion history with a substantial vacuum-energy contribution. Precise curvature information requires the much longer geometrical lever arm supplied by the cosmic microwave background.

We now change the question. Instead of asking only how $a(t)$ evolves, we ask how the universe's temperature evolves. That question begins with a more elementary one: under what conditions does a system have a temperature at all?

7.2 When a Universe Has a Temperature

Temperature is a property of thermal equilibrium. A collection of particles approaches equilibrium through collisions, and those collisions must act faster than the macroscopic environment changes. In an expanding universe the useful comparison is

$$\tau_{\rm scatt} \ll H^{-1}, \tag{7.2}$$

where $\tau_{\rm scatt}$ is a characteristic scattering time and H^{-1} is the expansion time. When this inequality fails, the universe changes before collisions can restore equilibrium.

Consider a perfectly reflecting box. Shine a laser pulse into it, close the opening, and suppose that the box contains only photons. The pulse bounces indefinitely, but ordinary low-energy photons scatter from one another extraordinarily weakly. Reflection alone does not redistribute their energy into a thermal spectrum. A photon gas therefore does not thermalize merely because it is confined.

Free charged particles change the situation. An electromagnetic wave accelerates an electron; the accelerated charge reradiates, so photons scatter efficiently. With enough free electrons and protons, repeated scattering redistributes energy among all three species until they form one thermal soup with a common temperature.

On large scales the universe is electrically neutral. In the simplest bookkeeping,

$$n_{e^-} = n_p, \tag{7.3}$$

because the electron and proton charges have equal magnitude and opposite sign. This equation states neutrality; it does not explain why charge is neutral or why the two charge magnitudes agree. Those are deeper facts about particle physics.

The distinction between a plasma and a neutral gas is decisive. Free electrons scatter radiation efficiently. Once electrons bind to protons to make neutral hydrogen, the dominant electric fields cancel at long distance and scattering becomes far less effective. At the low density of the present universe, neutral matter cannot bring the background photons into equilibrium within a cosmic lifetime.

At high enough temperature, energetic collisions repeatedly ionize hydrogen and maintain a substantial population of free charges. Electrons, protons, and photons then share one temperature. As the system cools, atoms survive, the free-electron density collapses, and the common equilibrium is lost. The temperatures subsequently assigned to matter and radiation need not agree.

> **Question from the audience.** If the microwave background has a temperature near 3 K, is that the temperature of the whole universe today?
>
> **Response.** No. The photon spectrum can be labeled by a blackbody temperature, but present matter is not in equilibrium with those photons. Different components have different kinetic or effective temperatures, so there is no single global thermodynamic temperature today.

7.3 Thermal Radiation and the Quantum Turnover

An equilibrium radiation field contains a spectrum of wavelengths and frequencies related by

$$\nu\lambda = c. \tag{7.4}$$

To avoid confusing radiance with energy density, let $u_\nu(T)$ denote energy per unit volume per unit frequency:

$$dE = u_\nu(T)\, dV\, d\nu. \tag{7.5}$$

Thus u_ν has dimensions of energy divided by volume and frequency.

Temperature itself is an energy scale. Historical temperature units obscure that fact, so Boltzmann's constant supplies the conversion:

$$E_{\text{thermal}} \sim k_B T. \tag{7.6}$$

One may set $k_B = 1$ in natural units, but retaining it keeps kelvin and energy visibly distinct.

Before quantum mechanics, dimensional analysis using $k_B T$, ν, and c gives the Rayleigh–Jeans spectral energy density,

$$u_\nu^{\text{RJ}}(T) = \frac{8\pi\nu^2}{c^3} k_B T. \tag{7.7}$$

The numerical factor depends on counting the electromagnetic modes, but the essential classical behavior is $u_\nu \propto T\nu^2$. It rises without limit as frequency increases, and therefore predicts

$$\int_0^\infty u_\nu^{\text{RJ}}\, d\nu = \infty. \tag{7.8}$$

This impossible result is the ultraviolet catastrophe. The measured spectrum instead rises, turns over, and falls rapidly enough to contain finite energy.

Planck's constant introduces the missing quantum scale. For the spectral energy density defined in equation (7.5), Planck's law is

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/(k_B T)} - 1}. \tag{7.9}$$

The exponent depends on the dimensionless ratio

$$x = \frac{h\nu}{k_B T}, \tag{7.10}$$

because $E_\gamma = h\nu$ and $k_B T$ are both energies.

At low frequency, $x \ll 1$, and

$$e^x - 1 = x + O(x^2). \tag{7.11}$$

Substitution into equation (7.9) gives

$$u_\nu(T) \simeq \frac{8\pi h\nu^3}{c^3} \frac{k_B T}{h\nu} \tag{7.12}$$

$$= \frac{8\pi\nu^2}{c^3} k_B T, \tag{7.13}$$

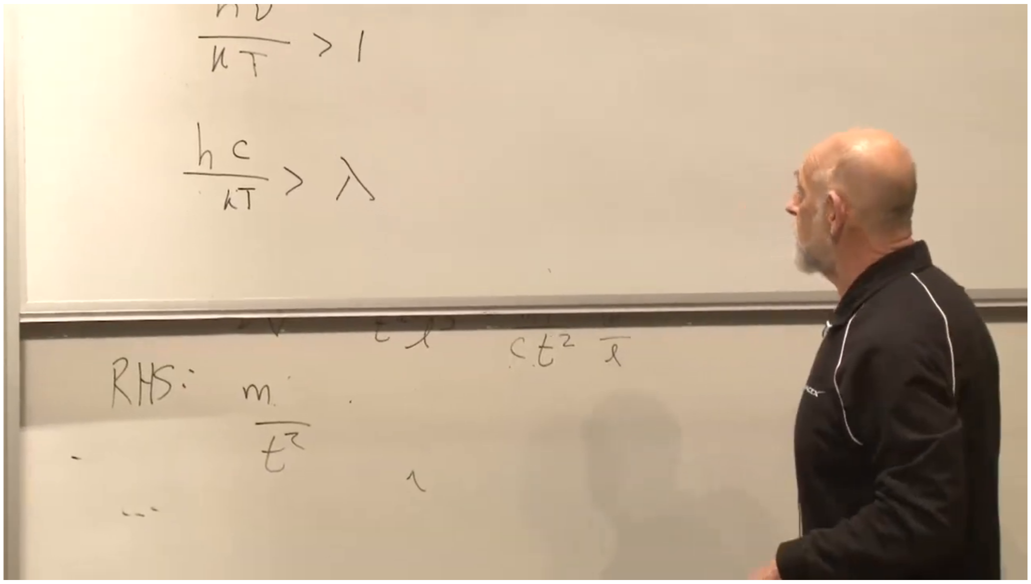

Figure 7.1: The thermal crossover on the board. The cropped upper expression is $h\nu/(kT) > 1$, and the clearly visible lower expression is $hc/(kT) > \lambda$. Faint remnants on the lower board are not used as equation evidence.

Lecture frame: 00:44:42.840.

so classical physics is recovered in its proper low-frequency domain. At high frequency, $x \gg 1$, the exponential dominates and suppresses the spectrum. The crossover occurs at

$$\frac{h\nu}{k_B T} \sim 1. \tag{7.14}$$

This is the point at which the quantum turnover replaces the classical rise.

Using $\nu = c/\lambda$, the crossover defines an order-of-magnitude thermal wavelength,

$$\boxed{\lambda_{\text{th}} \sim \frac{hc}{k_B T}}. \tag{7.15}$$

Shorter wavelengths lie in the exponentially suppressed tail; most of the power is carried by wavelengths of this order, up to a numerical factor that depends on whether the spectrum is plotted per unit frequency or per unit wavelength.

Question from the audience. Did Planck obtain the spectrum from the electron–proton plasma model used here?

Response. No. He combined empirical spectral information with Wien's displacement law, which already showed that the peak wavelength varies inversely with temperature. That scaling revealed a new constant h. Planck connected the result to oscillators in the material walls, while the full quantum interpretation of the radiation field came later. [1]

[1] Lecture timestamp: 00:37:06–00:39:25.

Question from the audience. Why did the dimensional check appear to miss one power of c?

Response. The quantity first called intensity was defined as energy per volume per frequency, whereas the familiar formula with $2h\nu^3/c^2$ is spectral radiance per area and solid angle. Spectral energy density carries c^{-3} and an angular factor, as in equation (7.9). The correction changes the normalization convention, not the dimensionless ratio $h\nu/(k_B T)$ or the thermal wavelength. [2]

Apart from its normalization, the Planck spectrum has a universal shape. Equation (7.9) can be written

$$u_\nu(T) = T^3\, F\left(\frac{\nu}{T}\right), \tag{7.16}$$

with constants absorbed into F. Raising T rescales the frequency axis and the overall height; the same dimensionless curve reappears. In wavelength language, raising the temperature squeezes the spectrum toward shorter wavelengths.

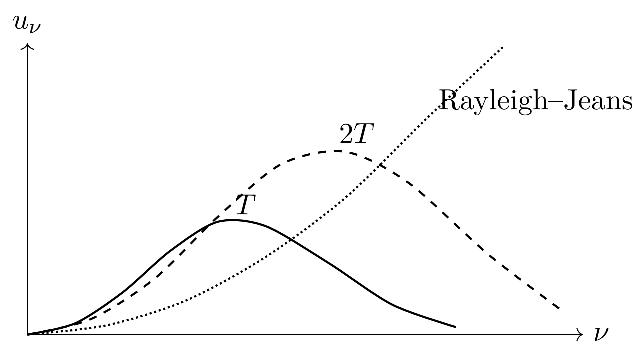

Figure 7.2: The classical spectrum rises without bound, while Planck spectra turn over. Increasing the temperature shifts the characteristic frequency upward and preserves the dimensionless shape.

The overwhelming majority of photons in the present universe belong to the microwave background and have characteristic wavelength of order a millimeter. Equation (7.15) associates that wavelength with a few kelvin, conventionally rounded here to 3 K. A wavelength alone would provide only a rough association. The stronger evidence is that measurements over many frequencies trace the complete blackbody curve with very high precision.

Question from the audience. How much space does one microwave-background photon occupy?

Response. A photon has no universal hard-edged size. It may be prepared as a wave packet whose spatial spread depends on the state, while its wavelength sets the scale of its oscillation. For the thermal background, both the characteristic wavelength and a natural wave-packet scale are of order a millimeter, but neither should be mistaken for the radius of a little classical object. [3]

[2] Lecture timestamp: 00:39:33–00:44:51.
[3] Lecture timestamp: 00:57:16–00:59:05.

7.4 Expansion Freezes the Blackbody Shape

Begin with a hot box containing ionized hydrogen and radiation in equilibrium. As the box expands, its contents do work, the energy density falls, and the temperature decreases. Eventually electrons and protons remain bound as neutral atoms. The formation of atoms is called *recombination*; the loss of efficient photon–matter scattering is called *decoupling*. The two processes are closely related but name different physical events.

After decoupling, matter and radiation no longer share a common temperature. Yet the photon spectrum retains its Planck form. Every freely propagating wavelength is stretched by the same expansion factor,

$$\lambda \propto a, \qquad \nu \propto a^{-1}. \tag{7.17}$$

Because the Planck curve depends on ν/T, this uniform redshift is equivalent to assigning the radiation a temperature

$$\boxed{T_\gamma \propto a^{-1}.} \tag{7.18}$$

Thus expansion preserves the shape even without collisions continually rebuilding it. Matter follows its own later thermal history.

The observed microwave background is therefore a fossil equilibrium distribution. Its blackbody form was established while free charges scattered efficiently and was then carried forward by uniform cosmological redshift. Starlight, gamma rays, and other later sources exist, but they are numerically small beside the relic photon population and have identifiable astrophysical origins.

Question from the audience. How can radiation remain blackbody after it is no longer in equilibrium with matter?

Response. Collisions are needed to create the equilibrium spectrum, but not to preserve it under perfectly uniform expansion. Multiplying every wavelength by the same factor maps one Planck spectrum into another with a lower temperature. The photons cease sharing matter's temperature while retaining a blackbody temperature of their own.

7.5 Why Recombination Happens So Late

The wavelength today and the wavelength at decoupling are related by

$$\frac{\lambda_0}{\lambda_{\rm dec}} = \frac{a_0}{a_{\rm dec}} = \frac{T_{\rm dec}}{T_0}. \tag{7.19}$$

Decoupling is not mathematically instantaneous, but it is short enough compared with the subsequent history that one may assign it a characteristic scale factor and temperature. Here T_0 means the CMB temperature, not the kinetic temperature of intergalactic matter.

Hydrogen's ground-state ionization energy is

$$\chi_H = 13.6\,\text{eV}. \tag{7.20}$$

A first guess would set $k_BT \sim \chi_H$, so that a typical photon can ionize an atom. That guess is much too high because a thermal spectrum contains a high-energy tail and the universe contains an enormous number of photons.

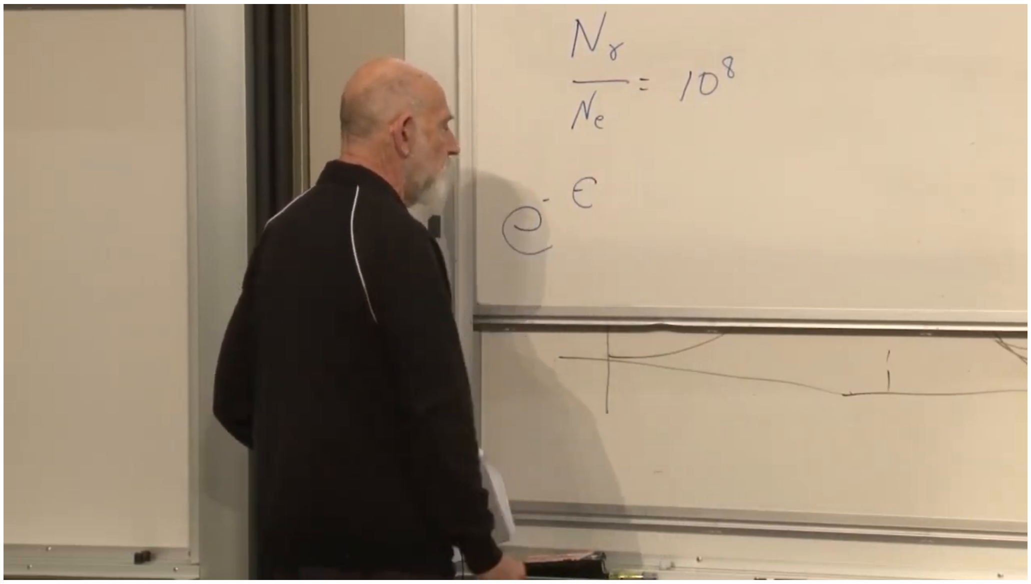

Figure 7.3: The board introduces the observed abundance ratio $N_\gamma/N_e = 10^8$ and the photon-energy symbol ϵ. Only the beginning of the exponential is visible; the completed Boltzmann factor follows from the spoken derivation.

Lecture frame: 01:06:10.631.

The rounded abundance ratio used here is

$$\frac{N_\gamma}{N_e} \sim 10^8. \tag{7.21}$$

It is inferred by comparing the measured CMB photon density with the ordinary-matter inventory. For photon energy ϵ, the high-energy tail has Boltzmann form

$$P(\epsilon) \propto e^{-\epsilon/(k_B T)}. \tag{7.22}$$

The number of ionizing photons available per atom is therefore estimated by

$$10^8 e^{-\chi_H/(k_B T)}. \tag{7.23}$$

Ionization remains effective until this quantity falls to order unity.

Taking logarithms gives

$$\frac{\chi_H}{k_B T} \sim \ln 10^8 \simeq 18.4, \tag{7.24}$$

or, at the intended accuracy,

$$k_B T \sim \frac{\chi_H}{20}. \tag{7.25}$$

The characteristic photon can lie far below threshold because one ionizing photon from the tail is enough for each atom.

Question from the audience. Does the tail estimate count photons exactly at the threshold or all photons above it?

Response. It counts photons at and above the threshold. Since the tail falls exponentially, most of that integral comes from energies near χ_H, so the simple exponential estimate captures the relevant order of magnitude. [4]

Question from the audience. Was the ratio N_γ/N_e also about 10^8 at decoupling?

Response. To the approximation used here, yes. From decoupling back to substantially hotter epochs, expansion dilutes both populations by the same volume factor and does not create or destroy enough of either species to alter their ratio. At still higher temperatures, pair-production reactions require new bookkeeping. [5]

Question from the audience. Do photons emitted by stars contribute appreciably to this count?

Response. No. Stars did not yet exist at recombination, and even today their photons are a tiny fraction of the CMB population. The ratio in equation (7.21) concerns the relic thermal photons. [6]

Question from the audience. When electrons recombine with protons, should the emitted photons leave a sharp hydrogen feature in the blackbody spectrum?

Response. While scattering and absorption remain fast compared with expansion, the radiation is continually reprocessed and remains close to thermal equilibrium. A sufficiently abrupt freeze-out could preserve a distortion, but slow near-equilibrium recombination prevents a large isolated line from replacing the blackbody form. [7]

The resulting classroom estimate is $T_{\text{dec}} \sim 4 \times 10^3$ K. A direct conversion of $\chi_H/20$ gives a somewhat higher number, leading to the suggestion that the effective denominator may be nearer 40. This is not a contradiction; it exposes the crudeness of replacing the full ionization balance by one tail count. A controlled equilibrium treatment uses the Saha relation,

$$\frac{x_e^2}{1-x_e} = \frac{1}{n_H}\left(\frac{m_e k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_H/(k_B T)}, \tag{7.26}$$

followed by nonequilibrium recombination kinetics. The rough argument is valuable because it identifies the dominant physics: the tiny baryon-to-photon ratio delays neutral-atom survival to a temperature far below χ_H/k_B.

[4] Lecture timestamp: 01:08:39–01:08:54.
[5] Lecture timestamp: 01:08:54–01:09:57.
[6] Lecture timestamp: 01:11:44–01:12:39.
[7] Lecture timestamp: 01:12:42–01:14:07.

7.6 Redshift to the Surface of Last Scattering

With $T_0 \simeq 3\,\mathrm{K}$ and $T_{\rm dec} \sim 4 \times 10^3\,\mathrm{K}$, equation (7.19) gives

$$\boxed{\frac{a_0}{a_{\rm dec}} \sim \frac{T_{\rm dec}}{T_0} \sim 10^3}. \tag{7.27}$$

Equivalently, the last-scattering redshift is of order 10^3. Before this epoch the ionized plasma was opaque; after it, photons propagated nearly freely.

This thermal event occurred only a few hundred thousand years after the beginning of the hot expansion, conventionally quoted in the exchange as roughly 380,000 years. That interval is tiny beside the subsequent cosmic age. Since decoupling, lengths have grown by about 10^3 and comoving volume elements by about 10^9.

Decoupling is one thermal landmark, not the only event in the intervening history. Galaxy formation, star formation, and black-hole formation all occur later, but they are set aside here so that the thermal landmarks can be followed cleanly backward.

Looking outward is looking backward. Nearby shells show recent epochs; more distant shells show earlier ones. Our past light cone eventually meets the epoch of recombination, and its intersection with that epoch is the surface of last scattering. Earlier electromagnetic radiation repeatedly scattered from free charges and could not travel directly to us.

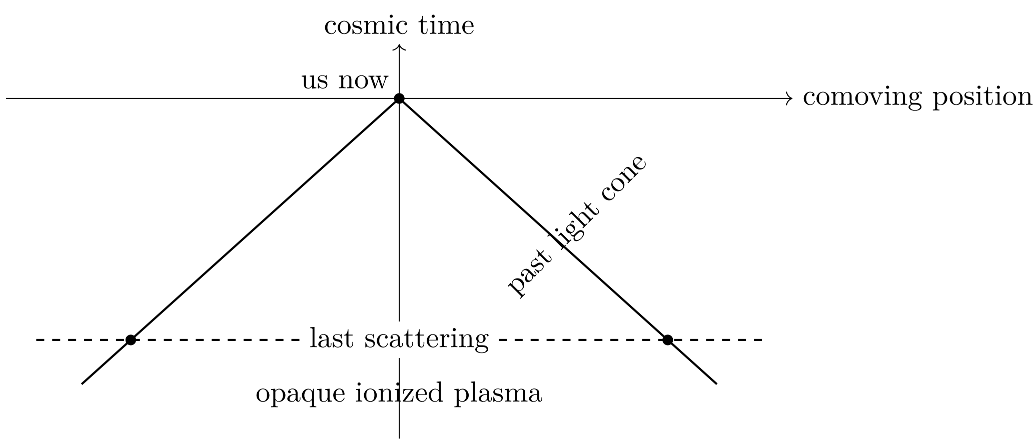

Figure 7.4: Electromagnetic observation reaches back along the past light cone only to the last-scattering surface. Earlier photons were repeatedly scattered by the ionized plasma.

Question from the audience. Does decoupling end every possible observation of earlier epochs?

Response. It ends direct electromagnetic viewing because the plasma was opaque to photons. More weakly interacting messengers could escape earlier: relic neutrinos in principle probe a deeper layer, and gravitational radiation could carry information from earlier still. Much of our knowledge beyond last scattering is therefore indirect but not arbitrary. [8]

[8] Lecture timestamp: 01:32:27–01:36:31.

Question from the audience. What was the absolute value of a at the beginning, and was the universe necessarily a tiny finite object?

Response. In flat space only ratios of scale factors have invariant meaning; the normalization of a is arbitrary. A negatively curved spatial slice may also be infinite. A closed $k = +1$ universe has a finite curvature radius, but observation still determines expansion ratios rather than a known primordial size. The hot big bang is not, by itself, the statement that all space was a tiny ball. [9]

Question from the audience. Because the CMB is older than galaxies, does seeing it tell us whether galaxies exist today beyond our observable region?

Response. No. The CMB shows a distant region as it was at decoupling, not as it is today. Light from present-day galaxies at sufficiently great distance has not had time to reach us. What favors continued galaxy populations beyond our horizon is the observed large-scale homogeneity and the absence of a visible edge, not a direct view supplied by the CMB. [10]

The resulting timeline runs backward from the present through galaxy formation and decoupling to matter–radiation equality. Earlier still, the shrinking scale factor means increasing temperature and the appearance of new particle species.

7.7 Matter–Radiation Equality

Moving backward beyond decoupling, another landmark appears. Nonrelativistic matter and radiation dilute differently:

$$\rho_m \propto a^{-3}, \qquad \rho_r \propto a^{-4}, \qquad \frac{\rho_m}{\rho_r} \propto a. \tag{7.28}$$

Matter dominates late, but radiation inevitably dominates at sufficiently small a. The expansion law changes correspondingly from the matter-era behavior $a \propto t^{2/3}$ toward the radiation-era behavior $a \propto t^{1/2}$.

A deliberately rough estimate starts with 10^8 photons per proton, characteristic CMB photon energy 10^{-4} eV, and proton rest energy 10^9 eV. Ignoring dark matter gives

$$\left(\frac{\rho_m}{\rho_r}\right)_0 \sim \frac{10^9}{10^8\,10^{-4}} \sim 10^5. \tag{7.29}$$

The argument then corrects itself: dark matter contributes substantially more mass than luminous baryons, raising the rounded estimate to 10^6. Within this arithmetic, equation (7.28) places equality at

$$\frac{a_{\rm eq}}{a_0} \sim 10^{-6}, \qquad \frac{T_{\rm eq}}{T_0} \sim 10^6. \tag{7.30}$$

These are scale-setting classroom estimates, not a precision cosmological parameter fit. Their purpose is to show how unequal powers of a locate successive thermal eras.

[9] Lecture timestamp: 01:30:05–01:32:22.
[10] Lecture timestamp: 01:59:01–02:00:58.

The radiation-dominated epoch lies behind the electromagnetic last-scattering wall, but its existence follows from the measured late-time densities and the known scaling laws. Continuing to higher temperature eventually invalidates the assumption that proton, electron, and photon numbers are separately fixed, because energetic collisions begin to create massive particle–antiparticle pairs.

Question from the audience. Does the statement that photon and proton numbers are conserved include starlight and arbitrarily high temperatures?

Response. No. It applies to the relic CMB population over the interval in which number-changing reactions are ineffective. Starlight was created later and is negligible in the photon count. At sufficiently high temperature, pair creation changes the particle content and separate number conservation fails. Recombination itself adds at most roughly one photon per electron, only one part in 10^8 of the existing CMB population. [11]

7.8 Pair Creation and the Almost Symmetric Soup

At a temperature of order

$$T \sim 10^{10}\,\mathrm{K}, \qquad \frac{a_0}{a} \sim 10^{14}, \tag{7.31}$$

the characteristic photon energy is around a mega-electron-volt, comparable to the rest-energy scale required to make an electron–positron pair. Two photons can then react through

$$\gamma + \gamma \longleftrightarrow e^- + e^+. \tag{7.32}$$

The exact threshold depends on collision kinematics, but the electron rest energy, $m_e c^2$, sets the relevant scale.

Pair creation and annihilation rapidly balance one another. If too few pairs are present, photon collisions replenish them; if too many are present, annihilation returns their energy to photons. The abundances of photons, electrons, and positrons are therefore determined by temperature and are all of comparable order in the relativistic plasma. They no longer retain a simple memory of their later numbers.

Question from the audience. How can this pair-filled plasma remain electrically neutral if there are already electrons needed to balance protons?

Response. Each reaction creates one negative and one positive charge, so the pairs add no net charge. They are an enormous nearly symmetric population superposed on a tiny electron excess. Individual electrons are indistinguishable; only the difference $N_{e^-} - N_{e^+}$ remembers the charge needed to balance the protons. [12]

Pair reactions change the sum but preserve the difference:

$$N_{e^-} + N_{e^+} \quad \text{changes}, \qquad N_{e^-} - N_{e^+} = \text{constant}. \tag{7.33}$$

[11] Lecture timestamp: 01:26:24–01:27:40.
[12] Lecture timestamp: 01:42:31–01:43:14.

In the hot soup $N_{e^-} + N_{e^+}$ is of the same order as N_γ, whereas the surviving excess is of the same order as the much smaller proton number. Hence the rounded ratio is

$$\boxed{\frac{N_{e^-} - N_{e^+}}{N_{e^-} + N_{e^+}} \sim 10^{-8}}. \tag{7.34}$$

There was roughly one excess electron for every 10^8 electron–positron pairs.

This reverses the original question. The striking fact is not simply that photons are numerous. At high temperature photons and pairs naturally have comparable abundances. The puzzle is why the negative and positive particle populations differ at all, and why their relative difference is tiny rather than exactly zero or of order unity. A symmetric initial state and symmetric pair production would give no excess. The microscopic origin of the nonzero asymmetry is the problem carried into the next stage of the discussion.

At temperatures another two or three orders of magnitude higher, proton structure can no longer be ignored. Collisions produce quarks and antiquarks, while protons dissolve into their constituents; photons, leptons, quarks, antiquarks, gluons, and whatever dark species are light enough form a hotter plasma. Vacuum energy is dynamically negligible there because radiation density grows as a^{-4}.

The corresponding quark excess is tied to the electron excess by overall charge neutrality. Three valence quarks make a baryon, so the bookkeeping differs by conventional factors, but the surviving baryonic matter and the surviving electrons are not independent charge imbalances. Their common smallness is a relic of the early universe.

Follow the electron–positron plasma forward again. Once the temperature falls below the pair-production scale, annihilation continues while the reverse reaction becomes rare. Almost every electron finds a positron, the pair energy becomes radiation, and only the tiny unmatched electron population remains. The matter around us is this residual imbalance, not a representative sample of the once enormous pair population.

Question from the audience. What were the protons doing during the electron–positron annihilation era?

Response. At that stage the temperature was high enough to create electron pairs but not yet high enough to create abundant proton–antiproton pairs or dissolve protons into quarks. The proton population was therefore small but persistent, and neutrality required $N_{e^-} - N_{e^+} \simeq N_p$. [13]

The visible universe appears to have the same sign of matter excess throughout. The classroom evidence offered is the absence of antinuclei in cosmic rays: secondary collisions can make occasional antiprotons, but an antigalaxy would be expected to supply complex antinuclei. This is a lecture-era observational argument and not a claim that regions beyond the observable horizon have been inspected.

Question from the audience. Could the observed matter excess be a local fluctuation, with antimatter regions beyond what we can see?

[13] Lecture timestamp: 01:55:35–01:56:53.

Response. The same sign across the enormous observable volume is far too coherent to be a random fluctuation. It demands a dynamical explanation. Beyond the observable region, however, direct evidence is unavailable; a transition to a different domain cannot be ruled out merely by looking farther than light has reached us. [14]

Question from the audience. Is the matter–antimatter explanation already a precise numerical theory, and does force unification occur at this temperature?

Response. There is a qualitative theory of how a small asymmetry can arise, but the rounded 10^{-8} value is not derived here from a unique microscopic model. Force unification belongs to temperatures much higher than the electron-pair epoch. Both questions require physics beyond the present thermal landmarks. [15]

7.9 What to Carry Forward

The thermal history is organized by rates and thresholds. Rapid scattering gives a common temperature; neutral-atom formation removes the scatterers; uniform expansion then redshifts a frozen Planck spectrum without destroying its shape. The CMB is therefore both a thermometer for the radiation and a visible surface marking the end of direct electromagnetic access to the earlier plasma.

The photon excess makes recombination a tail problem rather than an average-energy problem. The scale $T_{\mathrm{dec}}/T_0 \sim 10^3$ locates last scattering, while the unequal dilution laws for matter and radiation locate an earlier equality era. At still higher temperature, pair creation erases memory of the total electron and positron numbers but preserves their tiny difference. That residual asymmetry, mirrored in the baryonic sector, is the unexplained initial datum that the next investigation must confront.

[14] Lecture timestamp: 01:56:57–01:58:47.
[15] Lecture timestamp: 01:51:36–01:52:24.

Chapter 8

Baryogenesis, Cosmic Smoothness, and Hubble Friction

Overview. Two small dimensionless numbers organize the evening. The first is the surviving matter–antimatter excess, rounded in the course to one part in 10^8. Its dynamical explanation requires baryon-number violation, particle–antiparticle asymmetry, and departure from thermal equilibrium: the three Sakharov conditions. The second is the primordial density contrast, of order 10^{-5} at decoupling. Gravity amplifies rather than erases such ripples, so their smallness motivates inflation. We then prepare the mechanics of inflation by deriving the equation of motion of a homogeneous scalar field in an expanding universe and identifying $3H\dot{\phi}$ as Hubble friction.

8.1 The Entropy Question Turned Upside Down

We shall begin inflation tonight, but first we must finish baryogenesis: the creation of the excess of matter over antimatter. The framework is more secure than any detailed numerical model. That is why the subject can be summarized economically even though its central number has not been derived from first principles.

The historical route into the problem began in 1980, when Bob Wagoner asked Savas Dimopoulos and Leonard Susskind why the entropy of the universe is so large. Put cosmologically, the question was why there are so many photons for every surviving proton.

At very high temperature, photons, quarks, antiquarks, electrons, and positrons were all abundant and close to thermal equilibrium. Quarks and antiquarks each had number densities of the same broad order as the photon density. That observation already suggests turning the question around: why did annihilation leave any unmatched matter at all?

The priority of the structural argument belongs to Andrei Sakharov. Dimopoulos and Susskind independently arrived at closely related conditions in 1980, then found Sakharov's 1967 paper while preparing a historical contribution. The conditions are now justly known by Sakharov's name.

The connection to entropy can now be made explicit. For blackbody radiation, entropy is proportional to photon number up to a fixed factor of order unity. Indeed,

$$s_\gamma = \frac{4\rho_\gamma}{3T}, \qquad n_\gamma \propto T^3, \tag{8.1}$$

so a comoving thermal photon gas has $S_\gamma \propto N_\gamma$. The numerical constants are not the issue; the puzzle is the enormous photon-to-proton ratio.

> **Question from the audience.** How is the photon abundance related to entropy?
>
> **Response.** Blackbody radiation is thermal, and its entropy is proportional to its mean photon number. In this context, asking why the cosmic entropy is large is essentially asking why the cosmic microwave background contains so many photons per proton. The proportionality includes numerical constants, but they are irrelevant to the order-of-magnitude puzzle. [1]

As the universe cooled, particles and antiparticles annihilated, while the radiation remained and eventually decoupled. The surprising fact is therefore not that photons survived; it is that annihilation left a tiny unmatched population of matter.

The useful dimensionless number is therefore

$$\eta_B \equiv \frac{N_B - N_{\bar{B}}}{N_\gamma} \sim \frac{N_p - N_{\bar{p}}}{N_p + N_{\bar{p}}} \sim 10^{-8}, \tag{8.2}$$

where the last value is the rounded order used in the course, not a precision determination. Once an odd-looking number appears, "it simply began that way" ceases to be a satisfying stopping point. A quantitative mechanism should explain why the answer is small and why it has a definite sign.

Calling the excess *matter* rather than *antimatter* is partly conventional: matter is the member of the pair from which we are made. The physical statement is only that one sign dominates throughout the observable region.

8.2 Baryon Number and the First Condition

Baryon number is normalized so that a proton or neutron carries $B = 1$. In quark language,

$$B = \frac{N_q - N_{\bar{q}}}{3}. \tag{8.3}$$

The factor 1/3 is essential because each ordinary baryon contains three valence quarks. Other hadrons may carry baryon number, but the unstable ones eventually decay; the net quark excess is the more general bookkeeping device.

> **Question for the class.** Should the net quark number be multiplied or divided by three?
>
> **Response.** It must be divided by three. A proton has three quarks and baryon number one, so $3B = N_q - N_{\bar{q}}$. The correction on the board is part of the definition, not a change in the physical argument. [2]

Baryon number resembles electric charge only as a possible conserved quantity. Electric charge also sources a long-range Coulomb field; baryon number does not. A proton creates an electric field because it has electric charge, whereas a neutron has baryon number but no net electric charge. Thus conservation and force generation are logically separate properties.

[1] Lecture timestamp: 00:04:49–00:05:55.
[2] Lecture timestamp: 00:09:23–00:10:17.

Question from the audience. What is the relevant difference between baryon number and electric charge?

Response. Both can serve as bookkeeping charges, but only electric charge is known to source the long-range electromagnetic field. Baryon number does not produce a corresponding Coulomb force. [3]

If B were exactly conserved, an initial excess would remain frozen forever. Pair creation and annihilation could change N_B and $N_{\bar{B}}$ separately, but only by equal amounts, leaving their difference unchanged. A dynamical theory that begins from a balanced state therefore cannot work unless baryon number can change.

The cleanest schematic example is proton decay. Electric charge and energy allow us to imagine

$$p \longrightarrow e^+ + \gamma, \qquad \bar{p} \longrightarrow e^- + \gamma. \tag{8.4}$$

These channels are bookkeeping models for baryon violation, not predictions of the dominant experimental channel. They take $B = +1$ or -1 to zero while preserving electric charge.

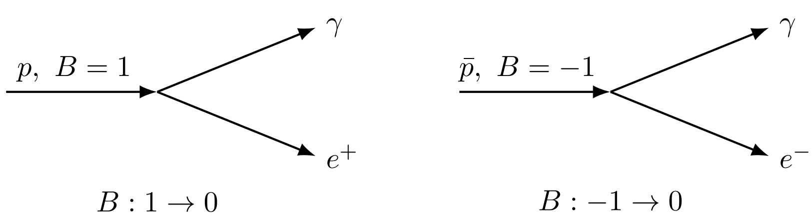

Figure 8.1: Charge-conjugate schematic channels. Either process violates baryon number, but equal rates would create no preferred sign.

Question from the audience. Could the proton instead decay into a positron and a neutrino?

Response. That two-fermion final state does not match the fermion parity of the one-fermion initial state. The schematic $e^+ + \gamma$ final state contains one fermion and is adequate for the symmetry argument. A realistic proton-decay theory may contain other particles and channels, but the present purpose is only to exhibit $\Delta B \neq 0$. [4]

Question from the audience. Would exact baryon conservation correspond to a symmetry?

Response. Yes. A conserved quantity may be associated with a continuous phase symmetry, although merely naming that symmetry does not independently prove that nature respects it. The operational test remains whether baryon-number-changing processes occur, so the argument becomes circular if the symmetry is inferred only from the conservation law. [5]

[3] Lecture timestamp: 00:11:03–00:12:15.
[4] Lecture timestamp: 00:14:39–00:15:07.
[5] Lecture timestamp: 00:15:08–00:15:46.

Question from the audience. Can proton–antiproton annihilation itself be regarded as proton decay?

Response. The terminology is secondary; the conservation law is not. In $p + \bar{p} \to$ photons, the initial baryon number is $1 - 1 = 0$, and the final baryon number is also zero. Annihilation removes two particles without changing net B, whereas the one-particle channel in equation (8.4) changes B. [6]

Present protons are extraordinarily stable: the lecture quotes bounds of order 10^{32}–10^{34} years, vastly longer than the age of the universe. That observation does not establish exact conservation; it tells us only that baryon-number-changing processes are exceedingly slow at present energies.

8.3 Discrete Symmetries and the Need for a Bias

The next question is whether the laws distinguish matter from antimatter. Three discrete transformations organize the answer:

$$C: \text{ particle} \longleftrightarrow \text{antiparticle}, \tag{8.5}$$
$$P: \mathbf{x} \longmapsto -\mathbf{x}, \tag{8.6}$$
$$T: t \longmapsto -t. \tag{8.7}$$

Here P reverses spatial handedness, and T reverses a microscopic movie. The macroscopic second law need not look time-reversal invariant, but the question concerns the underlying microscopic dynamics.

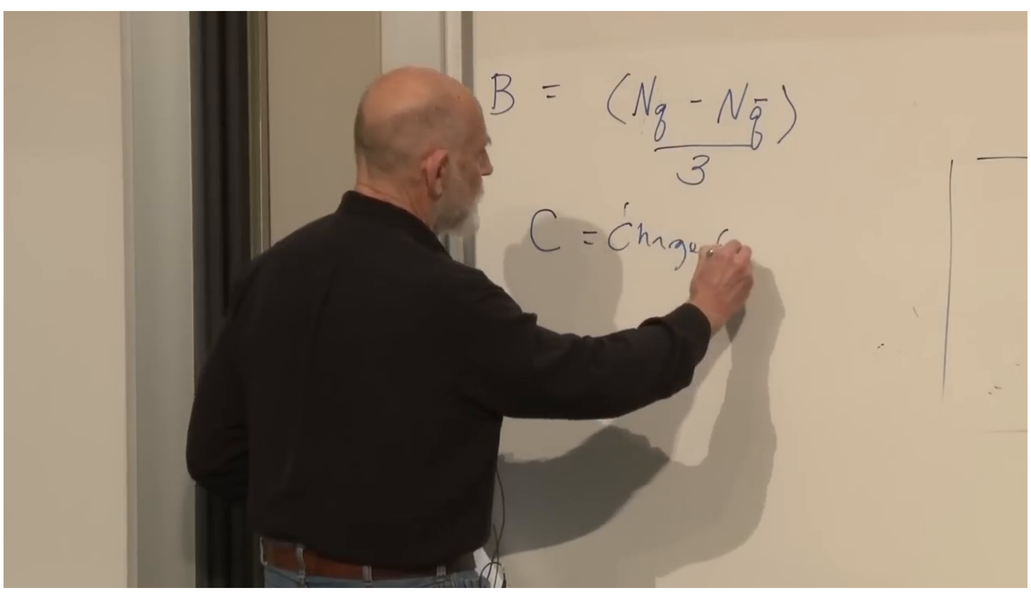

Figure 8.2: The board carries the corrected definition $B = (N_q - N_{\bar{q}})/3$ into the introduction of C, charge conjugation. The unfinished writing after the word charge is not used as independent evidence.

Lecture frame: 00:20:23.280.

[6] Lecture timestamp: 00:15:48–00:16:47.

Relativistic local quantum field theory does not require C, P, or T separately to be symmetries. Under its standard assumptions, it does require the combined transformation CPT. If a process is possible, replacing every particle by its antiparticle, reflecting the spatial configuration, and reversing the motion gives another allowed process.

Time reversal in quantum mechanics is subtler than merely changing the sign of t. For a spinless wavefunction in a time-reversal-invariant potential,

$$i\hbar\frac{\partial\psi}{\partial t} = H\psi, \qquad T:\ \psi(\mathbf{x},t) \longmapsto \psi^*(\mathbf{x},-t). \tag{8.8}$$

Changing $t \to -t$ reverses the time derivative; complex conjugation also sends $i \to -i$, restoring the equation. For particles with spin, the antiunitary time-reversal operator carries additional structure, but the complex conjugation is already visible in this elementary case.

> **Question from the audience.** For a wavefunction, does time reversal require complex conjugation as well as $t \to -t$?
>
> **Response.** Yes. The Schrödinger equation shows why: reversing the time derivative alone changes one side inconsistently, while complex conjugation reverses the sign of i as well. [7]

We can now state the first Sakharov condition. If the universe began with equal populations of particles and antiparticles, pair creation and annihilation would preserve zero net baryon number unless some process could change B:

$$\boxed{\text{baryon-number violation.}} \tag{8.9}$$

Without it, a symmetric initial state can never acquire a nonzero baryon excess.

The present stability of the proton is compatible with this condition. In grand-unified reasoning, baryon violation can be mediated by a very heavy particle of mass M_X. At energies far below M_X, the effective amplitude is suppressed schematically by inverse powers,

$$\begin{aligned} \mathcal{A}_{\Delta B\neq 0} &\propto \frac{E^2}{M_X^2}, \\ \Gamma_p &\sim \frac{m_p^5}{M_X^4}. \end{aligned} \tag{8.10}$$

The decay-rate estimate has the usual dimension-six scaling, up to couplings and channel-dependent factors. A very large M_X can make a cold proton effectively permanent while allowing baryon-changing reactions in a sufficiently energetic early plasma.

> **Question from the audience.** Is the extra collision energy already included in the heavy mass M_X?
>
> **Response.** No. M_X is the mediator's rest-mass scale; E is the energy available in the collision. The low-energy expansion in E/M_X is strongly suppressed when $E \ll M_X$, while the suppression weakens as the thermal collision scale approaches the heavy threshold. The literal denominator written heuristically at the board should be understood in this effective-field-theory sense. [8]

[7] Lecture timestamp: 00:23:58–00:25:04.
[8] Lecture timestamp: 00:32:32–00:34:02.

Suppose the initial state contained equal populations of particles and antiparticles. Baryon violation alone would still be unbiased. Every decay $p \to e^+ + \gamma$ would have a charge-conjugate partner $\bar{p} \to e^- + \gamma$, and the hot plasma would also drive the inverse reactions. Equal microscopic probabilities would produce traffic through baryon number but no average drift.

> **Question from the audience.** What is the baryon-number change in the antiproton diagram?
>
> **Response.** The incoming antiproton has $B = -1$, while the electron and photon have $B = 0$, so the channel changes baryon number from -1 to zero. Its conjugate changes $+1$ to zero. Equal occurrence of the two channels gives no net preference. [9]

Random fluctuations are far too small to account for the observed excess. If N independent events have equal probabilities for two outcomes, then

$$\Delta N \sim \sqrt{N}, \qquad \frac{\Delta N}{N} \sim N^{-1/2}. \tag{8.11}$$

Using the lecture's rough count $N \sim 10^{80}$ for baryons in the observable universe gives

$$\frac{\Delta N}{N} \sim 10^{-40}, \tag{8.12}$$

thirty-two orders of magnitude below 10^{-8}. A coin-toss fluctuation also gives no reason for the same sign to dominate every observed region.

> **Question from the audience.** Could an initially much larger population, perhaps 10^{160} pairs in the comoving region that became our observable universe, fluctuate by 10^{80}?
>
> **Response.** The square-root arithmetic permits that contrived count, but it does not explain the observed spatial coherence. An unrestricted statistical fluctuation would naturally produce matter- and antimatter-dominated patches, not one sign across the visible universe. [10]

The lecture-era observational argument uses two kinds of evidence. Ordinary nuclei, including helium, occur in cosmic rays, whereas antinuclei were not observed in the data being discussed. Antiprotons alone are less decisive because high-energy collisions can create secondary antiprotons. In addition, boundaries between matter and antimatter galaxies would generate conspicuous annihilation radiation. The absence of the required boundary signal strongly disfavors an equal mixture of galaxies and antigalaxies in the observable region. These are observational arguments within the lecture's historical context, not claims about regions beyond our horizon.

> **Question from the audience.** Are helium nuclei themselves measured in cosmic rays?
>
> **Response.** Yes. Charged helium nuclei are measured directly. The detailed cosmic-ray statistics are a separate specialty, but the contrast between common nuclei and the absence of a comparable antinuclear population is the point of the argument. [11]

[9] Lecture timestamp: 00:37:24–00:37:39.
[10] Lecture timestamp: 00:39:38–00:40:18.
[11] Lecture timestamp: 00:41:53–00:42:22.

We therefore need a microscopic bias: violation of C and, in realistic baryogenesis discussions, CP. Neutral B-meson systems provide a laboratory example. A neutral meson and its antiparticle can decay to conjugate final states with unequal partial rates,

$$\Gamma(B^0 \to f) \neq \Gamma(\bar{B}^0 \to \bar{f}). \tag{8.13}$$

The exact channel and numerical asymmetry were not fixed at the board; the robust lesson is the experimentally established existence of particle–antiparticle asymmetry in weak decays.

The CPT theorem nevertheless requires equal masses and equal total decay widths for a particle and its antiparticle in vacuum:

$$\Gamma_{\text{tot}}(p) = \Gamma_{\text{tot}}(\bar{p}). \tag{8.14}$$

This does not force each branching fraction to agree. Differences in one conjugate pair of channels can be compensated by differences in others, such as schematic electron and muon channels, while the sums remain equal.

Question from the audience. If proton and antiproton total decay rates are equal, how can a bias arise?

Response. Equality of total widths does not imply equality channel by channel. Moreover, baryogenesis occurs in a plasma: thermal occupation factors, inverse reactions, and scattering channels enter the kinetic equations. The decisive statement is not that a medium simply evades CPT, but that CP-asymmetric partial processes can generate a source term once detailed balance is lost. [12]

This is the second Sakharov condition:

$$\boxed{C \text{ and } CP \text{ violation.}} \tag{8.15}$$

It supplies directionality to baryon-number-changing reactions. The discovery of CP violation shortly before Sakharov's paper made this condition physically concrete.

The minimal Standard Model requires a careful distinction. It does not permit ordinary perturbative proton decay, but its electroweak anomaly allows nonperturbative violation of $B + L$; at high temperature, sphaleron transitions can make that violation efficient. This is the corrected content of the board's later self-revision. Whether Standard Model sources alone quantitatively reproduce the cosmic asymmetry is a different and much harder question.

8.4 Equilibrium and the Third Condition

One obstruction remains. In complete thermal equilibrium, every reaction is balanced by its inverse. If baryon number can change, its equilibrium chemical potential cannot preserve an arbitrary initial B, and detailed balance erases any generated excess. The lecture expresses the same point through CPT: a static equilibrium bath has no macroscopic arrow of time, so the T factor cannot provide the missing distinction between a process and its reverse.

[12] Lecture timestamp: 00:49:39–00:51:48; 00:53:22–00:57:14.

The actual universe is expanding and cooling. The macroscopic state therefore distinguishes forward from backward time. This matters only when the expansion competes successfully with microscopic relaxation:

$$\Gamma_{\text{reaction}} \lesssim H. \tag{8.16}$$

If $\Gamma_{\text{reaction}} \gg H$, collisions rapidly restore equilibrium despite the expansion. If the reaction rate falls below the Hubble rate, the plasma cannot readjust and the abundance freezes out.

Question from the audience. Why is expansion alone not automatically enough to take the system out of equilibrium?

Response. A very slowly expanding bath remains arbitrarily close to equilibrium because microscopic reactions continually readjust it. Departure becomes effective only when the expansion or cooling time H^{-1} is short compared with the relevant reaction time Γ^{-1}. Then a movie of the macroscopic state reveals which direction is forward. [13]

The three Sakharov conditions can now be assembled:

1. baryon-number violation, so B is able to change;
2. C and CP violation, so the change has a preferred sign;
3. departure from thermal equilibrium, so detailed balance cannot cancel the preference.

They are necessary for the mechanism under discussion and provide a structurally sufficient route to an excess. They do not determine its magnitude without a concrete high-energy model, reaction network, and cosmological history.

Question from the audience. Does the measured value of the asymmetry constrain high-energy theories?

Response. In principle, yes. In practice, the result depends on many unknown masses, couplings, phases, reaction rates, and features of the early cosmology. The single measured number is therefore difficult to turn into a narrow allowed region without committing to a detailed model. [14]

Question from the audience. How are the baryon and electron asymmetries connected?

Response. Electric neutrality links them. In the schematic channel, removing a proton creates a positron; reversing the baryon imbalance shifts the lepton imbalance in the compensating direction. More generally, baryogenesis and leptogenesis are coupled through charge conservation and Standard Model reactions, although the conserved combinations depend on the mechanism. [15]

[13] Lecture timestamp: 01:01:22–01:02:26.
[14] Lecture timestamp: 01:04:20–01:04:52.
[15] Lecture timestamp: 01:04:54–01:05:59.

Question from the audience. Is the observed Standard Model CP violation too small to explain the baryon excess?

Response. That is the standard conclusion of detailed electroweak-baryogenesis analyses, but the lecture emphasizes the model dependence of the calculation and the many unknowns at high energy. The defensible statement here is that known low-energy CP violation exists, while a successful numerical baryogenesis model generally requires additional ingredients. [16]

Question from the audience. Did today's electrons simply come from the destruction of antiprotons?

Response. The schematic channels make that bookkeeping tempting, but the early plasma contained independent, nearly symmetric lepton populations as well. One may formulate related mechanisms as baryogenesis or leptogenesis, with subsequent reactions transferring asymmetry between sectors. The historical word *baryogenesis* should not be mistaken for a unique microscopic genealogy of every electron. [17]

Question from the audience. Why is the discussion usually framed in terms of baryons rather than all leptons?

Response. Baryonic matter is inventoried through atomic spectra and other astronomical observations, whereas the cosmic neutrino sector is much harder to count directly. The naming reflects what was observationally accessible as well as the history of the subject. [18]

8.5 Annihilation Leaves the Excess Behind

At high temperature, pair creation and annihilation run in both directions,

$$\gamma + \gamma \longleftrightarrow p + \bar{p}, \qquad E_{\gamma\gamma} \gtrsim 2m_p c^2 \approx 2\,\mathrm{GeV}. \tag{8.17}$$

As the bath cools, photons energetic enough to create a baryon pair become exponentially rare. Annihilation remains energetically allowed, but inverse production can no longer maintain equilibrium. Almost every proton finds an antiproton, and the small excess in equation (8.2) is what remains. The energetic annihilation photons quickly share their energy with the thermal bath.

Question from the audience. Is the ratio 10^{-8} a present number, and when did the large annihilation occur?

Response. It is the course's present baryon-to-photon order of magnitude. Earlier, both baryons and antibaryons were numerous. Once inverse pair creation became ineffective, their symmetric component annihilated and left the net excess. [19]

[16] Lecture timestamp: 01:06:01–01:07:50.
[17] Lecture timestamp: 01:07:51–01:08:44.
[18] Lecture timestamp: 01:08:45–01:09:12.
[19] Lecture timestamp: 01:09:13–01:11:44.

Question from the audience. When did proton–antiproton annihilation stop?

Response. It stopped when antiprotons had been depleted to a negligible abundance. The lecture locates the event only broadly: at GeV-scale temperatures, well before nucleosynthesis and certainly long before recombination. No precise time is inferred from the discussion. [20]

The annihilation photons have not disappeared. They belong to the thermal radiation history that ultimately feeds the microwave background. Radiation nevertheless contributes little to the energy budget today because

$$\rho_{\text{rad}} \propto a^{-4}, \qquad \rho_m \propto a^{-3}. \tag{8.18}$$

Photon number is large, but each photon's energy redshifts; matter retains its rest-mass energy.

Question from the audience. If annihilation made so many photons, why is radiation dynamically negligible today?

Response. The photons are still present, chiefly as the microwave background, but their energy density loses an extra power of the scale factor. Expansion both dilutes their number and redshifts each photon. [21]

Question from the audience. Does a very heavy mediator mean the Standard Model is incomplete?

Response. A grand-unified proton-decay mediator would indeed be new physics beyond the minimal Standard Model. The later correction about anomalous Standard Model baryon violation does not supply such an ordinary proton-decay particle, nor by itself a complete quantitative baryogenesis model. [22]

Question from the audience. Could a large inequality between the two schematic decay channels convert an initially pure p-$\bar{p}$ population into the observed equal electron and proton counts?

Response. One can invent such bookkeeping, but a hot equilibrium plasma already contains large populations of all accessible particles and antiparticles. Moreover, observed CP asymmetries are not a license to make one channel arbitrarily dominant. A consistent model must evolve the full coupled reaction network rather than only the two arrows on the board. [23]

This is where the baryogenesis discussion honestly ends. The qualitative ingredients are known; the observed number is not derived. A more complete answer awaits both a more complete particle theory and a controlled history of the very early universe.

[20] Lecture timestamp: 01:11:45–01:12:13.
[21] Lecture timestamp: 01:12:14–01:13:18.
[22] Lecture timestamp: 01:13:20–01:13:31.
[23] Lecture timestamp: 01:13:33–01:15:10.

8.6 The Second Number: Primordial Lumpiness

Inflation addresses another question that became quantitative through the cosmic microwave background: why is the universe so homogeneous and isotropic on large scales? The CMB has an extraordinarily precise blackbody spectrum with temperature near 2.7 K, and that temperature is nearly the same in every direction. What had once been a broad cosmological impression became a high-precision fact.

Gravity makes the smoothness more surprising. Diffusion spreads a drop of ink until its concentration becomes uniform. Attractive gravity does the opposite: an overdense region pulls in surrounding matter, increasing its density while emptying neighboring regions. Small perturbations therefore seed galaxies, clusters, and voids by growing with time.

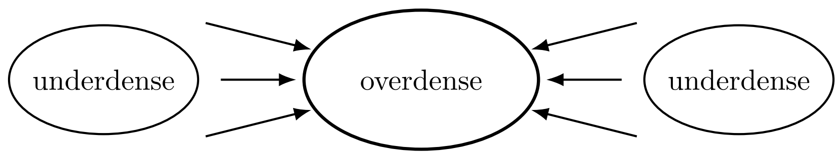

Figure 8.3: Gravitational instability transfers matter from underdense surroundings into an overdensity, amplifying an initial contrast rather than diffusing it away.

Running this growth backward implies a much smoother early universe. Exact homogeneity would also be unacceptable: exact symmetry would persist and no galaxies could form. The useful measure is the density contrast

$$\delta \equiv \frac{\delta\rho}{\rho}, \tag{8.19}$$

where $\delta\rho$ is the local departure from the background density. Structure-growth estimates associated with Jim Peebles and other early cosmologists required a primordial amplitude around

$$\boxed{\frac{\delta\rho}{\rho} \sim 10^{-5}} \tag{8.20}$$

at decoupling, with 10^{-4}–10^{-5} quoted as the historical range. Once again, the number sharpens the question without yet explaining its value.

The broad inflationary proposal, introduced by Alan Guth in 1980, is a period of enormous, approximately exponential expansion. Stretching a small smooth region by many orders of magnitude dilutes pre-existing curvature and pushes irregularities to scales larger than the observable patch. The crinkled-balloon analogy captures the smoothing intuition, though it should not be taken as a literal embedding of space.

Question from the audience. How can the universe be isotropic when nearby matter is visibly lumpy in different directions?

Response. Isotropy is a large-scale statistical statement, not a claim that this room, the Galaxy, or every line of sight looks identical. Earth is mountainous locally yet extremely smooth compared with its radius. Likewise, galaxies and clusters are local structure superposed on a very smooth large-scale background. [24]

[24] Lecture timestamp: 01:25:37–01:26:22.

Question from the audience. Wouldn't a random process occasionally create enormously overdense directions?

Response. The perturbations are not unrestricted independent fluctuations. They form a random field with a particular, nearly homogeneous and isotropic statistical spectrum. Their amplitudes are small, of order 10^{-5}, even when their wavelengths are large. The spectrum and its origin are the subjects to be developed, not assumptions of arbitrarily large coin-toss excursions. [25]

Question from the audience. Was inflation caused by the high temperature associated with early times?

Response. Temperature is not the main variable during inflation. The central phenomenon is the rapid exponential expansion driven by the stress-energy of a field. A thermal bath becomes important again when inflation ends and its energy is converted into particles. [26]

8.7 A Mechanical Prelude: Motion with Drag

To prepare the inflation equations, consider a unit-mass stone falling through a viscous fluid. Call its vertical coordinate ϕ, anticipating the scalar field that will soon replace the stone. If its potential energy is $V(\phi)$, the conservative force is

$$F(\phi) = -\frac{dV}{d\phi}, \tag{8.21}$$

and without drag Newton's equation is $\ddot{\phi} = F$.

For linear viscous drag, the additional force is $-\gamma\dot{\phi}$, with $\gamma > 0$. The equation becomes

$$\ddot{\phi} + \gamma\dot{\phi} = F(\phi) = -V'(\phi). \tag{8.22}$$

Drag vanishes at rest and always opposes the velocity. If ϕ increases upward, a falling stone has $F < 0$ and $\dot{\phi} < 0$; then $-\gamma\dot{\phi} > 0$, so drag points upward as required.

Question for the class. Why do the potential force and the drag force appear with the same minus sign? Should they not point in opposite directions?

Response. Their directions are determined by different quantities. The potential force is $-V'(\phi)$. Drag is $-\gamma\dot{\phi}$. During downward motion, both F and $\dot{\phi}$ are negative, so F points down while $-\gamma\dot{\phi}$ points up. The algebra therefore produces opposite physical directions without changing either formula. [27]

As the stone accelerates, the drag grows until the forces balance. For approximately constant F, terminal motion satisfies

$$0 = F - \gamma\dot{\phi}_{\text{term}}, \qquad \dot{\phi}_{\text{term}} = \frac{F}{\gamma}. \tag{8.23}$$

The sign is correct: a downward F gives a negative terminal velocity. A shallow potential and large γ therefore produce slow motion down the hill.

[25] Lecture timestamp: 01:26:26–01:28:23.
[26] Lecture timestamp: 01:28:25–01:29:04.
[27] Lecture timestamp: 01:34:16–01:35:27.

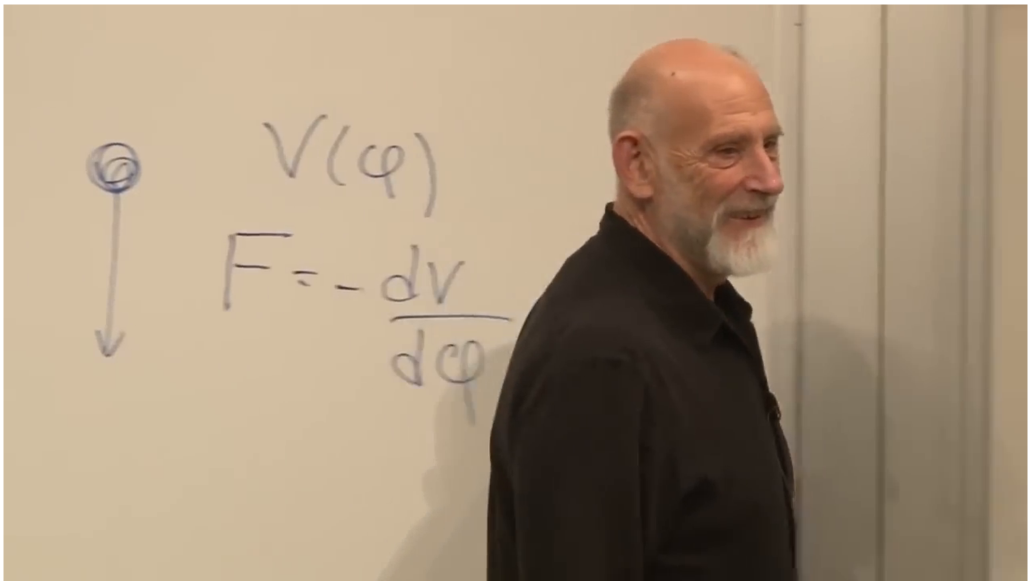

Figure 8.4: The readable board relation $F = -dV/d\phi$ for the falling-stone analogy. The downward arrow fixes the convention that increasing ϕ points upward.

Lecture frame: 01:35:09.513.

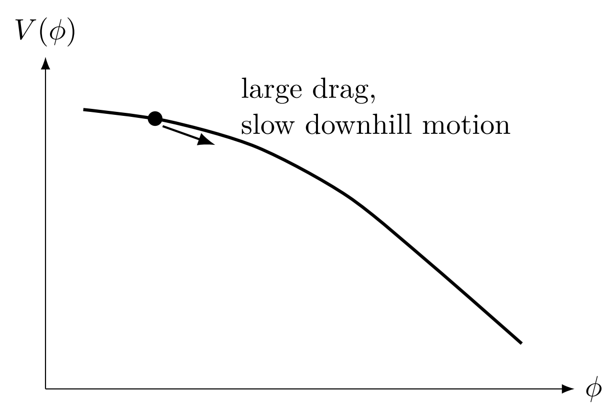

Figure 8.5: A shallow potential with strong drag. The field analogue will replace the stone's coordinate by a spatially homogeneous scalar amplitude.

8.8 The Inflaton and Hubble Friction

Inflation posits an additional scalar field ϕ, conventionally called the inflaton. Historically this was a bold hypothesis introduced to explain cosmic smoothness. We begin with a field approximately uniform in space, $\phi = \phi(t)$. Expansion stretches spatial variations, while the homogeneous problem exposes the essential dynamics.

For a canonical scalar field, the full local energy density is

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2a^2}(\boldsymbol{\nabla}\phi)^2 + V(\phi). \tag{8.24}$$

Under the homogeneous approximation $\boldsymbol{\nabla}\phi = 0$, this reduces to

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi). \tag{8.25}$$

The first term is kinetic energy in field configuration space: it measures temporal change, not translational motion through ordinary space.

Question from the audience. Does the dot on ϕ mean a partial derivative with respect to time?

Response. Yes. For the homogeneous field, $\dot{\phi} = d\phi/dt$. Spatial gradients would contribute a separate positive term, but they vanish under the present assumption. [28]

Follow a unit comoving coordinate box. Its physical volume is $a^3(t)$, so the scalar energy inside it is

$$E_\phi = a^3(t)\left[\frac{1}{2}\dot{\phi}^2 + V(\phi)\right]. \tag{8.26}$$

The explicit factor $a^3(t)$ is the new ingredient: the physical box expands even though its comoving coordinates are fixed.

Question from the audience. Does the scale factor a depend on spatial position?

Response. Not in the homogeneous FRW model. It is the same throughout space at a given cosmic time but may vary with time. If ϕ depended on position, its spatial-gradient energy would have to be restored. [29]

Question from the audience. Is the field value associated with each point in space even though we now write only one $\phi(t)$?

Response. Yes. A field assigns a value to every spacetime point. Homogeneity means those spatial values agree at each time, reducing the dynamics to one common function $\phi(t)$. [30]

[28] Lecture timestamp: 01:41:08–01:41:34.
[29] Lecture timestamp: 01:44:12–01:44:49.
[30] Lecture timestamp: 01:44:50–01:45:06.

Question from the audience. What physical motion does the field's kinetic term describe? Is the field becoming stronger or weaker?

Response. That is determined by the equation of motion, the initial value and velocity, and the sign of $V'(\phi)$. The analogy is a particle whose direction of motion cannot be known from its energy formula alone. [31]

The homogeneous Lagrangian is kinetic minus potential energy,

$$L_\phi = a^3(t)\left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right]. \tag{8.27}$$

Applying the Euler–Lagrange equation gives

$$\frac{d}{dt}\left(\frac{\partial L_\phi}{\partial \dot{\phi}}\right) = \frac{\partial L_\phi}{\partial \phi}, \tag{8.28}$$

$$\frac{d}{dt}\left(a^3\dot{\phi}\right) = -a^3 V'(\phi). \tag{8.29}$$

The product rule supplies two terms,

$$a^3\ddot{\phi} + 3a^2\dot{a}\,\dot{\phi} = -a^3V'(\phi). \tag{8.30}$$

Dividing by a^3 and defining $H = \dot{a}/a$, we obtain

$$\boxed{\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0} \tag{8.31}$$

or, with $F = -V'(\phi)$,

$$\ddot{\phi} + 3H\dot{\phi} = F. \tag{8.32}$$

Every line follows from the expanding volume factor in equation (8.27).

Question from the audience. Is $H = \dot{a}/a$ the same everywhere, and is it really a constant?

Response. In the homogeneous model it is the same at every spatial point on a cosmic-time slice. It need not be constant in time; "Hubble constant" often names its present value, while $H(t)$ is the Hubble rate throughout the evolution. [32]

Equation (8.31) makes the analogy precise. It has the same form as viscous motion, with drag coefficient

$$\gamma_{\text{cosmic}} = 3H. \tag{8.33}$$

Expansion therefore slows the evolution of a homogeneous scalar field. If H is large and $V(\phi)$ is sufficiently flat, the inertial term becomes small compared with the drag term and the field approaches the slow-roll relation

$$3H\dot{\phi} \simeq -V'(\phi). \tag{8.34}$$

Like a ball descending through cold, viscous oil, the field can take an extraordinarily long time to cross a shallow potential even though a force acts on it.

The next step is to insert the scalar energy density into the Friedmann equation and let the expansion rate respond to the slowly evolving field. That coupled system produces inflation. For now, the essential preparation is complete: the field stores kinetic and potential energy, the expanding volume generates Hubble friction, and a slowly rolling field can keep its potential energy nearly constant long enough to drive rapid expansion.

[31] Lecture timestamp: 01:45:07–01:46:08.
[32] Lecture timestamp: 01:50:32–01:50:57.

8.9 What to Carry Forward

The baryon asymmetry and the primordial density contrast play parallel roles. Neither 10^{-8} nor 10^{-5} is explained merely by naming it, but each converts a vague cosmological impression into a quantitative problem. The first leads to the Sakharov conditions; the second leads toward inflation.

The bridge into inflation is equally precise. A homogeneous scalar field in an FRW universe obeys

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0.$$

The middle term is not an inserted phenomenological viscosity. It follows from the growth of the physical volume $a^3(t)$. That single result will organize the next lecture's account of a universe driven by a field rolling slowly down its potential.

Chapter 9

Inflation from a Rolling Scalar Field

Overview. A question left from the previous meeting supplies the organizing equation, $\ddot{\phi}+3H\dot{\phi} = -V'(\phi)$: what moves, what exerts the force, and why does expansion resemble friction? We derive the equation first in a fixed box and then in an expanding one. Coupling it to the Friedmann equation gives the slow-roll mechanism, in which a high, shallow scalar potential drives nearly exponential expansion. Inflation dilutes monopoles and smooths geometry; its end reheats the universe and begins the conventional hot Big Bang. The final problem is then unavoidable: if inflation smooths so efficiently, where did structure come from? A one-dimensional caustic calculation shows how tiny velocity variations create sheets, filaments, and knots, while postponing the quantum origin of those variations to the next lecture.

9.1 What Is Acting on What?

The opening question recalls the equation

$$\ddot{\phi} + 3H\dot{\phi} = F(\phi) = -\frac{dV}{d\phi} \tag{9.1}$$

and asks what kind of force F can act on a field. That question is not peripheral. To answer it we must reconstruct the mechanics of the field, and in doing so we begin the subject of inflation itself.

A scalar field assigns one number $\phi(\mathbf{x}, t)$ to every point of space and time. Electric and magnetic fields carry energy; a scalar field can carry energy as well. The field needed here is hypothetical: no particular inflaton has been experimentally identified, so its energy as a function of field value is not known. We therefore represent it by an unspecified potential-energy density

$$V = V(\phi). \tag{9.2}$$

Different values of ϕ correspond to different amounts of energy stored per unit physical volume.

There is no reason for $\phi = 0$ to have zero energy. The origin of field space is a convention, particularly for an undiscovered field, and shifting the coordinate used to label field values need not place the minimum of V at the origin. The graph of $V(\phi)$ describes relative slopes and heights, not a privileged starting point for the field.

Potential energy is the part that remains when the field is instantaneously static. If the field changes with time, it also has kinetic energy. The quantity $\dot{\phi}$ is not the velocity of an object through ordinary space; it is the rate at which the value of the field changes, a velocity in field space. An elastic string gives a useful analogy. Its displacement is a field on a one-dimensional world, and a vibrating element of string carries kinetic energy proportional to the square of its time derivative.

The coefficient of that square can be absorbed into the definition of the field. If one begins with

$$\frac{1}{2}n^2\dot{\phi}^2, \tag{9.3}$$

then the change of variable

$$\chi = n\phi \tag{9.4}$$

turns the term into $\dot{\chi}^2/2$. We may therefore choose a canonically normalized field and write, in the homogeneous approximation,

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi). \tag{9.5}$$

The choice is a normalization of the coordinate in field space, not a claim that a literal particle mass has been set equal to one.

9.2 Mechanics in a Fixed Box

Follow a small box in which the field is nearly constant across space. Its energy density has the same mathematical form as the energy of a unit-mass particle: kinetic plus potential. The associated Lagrangian density has the opposite sign for the potential,

$$\mathcal{L}_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi). \tag{9.6}$$

The analogy is mathematical. The generalized coordinate is the field value, not the spatial position of a particle.

The Euler–Lagrange equation is

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}_\phi}{\partial \dot{\phi}}\right) = \frac{\partial \mathcal{L}_\phi}{\partial \phi}. \tag{9.7}$$

Here the velocity variable is $\dot{\phi}$, so

$$\frac{\partial \mathcal{L}_\phi}{\partial \dot{\phi}} = \dot{\phi}, \qquad \frac{\partial \mathcal{L}_\phi}{\partial \phi} = -V'(\phi), \tag{9.8}$$

we obtain

$$\ddot{\phi} = -V'(\phi) \equiv F(\phi). \tag{9.9}$$

The second derivative is the acceleration of the field coordinate, and the negative slope of the potential is its force.

> **Question from the audience.** What is the force, and what does it act on?
>
> **Response.** The evolving object is the scalar value ϕ. The quantity $-V'(\phi)$ is a force in field space: it accelerates the field value toward lower potential energy just as $-dV/dx$ accelerates a particle coordinate. It is not a new vector force pushing a material body through ordinary space. [1]

[1] Lecture timestamp: 00:00:27–00:00:49; 00:10:41–00:11:29.

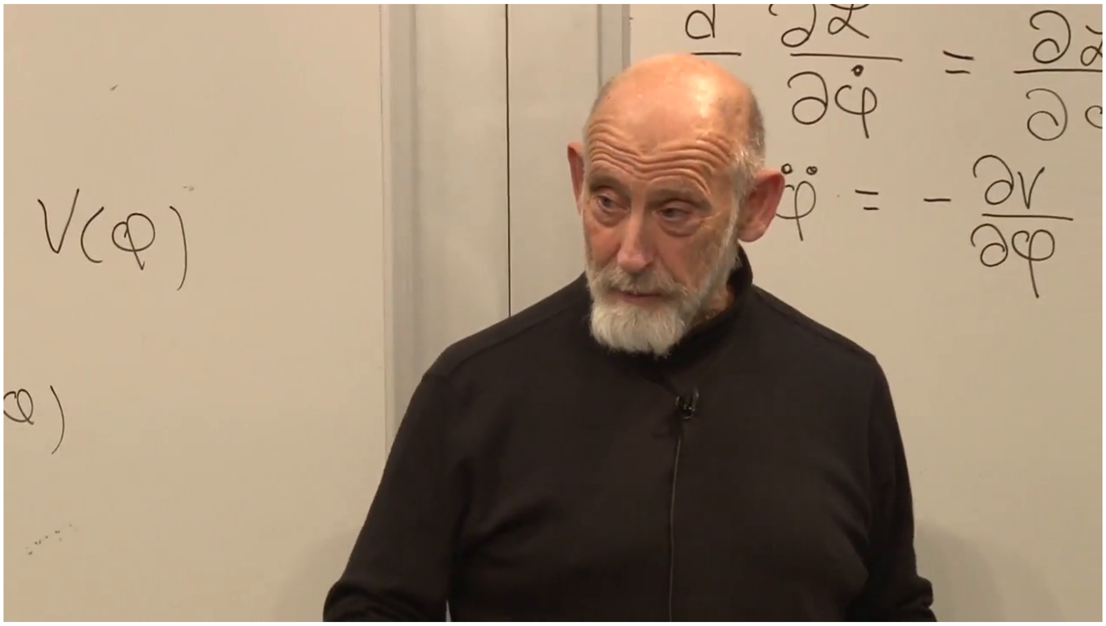

Figure 9.1: The readable part of the board carries $V(\phi)$ into the fixed-box equation $\ddot{\phi} = -\partial V/\partial\phi$. The upper Euler–Lagrange expression is partly cropped and obscured, so equation (9.7) is reconstructed from the derivation rather than quoted as a complete frame transcription.

Lecture frame: 00:10:53.836.

The full field can depend on space. For a canonical scalar in a spatially flat expanding geometry, its local energy density contains

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2a^2}|\nabla\phi|^2 + V(\phi). \tag{9.10}$$

We drop the gradient term because expansion stretches variations to long wavelengths, making the field nearly uniform inside the small box. This is an approximation justified by the cosmological setting, not a denial that scalar waves or spatial ripples can exist.

9.3 The Expanding Box

The first derivation answers the opening question, but it is not yet the cosmological derivation. Equations (9.5) and (9.6) are densities. A unit comoving box has physical volume proportional to $a^3(t)$, so its Lagrangian is

$$L_\phi = a^3(t)\left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right]. \tag{9.11}$$

The factor a^3 would be irrelevant in a static box. In cosmology it depends on time, and that dependence supplies the missing term.

Applying the Euler–Lagrange equation to L_ϕ gives

$$\frac{d}{dt}\left(a^3\dot{\phi}\right) = -a^3V'(\phi), \tag{9.12}$$

$$a^3\ddot{\phi} + 3a^2\dot{a}\,\dot{\phi} = -a^3V'(\phi), \tag{9.13}$$

$$\ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} = -V'(\phi). \tag{9.14}$$

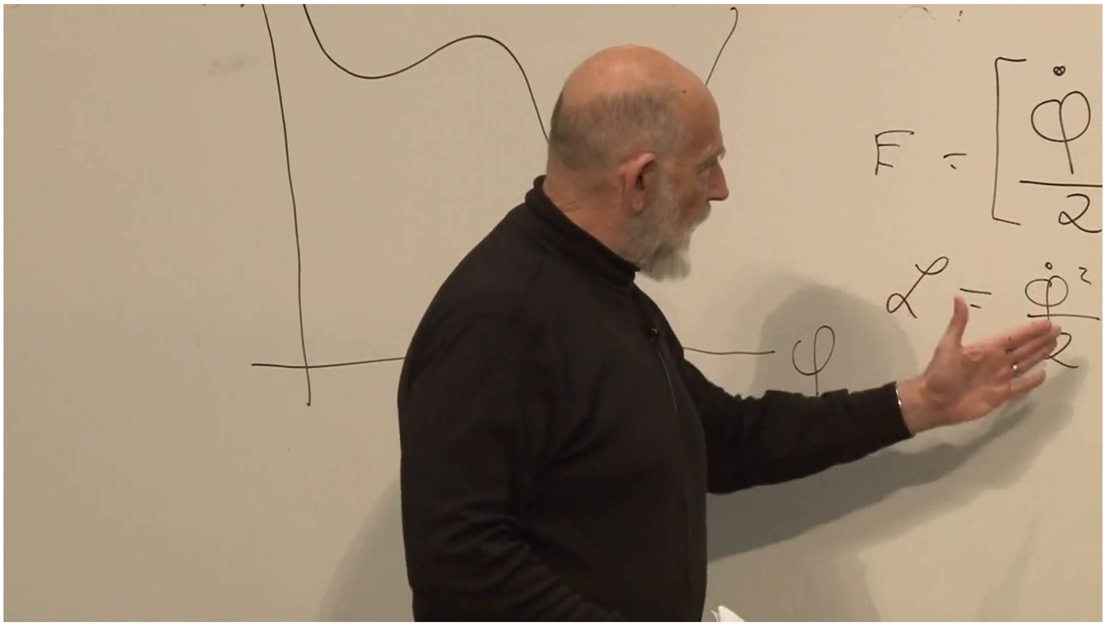

Figure 9.2: A hand-drawn potential curve is visible on the left while fragments of the energy and Lagrangian expressions appear on the right. The curve and the sign change are secure; the complete algebra in equation (9.11) comes from the spoken derivation because the lecturer obscures part of the board.

Lecture frame: 00:13:26.528.

With $H = \dot{a}/a$, the result is

$$\boxed{\ddot{\phi} + 3H\dot{\phi} = -V'(\phi)}. \tag{9.15}$$

The equation remembered at the beginning has now been derived rather than merely recalled.

> **Question from the audience.** Can the force F be visualized more concretely?
>
> **Response.** Draw $V(\phi)$ as a landscape whose horizontal coordinate is the value of the field. The slope determines the force, $F = -V'(\phi)$. A steep downward slope gives a large drive in field space; a shallow slope gives a small one. The picture is an aid to the one-dimensional field coordinate, not a literal hill located in physical space. [2]

9.4 Hubble Friction and Terminal Motion

Move the middle term in equation (9.15) to the right:

$$\ddot{\phi} = -V'(\phi) - 3H\dot{\phi}. \tag{9.16}$$

For an expanding universe $H > 0$, the second term always opposes the field velocity. It therefore behaves like viscosity. A rock released in thick honey initially accelerates, but drag grows with speed until the rock reaches terminal velocity. The mnemonic *H for honey* captures the analogy,

[2]Lecture timestamp: 00:18:26–00:19:18.

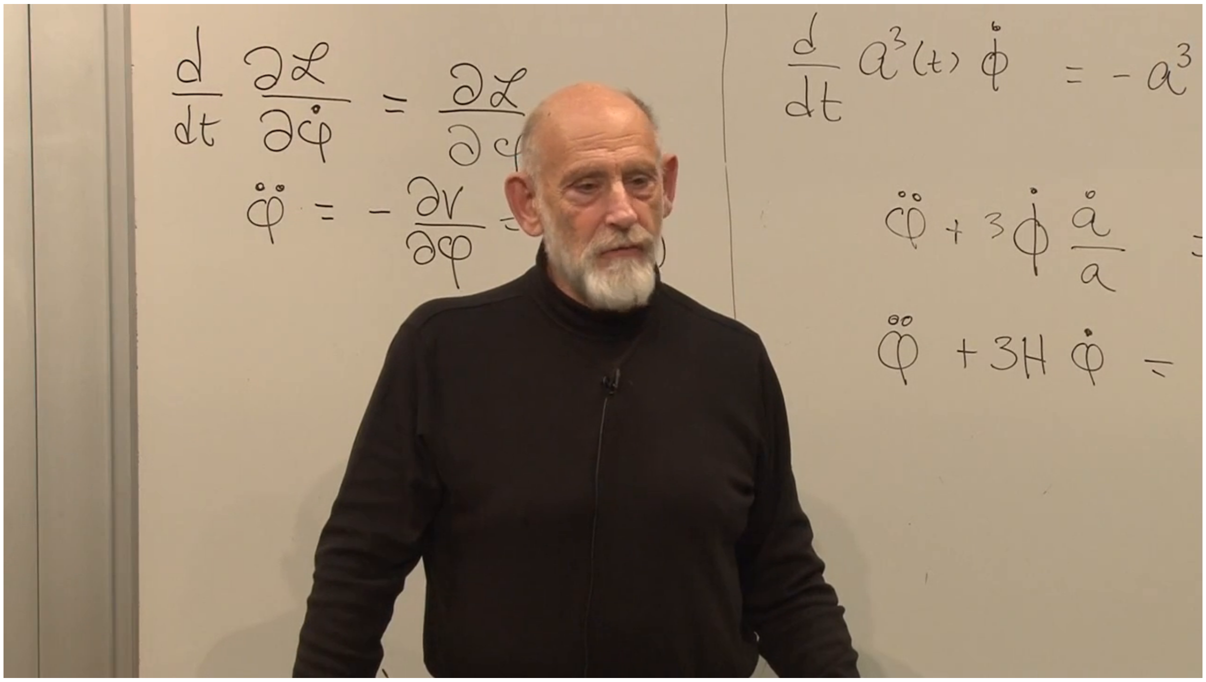

Figure 9.3: Fixed-box dynamics on the left and the expanding-box correction on the right. The visible board supports the sequence from $d(a^3\dot\phi)/dt$ to the $3\dot a/a$ and $3H$ forms; cropped right-hand sides are supplied by equation (9.12).

Lecture frame: 00:18:35.380.

while the equation supplies its actual content. If H changes with cosmic time, the effective viscosity changes too, much as ordinary viscosity changes with temperature.

When friction dominates inertia, $|\ddot\phi| \ll 3H|\dot\phi|$, and the field rapidly approaches

$$3H\dot\phi \simeq -V'(\phi), \qquad \dot\phi \simeq -\frac{V'(\phi)}{3H}. \tag{9.17}$$

A shallow potential supplies a small force, while a large H supplies strong drag. Both effects make $\dot\phi$ small and usually make the kinetic density $\dot\phi^2/2$ small compared with V.

Question from the audience. Does this derivation rely on the principle of least action?

Response. Yes. Equation (9.15) follows by extremizing the action built from equation (9.11). One may also take the field equation as the starting dynamical law, but the variational derivation explains exactly why the expanding volume contributes $3H\dot\phi$. [3]

Question from the audience. If the new term acts like friction, where does the lost energy go?

Response. A real rock in honey transfers mechanical energy into heat. The cosmological statement is subtler. The comoving-box Lagrangian depends explicitly on time through $a(t)$,

[3] Lecture timestamp: 00:25:40–00:26:15.

so the ordinary mechanical energy of that box is not conserved by time-translation symmetry. Local covariant stress-energy conservation still holds:

$$\dot{\rho}_\phi + 3H(\rho_\phi + p_\phi) = 0, \qquad p_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi).$$

For a homogeneous scalar this becomes $\dot{\rho}_\phi = -3H\dot{\phi}^2$: the field does work on the expanding geometry. What fails is a naive globally conserved energy for the evolving universe, not local conservation of stress-energy. [4]

9.5 The Field Drives the Expansion

So far expansion has acted on the scalar. That is only half of the feedback loop. The scalar's energy density also determines the expansion through the flat-space Friedmann equation,

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho. \tag{9.18}$$

For a nearly homogeneous scalar,

$$H^2 = \frac{8\pi G}{3}\left[\frac{1}{2}\dot{\phi}^2 + V(\phi)\right]. \tag{9.19}$$

Expansion damps the field, and the damped field can in turn maintain the energy density that drives expansion.

Question from the audience. Is this hypothetical scalar what is called the inflaton?

Response. Yes. The name describes the role assigned to it: a scalar field whose nearly constant potential energy drives inflation. Naming the field does not identify its particle physics. The detailed inflaton, its couplings, and even whether one elementary scalar alone played this role remain model-dependent. [5]

Question from the audience. Is there experimental evidence for the inflaton?

Response. There is no direct laboratory detection of a particular inflaton field. The evidence is indirect: inflationary dynamics provides a successful framework for the observed flatness and primordial perturbations, but many potentials can fit those broad facts. This indirect support should be separated from knowledge of the microscopic implementation. [6]

Question from the audience. Do all fields carry energy, and would a scalar excitation be a boson?

Response. Physical fields contribute energy and momentum. Quantizing a scalar field produces spin-zero bosonic quanta. The classical homogeneous background used in inflation is not best pictured as one ordinary particle, although after inflation its oscillations can be described in terms of inflaton quanta. [7]

[4] Lecture timestamp: 00:26:16–00:27:57.
[5] Lecture timestamp: 00:30:26–00:31:28.
[6] Lecture timestamp: 00:31:29–00:33:02.
[7] Lecture timestamp: 00:33:03–00:35:20.

Question from the audience. Could the Higgs field be the inflaton?

Response. A scalar is required, so the analogy is natural, but the observed Higgs with its ordinary Standard Model potential and couplings is not automatically the field described by this simple plateau. Models of Higgs inflation add nonminimal gravitational structure. Nothing in the present argument identifies the inflaton with the Higgs. [8]

Now imagine $V(\phi)$ high but gently sloping over a long interval. Equation (9.17) keeps the field moving slowly, and hence

$$\frac{1}{2}\dot{\phi}^2 \ll V(\phi). \tag{9.20}$$

The Friedmann equation reduces to

$$H^2 \simeq \frac{8\pi G}{3}V(\phi), \qquad H \simeq \sqrt{\frac{8\pi G}{3}V(\phi)}. \tag{9.21}$$

The derivative $V'(\phi)$ controls the slow motion of the field; the height $V(\phi)$ controls the rapid expansion of space.

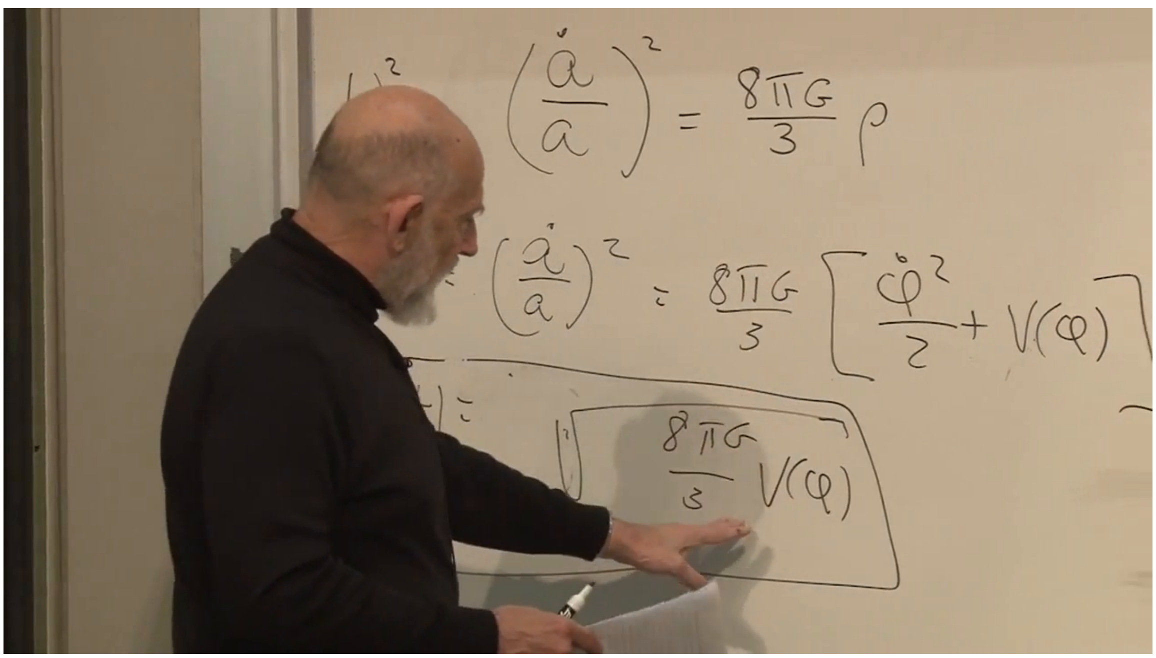

Figure 9.4: The Friedmann equation, the scalar energy density, and the boxed potential-dominated expression. The square-root term in $V(\phi)$ is clear; the leading $H \simeq$ in equation (9.21) is the standard completion supported by the spoken argument.

Lecture frame: 00:41:04.178.

If V changes only slightly while the field crosses the plateau, then H is nearly constant. Integrating $\dot{a}/a \simeq H$ gives

$$a(t) \simeq a_i e^{H(t-t_i)}. \tag{9.22}$$

This is inflation. A rough early-universe doubling time near 10^{-32} s conveys its violence; the number is illustrative and depends on the inflationary energy scale.

[8] Lecture timestamp: 00:35:21–00:37:11.

Question from the audience. How can a slowly moving field produce a rapidly expanding universe?

Response. It is ϕ, not a, that moves slowly. Hubble friction prevents the field from racing down the potential, so $V(\phi)$ remains almost constant. A large, almost constant V gives a large, almost constant H, and the scale factor then grows exponentially. The weak slope controls field motion; the high plateau controls cosmic expansion. [9]

9.6 Why Introduce Inflation?

Inflation was proposed by Alan Guth around 1980 to cure concrete cosmological problems. Subsequent work, especially by Andrei Linde and others, developed versions in which the inflating phase ends gracefully. The broad framework fits important observations, but no unique potential has been established; one should not turn the success of the mechanism into a claim that every detail is experimentally settled.

9.6.1 The Monopole Problem

Grand-unified theories commonly permit magnetic monopoles. In a rough estimate they are fantastically heavy, perhaps many orders of magnitude above the proton mass; the precise mass is theory-dependent and need not be exactly the Planck mass. A sufficiently hot early universe would thermally produce monopole–antimonopole pairs. Magnetic charge would then protect a relic population after cooling.

A substantial relic density is not observed. In particular, freely moving magnetic charges would efficiently drain galactic magnetic fields, just as mobile electric charges screen or discharge electric fields. The continued existence of galactic magnetism therefore places a severe limit on cosmic monopoles. Inflation solves the problem by increasing the volume by an enormous factor: any monopoles present before inflation are diluted until essentially none remain in our observable region.

9.6.2 Smoothness and Flatness

The cosmic microwave background is remarkably uniform. Without inflation, widely separated regions of the last-scattering surface have too little ordinary causal contact in the simplest hot-Big-Bang extrapolation to explain why their temperatures agree so closely. Inflation begins with a much smaller causally connected patch and stretches it to encompass the observable universe. It also drives spatial curvature toward zero, in much the same way that a small patch on an enormously inflated sphere looks flat.

The amount of expansion is measured by the number of e-folds,

$$N \equiv \ln \frac{a_{\text{end}}}{a_{\text{start}}} = \int_{t_{\text{start}}}^{t_{\text{end}}} H\, dt. \tag{9.23}$$

We shall use $N \gtrsim 60$ as the useful order-of-magnitude lower bound. The exact requirement depends on the inflationary scale and reheating history; estimates in the broad neighborhood of 40–80 occur in different models. More initial roughness or unwanted relic density can demand more inflation.

[9] Lecture timestamp: 00:40:06–00:41:39.

Question from the audience. If the initial universe were completely chaotic, would sixty e-folds still be enough?

Response. Not necessarily. Sixty is a conventional lower-bound estimate for solving the standard horizon and flatness problems under specified assumptions. A rougher starting state can require a longer smooth plateau. The claim is not that one universal integer erases every conceivable initial condition. [10]

Question from the audience. Is there an observational upper bound on the duration of inflation?

Response. No comparable direct upper bound follows from the present discussion. Once inflation lasts long enough, additional e-folds push curvature and pre-existing relics farther beyond observation. Producing an extremely long phase may require a correspondingly broad and finely arranged potential, but that is a model-building cost rather than a measured upper limit. [11]

If inflation lasted only barely long enough, a small residual curvature might remain within reach of future observations. Continued improvement in curvature measurements therefore probes whether the inflationary stretch was close to the minimum or vastly longer, although a null result cannot determine an arbitrarily large N.

Question from the audience. What happens to the temperature while space inflates?

Response. Ordinary particles and radiation are diluted and redshifted, so the universe becomes extremely cold and empty in the thermal sense. It is not empty of energy: the nearly constant scalar potential remains and drives the expansion. The hot thermal universe must be created again when inflation ends. [12]

9.7 Leaving the Plateau

An eternally flat plateau would leave the universe inflating forever and empty of ordinary matter. The potential must therefore change character. A useful schematic has a long, shallow region followed by a steeper descent and a minimum:

As ϕ approaches the descent, V and hence H decrease, while $|V'(\phi)|$ grows. The honey becomes thinner just as the downhill force grows stronger. The slow-roll approximation fails, the field accelerates, and inflation ends. Why nature should provide such a potential, and why the field should initially occupy its plateau, are not explained by drawing the curve. They remain questions for the microscopic theory and its initial conditions.

[10] Lecture timestamp: 00:55:23–00:56:48.
[11] Lecture timestamp: 00:56:49–00:58:45.
[12] Lecture timestamp: 00:59:49–01:01:10.

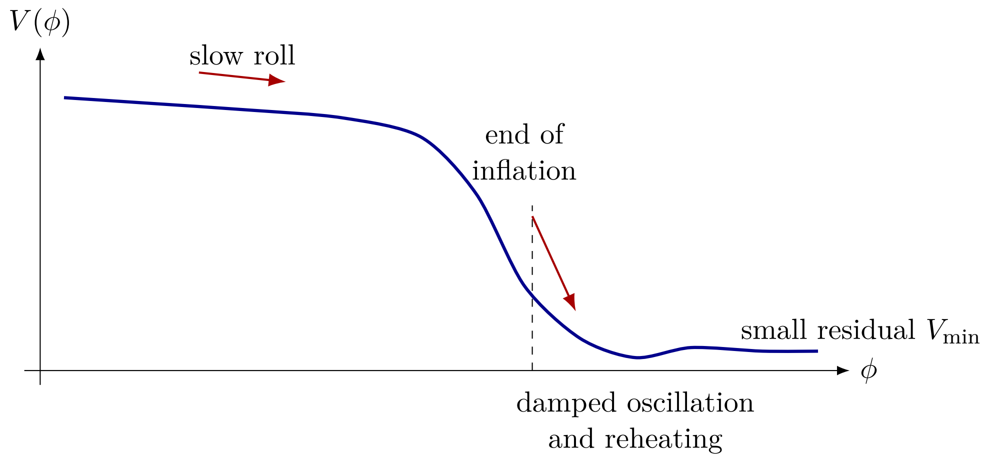

Figure 9.5: A schematic inflationary potential: a high shallow plateau, a steep exit, and a damped approach to a minimum. Its detailed shape is unknown; the curve records only the qualitative stages required by the argument.

Question from the audience. Did inflation begin near photon decoupling?

Response. No. Inflation belongs to an extraordinarily early epoch. Decoupling occurred hundreds of thousands of years into the later hot Big Bang, after reheating, nucleosynthesis, and a long period of expansion. The two events are separated by an enormous span of cosmic history. [13]

9.7.1 Reheating

When the field descends, its potential energy becomes coherent kinetic energy and oscillations about the minimum. Couplings to other fields can transfer that energy into particles, which scatter and approach a thermal distribution. This process is called reheating. It supplies photons, electrons, positrons, quarks, and other species and thereby establishes the initial hot plasma of the conventional Big-Bang account. The hot Big Bang is therefore not the inflationary state itself; it is the aftermath of inflation.

Question from the audience. Is reheating like the Higgs field producing photons and matter?

Response. The useful analogy is the decay of an excited field into quanta of fields to which it couples. Inflaton oscillations can create inflaton quanta, and those excitations can decay into lighter species. Which particles appear directly, which emerge through later reactions, and whether the Higgs participates are all model-dependent. [14]

Question from the audience. Does inflation mean that space began as a single point and expanded into something outside it?

[13] Lecture timestamp: 01:03:33–01:04:39.
[14] Lecture timestamp: 01:06:02–01:08:18.

Response. No external arena is required. One may picture a tiny closed three-dimensional space as an aid, but the universe is not an ordinary ball expanding into a surrounding room. Inflation magnifies physical distances within space. It also does not by itself establish an absolute beginning; it describes the evolution of a patch once the inflationary conditions hold. [15]

9.8 At the Minimum: Vacuum Energy and Open Questions

If the minimum retains a tiny positive value $V_{\min}$, it behaves like a cosmological constant. The same broad scalar-potential picture can then place primordial inflation and the present accelerated era on one diagram. This is an economical classroom possibility, not evidence that today's dark energy is literally the residual energy of the primordial inflaton.

Question from the audience. If monopoles are not found, does that force the minimum number of e-folds upward?

Response. The prospective observation discussed here is residual spatial curvature, not a near-term monopole detection. Existing magnetic-field arguments already make an abundant monopole population implausible. A tighter null constraint on curvature pushes the lower bound on inflation upward because more expansion makes space flatter. These are related consequences of inflation, but they are not the same measurement. [16]

Question from the audience. Is the small energy at the bottom today's vacuum energy?

Response. In this schematic picture, yes: $V_{\min}$ is the present vacuum-state energy and acts as the cosmological constant. The hierarchy is staggering, roughly 120 orders of magnitude below a natural Planck-scale estimate and enormously below a typical inflationary density. The appealing unified picture does not explain that number. [17]

Question from the audience. If inflation first emptied the universe, what exactly was smoothed, and when were monopoles removed?

Response. Inflation smoothed and flattened spatial geometry while diluting any particles or monopoles that already existed. Trouble returns if the subsequent release of energy heats the universe above the monopole-production threshold. The chronology and energy scales must therefore be arranged together rather than treating smoothing and reheating as independent episodes. [18]

Reheating must thread a needle. It has to be hot enough to make the particles needed for the subsequent universe, yet below any temperature that would regenerate the monopoles inflation just diluted. This requirement is one reason the end of inflation cannot be specified independently of particle physics.

[15] Lecture timestamp: 01:08:19–01:11:49.
[16] Lecture timestamp: 01:12:37–01:14:06.
[17] Lecture timestamp: 01:14:07–01:14:56.
[18] Lecture timestamp: 01:14:58–01:16:38.

Question from the audience. Are magnetic monopoles particles, and must they have existed before inflation?

Response. They are extremely heavy particle-like excitations carrying magnetic charge. They may have existed and then been diluted, or the pre-inflationary state may never have become hot enough to produce them. What must be avoided is a post-inflationary thermal bath hot enough to create them abundantly again. [19]

The field can overshoot the minimum and ring back and forth, but Hubble friction and particle production damp those oscillations. In the simple picture it settles near the bottom rather than climbing the far side indefinitely.

Question from the audience. Why does the potential stop being flat and suddenly descend?

Response. That is unknown. The curve parametrizes the dynamics required by observation; it does not explain its own shape. Anthropic selection has been proposed as one possible way to discuss why habitable regions have suitable vacuum energy and cosmic history, but it is not known to be the correct explanation and supplies no derived inflaton potential here. [20]

Question from the audience. Does saying the cause is unknown mean that the change had no cause?

Response. No such conclusion follows. It means only that the responsible microscopic theory is unknown. Ignorance of a cause is not evidence for a causeless transition. [21]

Question from the audience. Is time the horizontal axis of the potential graph, and where is $\phi = 0$?

Response. The horizontal coordinate is ϕ, and its zero may be placed wherever convenient, perhaps at today's minimum. Time parameterizes the trajectory $\phi(t)$ along the curve. During nearly constant-H slow roll, $N \simeq H\Delta t$, so sixty e-folds means roughly sixty Hubble times, not sixty units of field value. The descent may take longer than one early Hubble time while still being extremely rapid on ordinary scales. [22]

Question from the audience. Could the field climb the far side of the potential again?

Response. It may overshoot and ring a few times, but the $3H\dot{\phi}$ term removes its motion. A rock resting at the bottom of a viscous sea does not spontaneously rise to the surface; similarly, the damped field does not classically return to the plateau unless some additional disturbance supplies energy. [23]

[19] Lecture timestamp: 01:16:41–01:17:09.
[20] Lecture timestamp: 01:17:33–01:19:22.
[21] Lecture timestamp: 01:19:23–01:19:59.
[22] Lecture timestamp: 01:20:00–01:21:45.
[23] Lecture timestamp: 01:21:46–01:22:58.

Question from the audience. How hot did reheating make the universe?

Response. The detailed temperature is unknown. It had to be high enough to produce quarks and abundant particle–antiparticle pairs, so 10^{11}–10^{12} K is only a rough low-end orientation in the classroom estimate, not a measurement. It could have been much hotter, while still remaining below the threshold that would repopulate dangerous monopoles. [24]

Question from the audience. Are there bosons associated with the inflaton field, and what do they decay into?

Response. Quantizing the scalar gives inflaton bosons. During the descent and oscillatory stage, their couplings can produce photons, electrons, quarks, other ordinary species, and possibly dark matter. The list and branching fractions belong to the unknown microscopic model, but their decay is the mechanism by which field energy becomes a thermal bath. [25]

9.9 A Universe That Hides Its Past

At the present epoch the field is pictured as sitting near the minimum. If its small positive energy persists, the late universe approaches de Sitter-like expansion. The rate is slow compared with primordial inflation, but over immense times it carries distant, unbound galaxies beyond causal reach. The microwave background cools below practical detectability, and future observers in the local bound remnant may see an apparently empty cosmos with little direct evidence of the larger population of galaxies.

Question from the audience. Did the universe begin as a point source inside some larger space, and could there have been several such sources?

Response. Calling it a point source tempts us to imagine an embedding space in which the source sits, and that is the wrong picture. A tiny closed three-space is only a heuristic initial geometry; it is not a ball inside an external room. Whether there were several causally separate inflating regions is possible in broader theories but is not answered by this model. [26]

Question from the audience. Is that far-future isolation analogous to regions we cannot see beyond our present horizon?

Response. Yes. In both cases a causal boundary limits the available signals. The far-future version is more extreme for local observers because almost every galaxy not gravitationally bound to them has passed beyond that boundary. [27]

[24] Lecture timestamp: 01:22:58–01:25:07.
[25] Lecture timestamp: 01:25:08–01:25:49.
[26] Lecture timestamp: 01:30:10–01:30:57.
[27] Lecture timestamp: 01:31:00–01:31:18.

Question from the audience. Would old radio transmissions from distant civilizations remain detectable?

Response. Signals already en route continue to propagate, but expansion stretches their wavelengths and lowers their energy and arrival rate. Beyond the event horizon, transmissions emitted sufficiently late never reach us. Even the oldest broadcasts would eventually be redshifted beyond practical recovery. [28]

Question from the audience. How heavy is an inflaton quantum?

Response. For a canonically normalized field, the small-oscillation mass is set by the curvature of the potential near the minimum,

$$m_\phi^2 = V''(\phi_{\min}).$$

That curvature is unknown. The broad empirical statement is only that no sufficiently light, ordinarily coupled inflaton has appeared in existing accelerator experiments. [29]

Question from the audience. Is ϕ also a function of position, and are its original ripples quantum fluctuations?

Response. In general $\phi = \phi(\mathbf{x}, t)$. Classical spatial ripples can be stretched to such long wavelengths that the field looks homogeneous within a local region, which justified dropping $\nabla\phi$ earlier. Quantum fluctuations are an additional effect, not the classical ripples used in that first smoothing argument. Their role is deliberately postponed because they will supply the seeds of structure. [30]

Question from the audience. Should we see galaxies visibly slipping across the horizon today?

Response. Not as nearby objects crossing a luminous surface. Looking to the greatest observable distances also means looking far into the past, back toward the opaque surface of last scattering before galaxies formed. A distant galaxy's later signals become progressively redder and dimmer; the horizon is a causal boundary, not a material shell. [31]

Question from the audience. Could sufficiently patient future observers recover evidence for cosmic expansion?

Response. Direct astronomy would be extraordinarily difficult once only one bound galactic remnant remained. In principle, probes and long-baseline signal exchanges could measure geometry, recession, and curvature over billions of years. A probe must turn around before crossing the event horizon, however, or no report can return. [32]

[28] Lecture timestamp: 01:31:18–01:31:42.
[29] Lecture timestamp: 01:31:58–01:32:25.
[30] Lecture timestamp: 01:32:32–01:33:51.
[31] Lecture timestamp: 01:33:52–01:34:50.
[32] Lecture timestamp: 01:34:51–01:37:11.

Question from the audience. Does dark energy destroy information?

Response. No such destruction follows from this classical argument. Dark energy can make information operationally inaccessible to a particular observer by carrying its sources beyond a horizon. Fundamental information loss is a separate quantum-gravity question. [33]

Question from the audience. Will Andromeda and other nearby galaxies disappear too?

Response. Objects that are gravitationally bound do not participate in the asymptotic Hubble flow. The Milky Way and Andromeda should form one local remnant; more distant unbound structures recede. Exactly which neighboring systems remain bound requires detailed local dynamics, so the classroom boundary should not be read as a precise census. [34]

Question from the audience. What is the field doing on the potential today, and can $\phi(t)$ be written in a simple formula?

Response. In this picture it is essentially sitting at the minimum. Its complete history follows the coupled nonlinear equations for $\phi(t)$ and $a(t)$: slow drift across the plateau, a rapid descent, damped oscillations, and rest near the bottom. No single linear formula captures all those regimes. [35]

For nearly constant H, the characteristic Hubble length is

$$R_H = \frac{c}{H}. \tag{9.24}$$

It is also the event-horizon scale in exact de Sitter space. In a general time-dependent cosmology, the Hubble radius and event horizon are distinct. Rough classroom distances such as fifteen, fifty, or sixty billion light-years also mix different cosmological distance conventions; the reliable relation here is the scale c/H, not any one quoted number.

Question from the audience. Is the horizon an edge of space, and does the region inside contain less space as galaxies leave?

Response. No. The horizon limits causal communication, not spatial existence. There can be abundant space with progressively less observable matter in it. In exact de Sitter expansion the horizon scale remains fixed for a comoving observer even while unbound material redshifts away. [36]

Question from the audience. What will the surviving galaxy itself look like after such an enormous time?

Response. That requires stellar and galactic evolution beyond the present argument. A bound remnant, stars, and perhaps planets may persist for very long periods, but its exact density, stellar population, and habitability are uncertain. The cosmological claim concerns isolation, not a detailed portrait of the remnant. [37]

[33] Lecture timestamp: 01:37:12–01:37:20.
[34] Lecture timestamp: 01:37:21–01:38:03.
[35] Lecture timestamp: 01:38:03–01:38:54.
[36] Lecture timestamp: 01:39:49–01:41:30.
[37] Lecture timestamp: 01:41:37–01:42:33.

The potential history is therefore an empirically motivated outline rather than a microscopic explanation. It is constrained by the observable universe, yet the deeper reasons for its initial plateau, exit, reheating, and tiny final energy remain unknown.

9.10 If Inflation Smooths, Why Is There Structure?

The universe is not a featureless gas. Galaxy surveys reveal filaments, sheets, knots, and large voids. Numerical simulations show a similar web, and sunlight on the bottom of a swimming pool produces a strikingly related network of bright curves. The comparison is not between individual galaxies and individual ripples. It is between the geometry of large-scale concentrations.

To isolate the mechanism, begin with a force-free model. Let many particles start with nearly uniform density but slightly different velocities from place to place. A simulation can be initialized with such a velocity field and allowed to stream. The point is not that gravity is absent from real structure formation; it is that kinematics alone already explains why folds and intersections naturally make web-like concentrations.

Question from the audience. Are any forces acting in this first simulation?

Response. No. In the stripped-down example each particle keeps its assigned velocity. Gravity is deliberately omitted so that the caustic mechanism can be seen by itself. Real cosmic structure subsequently requires gravity, especially on smaller scales and after streams cross. [38]

The swimming-pool picture supplies the optical version. Small waves on the water surface refract neighboring light rays through slightly different angles. Depth below the surface then plays the role of elapsed time: rays that began nearly parallel cross and concentrate at selected points on the bottom. The particle model uses the same geometry, replacing ray angle by initial velocity and depth by time. In one dimension the concentrations are spots; in two dimensions they become the familiar bright curves.

9.10.1 One-Dimensional Derivation

Label each particle by its initial position x. Choose the initial line density to be uniform and set its normalization to one,

$$\frac{dn}{dx} = 1. \tag{9.25}$$

Let the particle initially at x have velocity $v(x)$. With no force, its later position is

$$y(x,t) = x + v(x)t. \tag{9.26}$$

Here x remains a particle label, while y is its physical position at time t.

Differentiate the map with respect to the initial label:

$$\frac{\partial y}{\partial x} = 1 + v'(x)t. \tag{9.27}$$

[38] Lecture timestamp: 01:50:31–01:50:47.

Particle number is conserved under the map, so the later density follows from the Jacobian:

$$\frac{dn}{dy} = \frac{dn}{dx}\frac{dx}{dy} \tag{9.28}$$

$$= \frac{1}{1+v'(x)t}. \tag{9.29}$$

The density becomes large where neighboring trajectories converge and the denominator approaches zero,

$$1 + v'(x)t = 0. \tag{9.30}$$

Equivalently, the first caustic appears at a sufficiently large positive value of $-v'(x)t$.

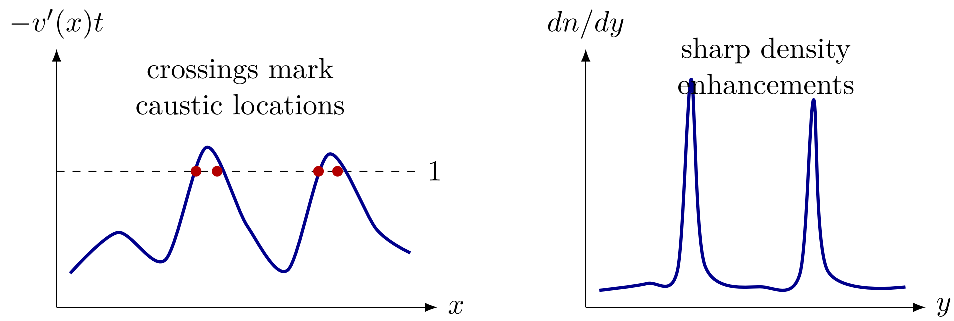

Figure 9.6: One-dimensional caustic formation. As time increases, peaks of $-v'(x)t$ cross unity; the Jacobian vanishes and the geometric single-stream density develops sharp enhancements.

The formula predicts an infinite density at the fold only in the ideal geometric, zero-dispersion, pre-shell-crossing description. Waves diffract, and particles possess finite velocity dispersion and interactions, so a physical caustic is sharp but not literally infinite. As t grows, the highest peak of $-v'$ reaches unity first; one crossing is born and then splits into two as the horizontal level cuts both sides of the peak. Further peaks generate additional pairs.

9.10.2 From Spots to the Cosmic Web

Dimensionality controls the appearance of the caustics. In one spatial dimension they are points. In two dimensions they are curves, with especially strong concentrations where curves meet. In three dimensions the generic folds are surfaces; surfaces meet along filaments, and filaments meet at knots. The regions between them are comparatively empty. This is the geometry seen in swimming-pool light patterns and, at a qualitative level, in the cosmic web.

The force-free construction is not a replacement for cosmological perturbation theory. Gravity is crucial to the growth of density contrast, halo formation, galaxies, and nonlinear structure. The toy model isolates one robust geometrical fact: a slightly varying initial flow naturally folds and concentrates. Small initial density variations could be included as well without changing the central Jacobian argument.

We have therefore moved the mystery rather than eliminated it. Inflation was introduced to stretch a region until it became extraordinarily flat and uniform. The caustic calculation assumes an initial field of tiny velocity variations. Where did that inhomogeneous seed come from? The next lecture turns to quantum fluctuations of fields during inflation and asks how they become the initial conditions for the later cosmic web.

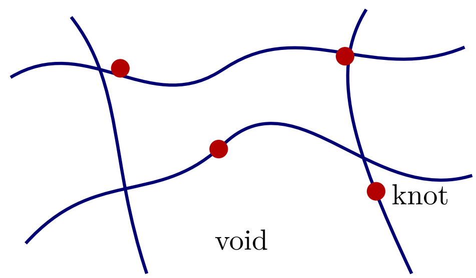

Figure 9.7: A clean two-dimensional analogy for the caustic network: line-like concentrations enclose voids and meet at brighter knots. In three dimensions the primary folds are sheets whose intersections form filaments and nodes.

9.11 What the Argument Has Established

A homogeneous scalar field behaves like a coordinate moving in the potential $V(\phi)$. In a fixed box it obeys $\ddot{\phi} = -V'(\phi)$. In an expanding comoving box, the physical volume a^3 enters the action and produces

$$\ddot{\phi} + 3H\dot{\phi} = -V'(\phi).$$

Hubble friction can hold the field to a slow terminal motion on a high, shallow potential. Its kinetic energy then becomes negligible, its potential energy keeps H nearly constant, and $a(t)$ grows approximately exponentially.

Expansion dilutes unwanted monopoles, smooths the geometry, and drives curvature toward zero. Yet inflation must end. The field leaves the plateau, oscillates, and transfers its energy to the hot particles of reheating. Several questions remain open: the form of the potential, the identity and initial state of the inflaton, its reheating couplings, and the tiny vacuum energy observed today.

Inflation's success creates the final question. A perfectly smoothed universe would contain no galaxies. The caustic map

$$y = x + v(x)t, \qquad \frac{dn}{dy} = \frac{1}{1 + v'(x)t}$$

shows how tiny variations in an initial flow can generate sharp sheets, filaments, and knots. It does not explain the origin of those variations. Their quantum origin is the subject to which the argument now turns.

Chapter 10

Quantum Fluctuations and the Origin of Structure

Overview. The final lecture asks how a universe made smooth by inflation can nevertheless acquire the tiny irregularities from which galaxies grow. The answer is built in stages. Universal constants first define the Planck scales and expose the enormous gap between familiar particle physics and quantum gravity. The microwave sky then supplies the target: temperature and density variations of order 10^{-5}, spread over many scales. Zero-point motion supplies unavoidable fluctuations, damping explains how an oscillation can freeze, and the expanding-universe wave equation turns every Fourier mode of the inflaton into just such a damped oscillator. Modes freeze when their physical wavelength reaches the Hubble scale. When inflation ends, the frozen variation makes neighboring regions reheat at slightly different times, converting microscopic quantum noise into the primordial density pattern.

10.1 Which Constants Are Universal?

We possess an extraordinarily accurate quantum theory of elementary particles. Its parameters include quark and lepton masses, coupling constants, mixing data, charges, and the list of particles themselves. Given unlimited calculating power, the theory would account not only for accelerator experiments but also for atoms, molecules, and chemistry; in principle the consequences continue into biology and the physical processes underlying thought. That does not make such systems easy to calculate. It says that the Standard Model is intended as a microscopic foundation for ordinary matter. Gravity is described separately, at the classical level, by general relativity. Both theories are powerful; neither makes every number appearing in low-energy physics look inevitable.

A proton-to-electron mass ratio is a perfectly good physical number, but it is tied to two particular particles. Replacing the proton by a neutron or the electron by a muon produces a different ratio. Such parameters may encode deep physics that we do not yet know, but they do not advertise their universality. Three dimensional constants are different:

$$c, \qquad \hbar, \qquad G.$$

Their defining laws repeatedly contain the word *all*. All matter and information respect the limiting speed c. All quantum systems obey uncertainty relations governed by $\hbar$. All forms of energy gravitate with the same Newton constant G. That breadth is what makes these constants fundamental in the present argument.

Planck recognized the importance of this combination before the full meaning of quantum mechanics was known. His constant first entered the description of blackbody radiation. Even then, the dimensional fact was visible: from c, $\hbar$, and G one can construct units of length, time, and mass without referring to any particular atom or particle.

10.2 Building the Planck Scales

Let L, T, and M denote the dimensions of length, time, and mass. The speed of light has dimensions

$$[c] = LT^{-1}.$$

The uncertainty relation $\Delta x\,\Delta p \gtrsim \hbar$, together with $p = mv$, gives

$$[\hbar] = ML^2T^{-1}.$$

Finally, Newton's equations

$$F = \frac{Gm_1m_2}{r^2}, \qquad F = ma$$

give

$$[G] = L^3M^{-1}T^{-2}.$$

This derivation matters because dimensional analysis is not guesswork: each exponent is fixed by requiring unwanted dimensions to cancel.

Seek a length of the form

$$\ell = c^{\alpha}\hbar^{\beta}G^{\gamma}. \tag{10.1}$$

The mass exponent must vanish, so

$$\beta - \gamma = 0.$$

The time exponent must vanish, and the length exponent must equal one:

$$-\alpha - \beta - 2\gamma = 0, \qquad \alpha + 2\beta + 3\gamma = 1.$$

Using $\gamma = \beta$, these become

$$\alpha + 3\beta = 0, \qquad \alpha + 5\beta = 1.$$

Their difference gives $2\beta = 1$, hence

$$\beta = \gamma = \frac{1}{2}, \qquad \alpha = -\frac{3}{2}.$$

Therefore the unique length made from the three universal constants is

$$\ell_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^3}} \approx 1.6 \times 10^{-35}\,\mathrm{m}. \tag{10.2}$$

The lecture keeps only the order of magnitude, 10^{-35} m, but the dimensional derivation fixes the combination itself.

The associated time is the light-crossing time:

$$t_{\mathrm{P}} = \frac{\ell_{\mathrm{P}}}{c} = \sqrt{\frac{\hbar G}{c^5}} \approx 5.4 \times 10^{-44}\,\mathrm{s}. \tag{10.3}$$

At order-of-magnitude accuracy this is 10^{-43} s. It is not an independent construction once the Planck length and the universal speed are known.

For the mass, use the dimensional relation $Et \sim \hbar$ and $E = mc^2$. Taking $t = t_{\mathrm{P}}$ gives

$$m_{\mathrm{P}} = \frac{\hbar}{c^2 t_{\mathrm{P}}} = \sqrt{\frac{\hbar c}{G}} \approx 2.2 \times 10^{-8}\,\mathrm{kg}. \tag{10.4}$$

This is surprisingly macroscopic by particle standards, roughly the mass of a fine dust mote. Its rest energy is of order 10^9 J, comparable in scale to the chemical energy in a tank of fuel. A car can travel hundreds of miles on only a few tanks because combustion releases merely a minute fraction of the fuel's rest energy. Converting one Planck mass completely into energy would be an entirely different affair. A tiny mass can represent an enormous energy when multiplied by c^2.

The contrast with accelerator physics is severe. A distance of order 10^{-18} m, characteristic of the shortest scales probed in high-energy collisions at the time of the lecture, is still about 10^{17} Planck lengths. The unknown territory between the electroweak and Planck scales is therefore vast. Nuclear physics offers the right analogy. A nucleus looks elementary from far away, but at shorter distances it resolves into protons and neutrons, and these in turn resolve into quarks and gluons. Low-energy particles may be simple compared with nuclei yet still be complicated emergent objects relative to whatever lies at the Planck scale. The many parameters of the Standard Model are less surprising when viewed across that hierarchy, though their observed values and fine tunings remain unexplained.

Question from the audience. Is the Higgs lighter or heavier than expected, and why is its mass puzzling?

Response. There are two comparisons. Relative to the Planck mass, a 125 GeV Higgs is extraordinarily light. In a theory with no protective structure, quantum corrections tend to make a scalar mass sensitive to much higher scales, so obtaining the observed value looks like balancing parameters on a narrow edge. Relative to some simple supersymmetric expectations discussed at the time, the observed Higgs could instead be called somewhat heavy. The apparent contradiction comes from comparing it with two different reference scales; the underlying hierarchy problem remains. [1]

Question from the audience. If the Higgs helps give particles mass, should dark-matter particles have masses near the Planck scale?

Response. No. A WIMP is a *weakly interacting massive particle*; the phrase points to an interaction scale that might make laboratory detection possible, not to the Planck mass. Under the usual WIMP assumptions, masses can be well above the proton mass and still be fantastically below m_{P}. If the interaction with ordinary matter is allowed to be arbitrarily weaker, cosmology permits a much wider mass range, but direct discovery correspondingly becomes harder. The proton and neutron also provide a useful warning: most of their mass is binding energy from the strong interaction, not simply a sum of Higgs-generated quark masses. [2]

[1] Lecture timestamp: 00:26:45–00:30:54.
[2] Lecture timestamp: 00:30:55–00:33:11.

Question from the audience. What does a Planck mass signify, and is there a fundamental energy per bit?

Response. A useful gravitational interpretation is that a black hole cannot be made parametrically smaller than the Planck scale while remaining a conventional semiclassical black hole. Such an object has mass of order m_{P}, horizon area of order one Planck area, and entropy of order one unit. In that restricted black-hole sense, one may associate an order-Planck energy with order-one information. This is not a universal claim that every stored bit in every device costs a Planck energy; ordinary particles and ordinary memories can be far lighter. [3]

From here it is convenient to use Planck units,

$$c = \hbar = G = 1. \tag{10.5}$$

Lengths, times, masses, and energies can then be quoted as pure numbers measured against equations (10.2)–(10.4).

10.3 The Nearly Uniform Microwave Sky

The cosmological problem begins with a microwave map of the whole sky. WMAP, the Wilkinson Microwave Anisotropy Probe, measured radiation released near the epoch when the primordial plasma became transparent. The photons have since been redshifted by roughly a factor of one thousand and now form an almost perfect blackbody at about 2.7 K. We do not see the universe at one common present time when we look outward. Greater distance means earlier emission, and the microwave background is the distant shell from which photons first streamed freely toward us.

Question from the audience. What exactly does the oval WMAP image represent?

Response. Every point in the oval represents a direction on the sky, with the sphere flattened by a map projection. The left and right edges join, the upper and lower portions represent opposite galactic hemispheres, and the galactic plane runs across the middle. Foreground microwave emission from our own Galaxy must be modeled and subtracted. What remains is not a map of nearby stars but a directional map of the microwave temperature on the last-scattering shell. [4]

To first approximation the result is isotropic:

$$T(\hat{\mathbf{n}}) \simeq 2.7\,\mathrm{K}$$

for every viewing direction $\hat{\mathbf{n}}$. Perfect isotropy around every point would imply perfect homogeneity. But a perfectly homogeneous universe would not spontaneously produce galaxies, clusters, and the caustic-like cosmic web discussed in the previous lecture. Some inhomogeneity must have been present, however small its amplitude.

Write the directional temperature departure as

$$\delta T(\hat{\mathbf{n}}) = T(\hat{\mathbf{n}}) - \bar{T}.$$

[3]Lecture timestamp: 00:33:11–00:35:10.
[4]Lecture timestamp: 00:37:49–00:40:49.

The measured fractional variation is of order

$$\frac{\delta T}{T} \sim 10^{-5}. \tag{10.6}$$

One route to this number was dynamical: evolve today's matter distribution backward and ask what initial contrast is needed at decoupling. The other was direct: amplify the nearly featureless microwave map by five orders of magnitude and measure the mottling itself. The two routes agree at the level needed for this argument.

The mottling occurs on many angular scales. Large patches contain smaller patches, and magnifying a region gives a statistically similar texture. The term *scale invariant* does not mean that every scale is literally identical; it means that the fluctuation power changes only weakly with scale over the relevant range. That near scale invariance is part of the evidence the theory must explain, not merely the overall amplitude.

Why should temperature trace matter? A photon escaping a gravitational potential well loses energy. Radiation arriving from a denser region can therefore be shifted relative to radiation from a less dense region. At the lecture's order-of-magnitude level this motivates

$$\frac{\delta \rho}{\rho} \sim 10^{-5} \tag{10.7}$$

near decoupling. A precise CMB calculation includes acoustic dynamics, gravitational potentials, and transfer functions, so equation (10.7) is a scale-setting shorthand rather than an exact equality between $\delta T/T$ and $\delta\rho/\rho$.

Question from the audience. Could the mottling be ordinary microscopic noise generated at last scattering?

Response. Not on the observed angular scales. The patches corresponded to enormous physical regions by decoupling, not atomic-scale disturbances in the recombining plasma. Tracing them much farther backward, however, makes their physical wavelengths smaller. Inflation supplies the missing magnification: fluctuations that began as quantum, microscopic disturbances can be stretched to astronomical scales. [5]

The mechanism can be understood from three pieces of physics: zero-point motion of a harmonic oscillator, damping of an oscillator, and the wave equation for a scalar field in an expanding universe. We shall assemble them in that order.

10.4 Zero-Point Motion of an Oscillator

For a unit-mass harmonic oscillator,

$$E = \frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2, \qquad \ddot{x} + \omega^2 x = 0. \tag{10.8}$$

Kinetic and potential energies continually exchange. Averaged over an oscillation, each carries half the total energy.

[5] Lecture timestamp: 00:46:42–00:48:02.

Quantum mechanics gives the energy levels

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega. \tag{10.9}$$

The ground state cannot have both $x = 0$ and $\dot{x} = 0$, because that would assign exact position and momentum simultaneously. Its irreducible energy,

$$E_0 = \frac{1}{2}\hbar\omega,$$

is the zero-point energy. The word *oscillation* should be understood statistically: a stationary ground state does not trace a classical sinusoidal orbit, but it has nonzero variances in position and momentum.

Half the ground-state energy is in the potential term on average:

$$\frac{1}{2}\omega^2\langle x^2\rangle = \frac{1}{4}\hbar\omega.$$

Thus

$$\langle x^2\rangle = \frac{\hbar}{2\omega} \sim \frac{\hbar}{\omega}. \tag{10.10}$$

Ignoring factors of two, a complex notation for the scale of the fluctuation is

$$x(t) \sim \sqrt{\frac{\hbar}{\omega}}\, e^{i\omega t}.$$

Differentiation then gives

$$\langle \dot{x}^{\,2}\rangle \sim \hbar\omega, \tag{10.11}$$

consistent with equal average kinetic and potential energies. Cooling can remove excitations with $n > 0$, but it cannot remove these ground-state variances.

10.5 Damping, Criticality, and Freezing

Immerse the oscillator in a viscous medium. A spring and mass moving through honey obey

$$\ddot{x} + \gamma\dot{x} + \omega^2 x = 0, \tag{10.12}$$

where $\gamma > 0$ measures the drag. Looking for $x = e^{\alpha t}$ yields

$$\alpha^2 + \gamma\alpha + \omega^2 = 0, \tag{10.13}$$

and therefore

$$\alpha_\pm = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega^2}}{2}. \tag{10.14}$$

The two roots provide the two independent solutions required by a second-order differential equation.

Question from the audience. Should the drag term in equation (10.12) carry a minus sign?

Response. The drag force itself is $F_{\text{drag}} = -\gamma\dot{x}$. Newton's equation is $\ddot{x} = -\gamma\dot{x} - \omega^2 x$. Moving both force terms to the left gives equation (10.12) with plus signs. The sign in the differential equation and the direction of the physical force are therefore consistent. [6]

[6] Lecture timestamp: 00:59:38–01:00:10.

If $\gamma^2 < 4\omega^2$, the roots are complex. Define

$$\Omega = \sqrt{\omega^2 - \frac{\gamma^2}{4}}.$$

Then

$$x(t) = e^{-\gamma t/2} \left(A \cos \Omega t + B \sin \Omega t\right). \tag{10.15}$$

The oscillator still crosses the equilibrium point repeatedly, but its envelope decays. This is the underdamped regime.

If $\gamma^2 > 4\omega^2$, both roots in equation (10.14) are real and negative:

$$x(t) = Ae^{\alpha_+ t} + Be^{\alpha_- t}. \tag{10.16}$$

There is no sustained oscillation; both components decay. This is the overdamped regime. At

$$\gamma = 2\omega \tag{10.17}$$

the roots coincide, marking critical damping, the crossover between the two behaviors.

One limiting case is especially important. With no restoring force, $\omega = 0$,

$$\alpha(\alpha + \gamma) = 0,$$

so

$$x(t) = c_0 + c_1 e^{-\gamma t}. \tag{10.18}$$

The velocity dies, but the coordinate does not have to return to zero. It freezes at the constant c_0, selected by the initial conditions. A stone pushed through honey with no spring attached comes to rest wherever the drag exhausts its motion.

Now let the frequency vary slowly. Begin with $\omega \gg \gamma$, so the motion oscillates. Reduce $\omega(t)$ while keeping the damping scale approximately fixed. The oscillation slows, crosses $\gamma = 2\omega$, becomes overdamped, and approaches the force-free form as $\omega \to 0$. Equivalently, one may raise the damping relative to the restoring force. In the cosmological application the decisive change will be the fall of the effective frequency while the Hubble damping remains nearly constant. The final frozen value cannot be inferred from the crossover alone; it follows from the complete earlier evolution and initial quantum state.

10.6 Scalar Waves in an Expanding Universe

Return to the inflaton ϕ. Its potential is assumed to contain a high, broad plateau followed by a steep descent into a lower minimum. On the plateau the important approximation is

$$V'(\phi) \simeq 0, \tag{10.19}$$

not $V(\phi) \simeq 0$. The slope is small while the energy density can remain large enough to drive inflation. The detailed origin of this shape is not supplied; it is a phenomenological requirement of the model.

For a nearly free scalar in flat spacetime,

$$\frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi = 0. \tag{10.20}$$

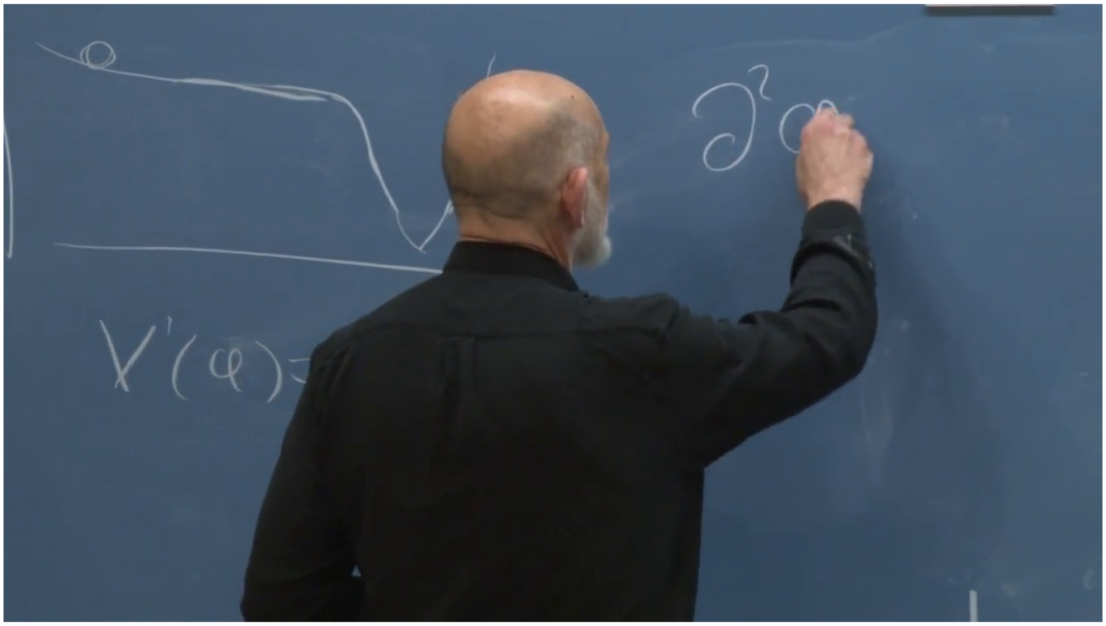

Figure 10.1: The inflaton plateau and cliff remain visible at left while the flat-space wave equation is begun at right. The board securely supports the approximate $V'(\phi) = 0$ condition and the start of a second-derivative term, but the lecturer obscures the unfinished equation; equation (10.20) is reconstructed from the spoken derivation.

Lecture frame: 01:17:17.865.

A nonflat potential would add $-V'(\phi)$ on the right. For the plateau approximation we omit it. If the field were perfectly homogeneous, the spatial derivative would vanish. In an expanding universe the remaining homogeneous equation would be

$$\ddot{\phi} + 3H\dot{\phi} = 0,$$

the Hubble-friction equation of the previous lecture.

For an inhomogeneous field, coordinate distance and physical distance must be separated. Choose dimensionless comoving labels $\mathbf{x}$, so an infinitesimal proper distance is

$$d\boldsymbol{\ell} = a(t)\,d\mathbf{x}.$$

At fixed time,

$$\nabla_{\text{proper}} = \frac{1}{a}\nabla_{\mathbf{x}}, \qquad \nabla^2_{\text{proper}} = \frac{1}{a^2}\nabla^2_{\mathbf{x}}.$$

The scalar equation is therefore

$$\ddot{\phi} + 3H\dot{\phi} - \frac{1}{a^2}\nabla^2\phi = 0, \tag{10.21}$$

where ∇ differentiates with respect to the comoving labels. If instead one chooses a dimensionless scale factor, the comoving coordinates carry length; the physics is unchanged.

Question from the audience. Should the spatial term in equation (10.21) be positive?

Response. No. The flat-space wave operator already has the relative sign $\partial_t^2 - \nabla^2$. Expansion changes the spatial derivative by the factor a^{-2}, but it does not reverse that sign. After a Fourier

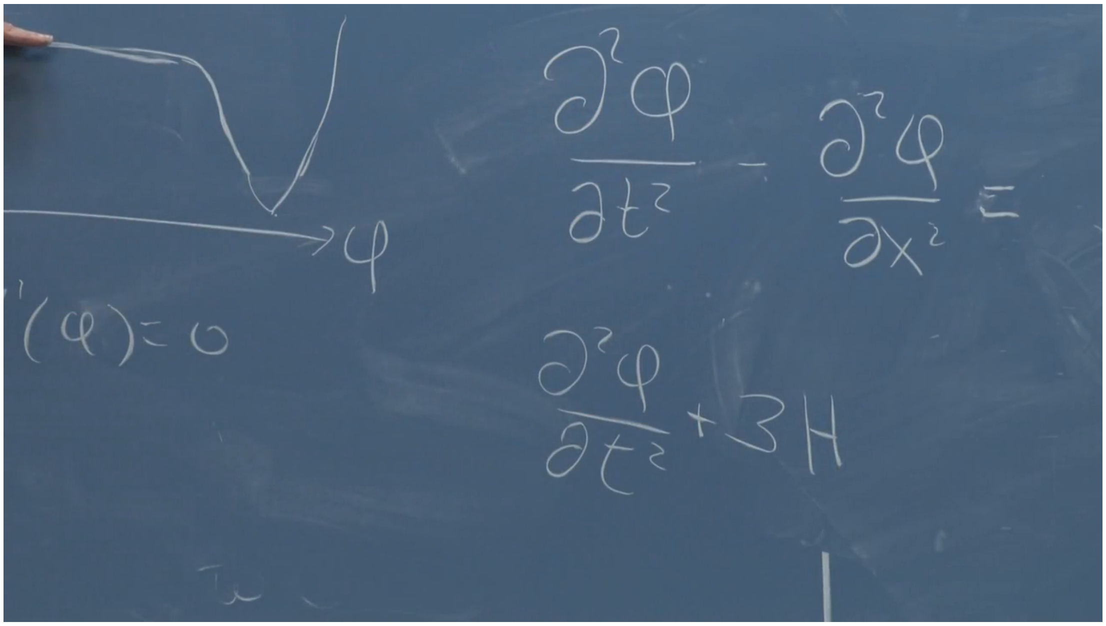

Figure 10.2: The plateau is indicated at left while successive stages of the wave equation appear at right. The upper flat-space operator is readable; the lower line reaches $+3H$ but is unfinished, so the complete damping term is supplied from the contemporaneous explanation.

Lecture frame: 01:18:57.554.

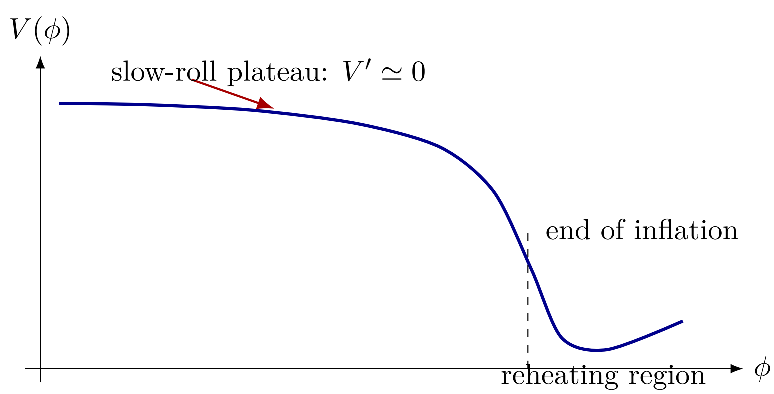

Figure 10.3: Clean reconstruction of the potential used throughout the argument. A shallow slope permits slow roll while the high value of V keeps H nearly constant; the steep descent ends inflation and releases the stored energy.

substitution the two minus signs combine, and the mode equation acquires the expected positive restoring term $+k^2\phi_k/a^2$. [7]

10.7 Each Fourier Mode Is a Damped Oscillator

Study one spatial wave at a time:

$$\phi(\mathbf{x}, t) = \phi_{\mathbf{k}}(t)e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{10.22}$$

The complex exponential is bookkeeping for sines and cosines. An arbitrary real profile is built by superposing modes with the appropriate reality relation between $\mathbf{k}$ and $-\mathbf{k}$.

Time derivatives act only on $\phi_{\mathbf{k}}(t)$, while

$$\nabla^2 e^{i\mathbf{k}\cdot\mathbf{x}} = -k^2 e^{i\mathbf{k}\cdot\mathbf{x}}.$$

Substitution into equation (10.21) gives

$$\ddot{\phi}_{\mathbf{k}} + 3H\dot{\phi}_{\mathbf{k}} + \frac{k^2}{a^2}\phi_{\mathbf{k}} = 0. \tag{10.23}$$

Thus every wave number behaves like a damped oscillator with

$$\gamma = 3H, \qquad \omega_k(t) = \frac{k}{a(t)}. \tag{10.24}$$

The physical wave number k/a redshifts as the universe expands. Consequently the effective restoring force k^2/a^2 decreases toward zero.

A mode that begins at short physical wavelength has $\omega_k \gg H$ and oscillates. Its wavelength is stretched, its frequency falls, and Hubble damping eventually dominates. It then approaches a nonoscillating value rather than being pulled back to zero. Different $\mathbf{k}$ modes cross over at different times: a larger k, corresponding at fixed a to a shorter wavelength, must wait for more expansion before its physical frequency falls to the Hubble scale.

10.8 Horizon Exit and Frozen Vacuum Fluctuations

Using the oscillator's critical condition as an order-one estimate,

$$\gamma = 2\omega_k,$$

equation (10.24) gives

$$3H = \frac{2k}{a}, \qquad a_{\text{cross}} = \frac{2k}{3H}. \tag{10.25}$$

The numerical factors are not the conceptual point. The crossover occurs when the physical wave number k/a is of order H.

Let

$$\lambda_{\text{com}} = \frac{2\pi}{k}, \qquad \lambda_{\text{phys}} = a\lambda_{\text{com}} = \frac{2\pi a}{k}.$$

[7] Lecture timestamp: 01:22:18–01:23:06.

A dimensional check fixes the placement of a: with dimensionless $\mathbf{x}$, both k and $\mathbf{k}\cdot\mathbf{x}$ are dimensionless, while a and λ_{phys} carry length. Equation (10.25) therefore implies

$$\lambda_{\text{phys}} \sim H^{-1}, \tag{10.26}$$

up to factors such as 2, 3, and 2π.

The Hubble law $v = Hd$ associates the distance H^{-1} with a recession speed of order $c = 1$. During inflation, H is large and nearly constant, so this Hubble length is microscopic even though it plays the role of a causal scale. Calling the event *horizon exit* is conventional: no material object passes through a physical wall. The mode's physical wavelength grows beyond the approximately fixed Hubble scale, and its dynamics changes from rapid oscillation to an almost frozen amplitude.

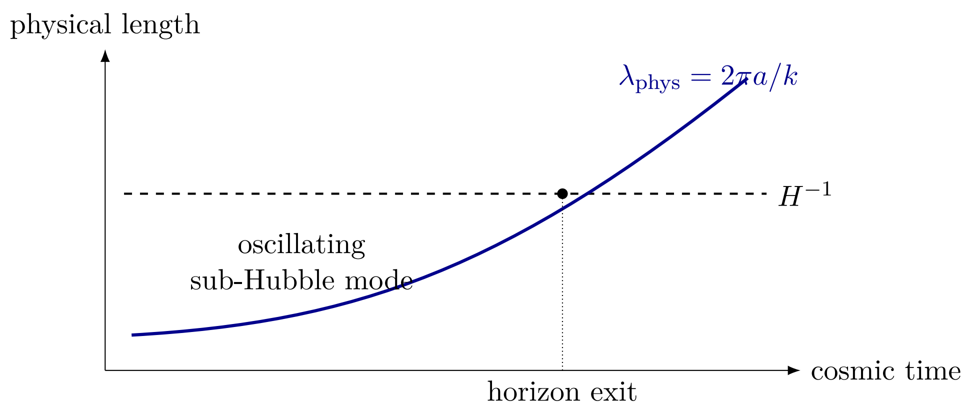

Figure 10.4: During nearly exponential expansion, a physical wavelength grows while the Hubble length remains approximately fixed. Their crossing marks the transition from rapid oscillation to a frozen mode, up to order-one factors.

Vacuum fluctuations prevent inflation from simply erasing every wave. Every short-wavelength mode begins with zero-point motion. Expansion stretches one mode to the Hubble scale and freezes it, while still shorter vacuum modes continue to oscillate. They in turn stretch and freeze. Repetition builds a field profile containing frozen contributions over a broad range of wavelengths and orientations. The result is stochastic but not structureless: its statistics follow from the quantum state and the inflationary background.

Question from the audience. Do the modes freeze at a point or at a wavelength, and is H itself growing exponentially?

Response. They freeze when a wavelength reaches the Hubble scale. On the nearly flat plateau, V and hence H are approximately constant; it is the scale factor $a(t)$, and therefore the physical wavelength a/k, that grows exponentially. Each comoving mode reaches the condition $k/a \sim H$ at its own time. [8]

Question from the audience. Are these electromagnetic waves?

Response. No. They are fluctuations of the inflaton scalar field. Electromagnetic vacuum fluctuations provide a familiar analogy, but the variable in equation (10.23) is ϕ. [9]

[8] Lecture timestamp: 01:45:06–01:45:39.
[9] Lecture timestamp: 01:45:39–01:46:04.

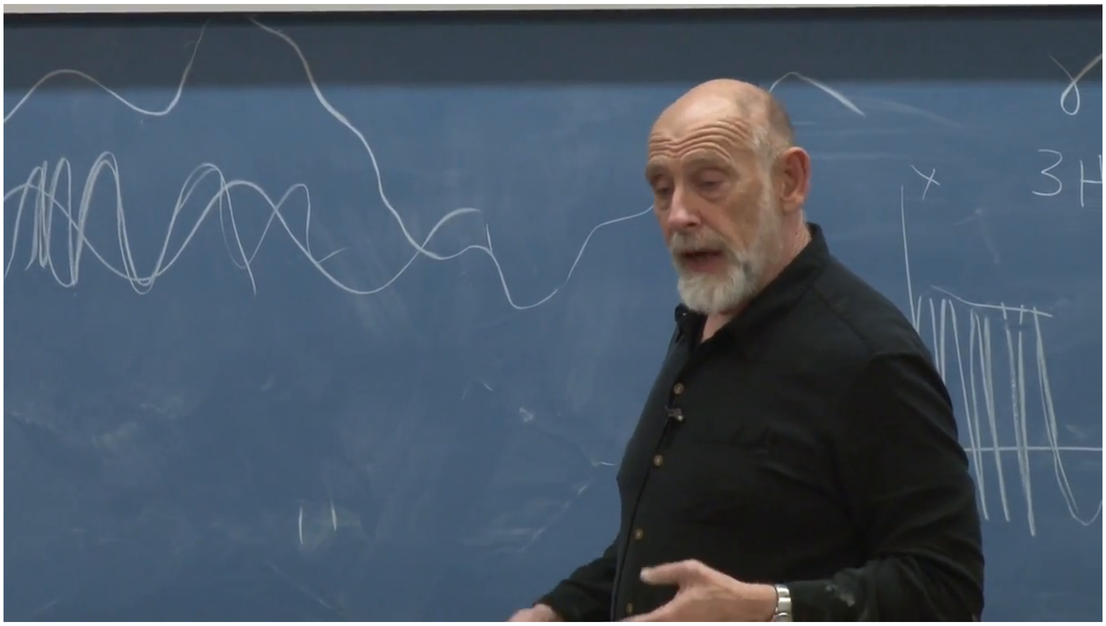

Figure 10.5: A board-wide superposition of long and short field waves, with a smaller damping sketch beside the labels γ and $3H$. The frame documents the qualitative mode picture; it is not used to read off an exact function.

Lecture frame: 01:45:28.476.

Question from the audience. Is there a continuum of wavelengths?

Response. For the infinite spatial idealization, yes. The field is represented by a Fourier integral,

$$\phi(\mathbf{x},t) = \int \frac{d^3k}{(2\pi)^3}\,\phi_{\mathbf{k}}(t)e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{10.27}$$

Every wave number undergoes the same redshifting and freezing mechanism. [10]

Question from the audience. Would a finite universe have only discrete modes?

Response. Yes. Boundary conditions on a finite spatial volume quantize the allowed wave numbers. For a volume much larger than the scales under discussion, the spacing can be so fine that the sum is well approximated by the integral in equation (10.27). The continuous picture is an idealization, not a different local mechanism. [11]

Two motions now coexist. The spatially averaged field $\bar{\phi}(t)$ rolls slowly and classically down the potential. On top of it sits a small quantum fluctuation,

$$\phi(\mathbf{x},t) = \bar{\phi}(t) + \delta\phi(\mathbf{x},t),$$

assembled from the frozen modes. The fluctuation energy itself is too small to be the observed matter contrast. Its importance is that it perturbs the time at which inflation ends.

[10] Lecture timestamp: 01:46:04–01:46:34.
[11] Lecture timestamp: 01:46:34–01:47:01.

A full single-field quantum calculation sharpens this oscillator argument. A light scalar acquires a typical fluctuation per logarithmic interval of wave number of order

$$\delta\phi_{\rm exit} \sim \frac{H}{2\pi}. \tag{10.28}$$

A derivation of the normalization requires quantizing the field modes in the inflationary background. The oscillator picture has already supplied the essential mechanism: vacuum fluctuations exist even in the ground state, and expansion converts them into a classical-looking frozen pattern.

10.9 The Cliff Converts Field Noise into Density Noise

First suppress $\delta\phi$. At any fixed time the field then has one value everywhere, represented on the board by a vertical line when space x is drawn upward and field value ϕ horizontally. As time advances, the line moves toward the cliff. The whole universe reaches the drop together, inflaton potential energy becomes particles, radiation, and heat, exponential expansion ends, and the resulting energy density begins to dilute with ordinary expansion.

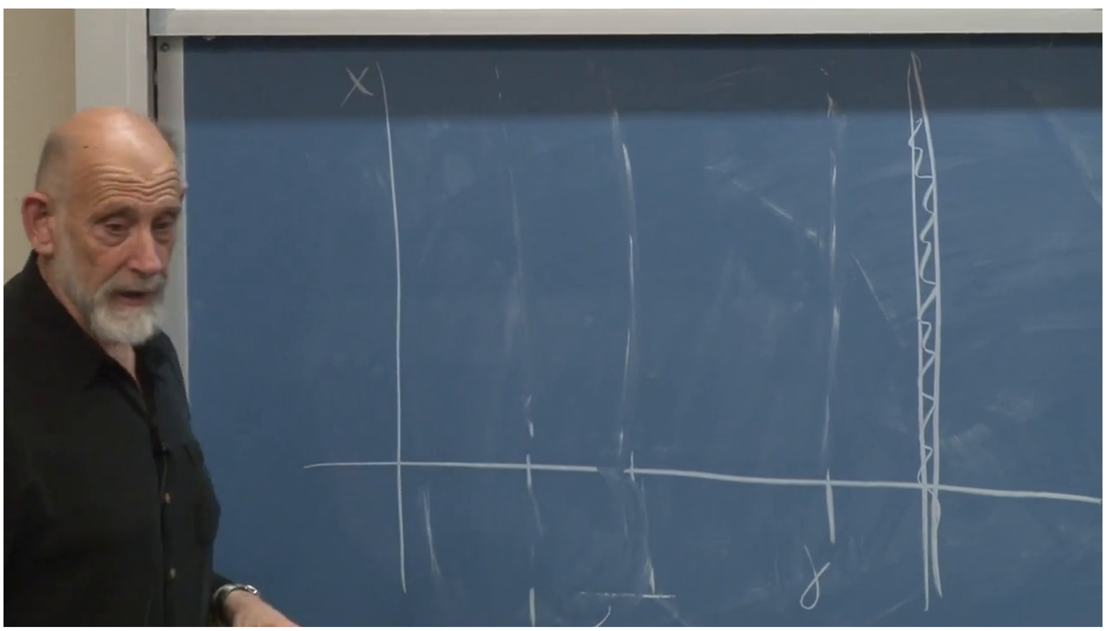

Figure 10.6: The blackboard's space-versus-field diagram. The vertical axis is x, the horizontal direction is ϕ, faint near-vertical profiles mark successive field configurations, and the hatched boundary at right is the cliff where inflation ends.

Lecture frame: 01:50:54.502.

Restore the frozen variation. The line is now slightly wavy: neighboring places have slightly different values of ϕ. Think of a rank of people trying to march shoulder to shoulder toward a cliff. A few are a little ahead, a few a little behind. The ones who are ahead go over first.

That timing offset is the real source of the density contrast. A region that goes over first reheats first. Its potential energy is dumped into particles and heat, and then that energy has time to dilute while the universe expands in the post-inflationary era. A nearby region that goes over slightly later reheats later, so when the two are compared, the later region is denser simply because it has had

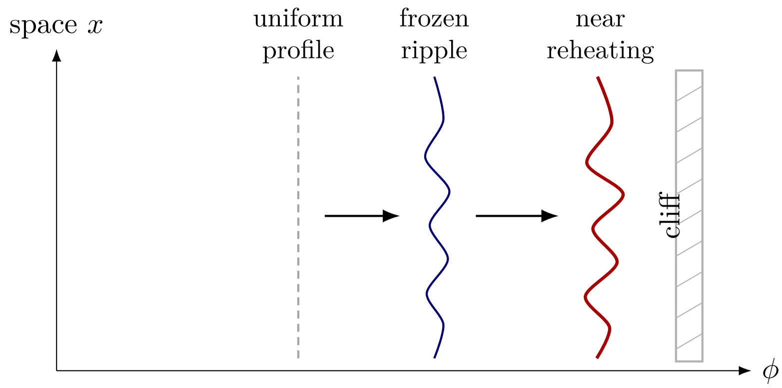

Figure 10.7: Clean reconstruction of the space-versus-field picture. The mean field moves toward the cliff while a frozen spatial ripple persists, so different positions reach the end of inflation at slightly different times.

less time to dilute. The first set of crossings therefore leaves relatively dense pockets. While those pockets expand, shorter-wavelength ripples bring another set of positions over the edge and make smaller pockets inside the larger pattern. Repeating this across the full hierarchy of frozen modes produces structure within structure. Thus the tiny field fluctuation is not important because of its own energy; it is important because it shifts the local time at which reheating happens.

The timing relation can be written schematically. If the homogeneous field reaches the end value ϕ_{end} with velocity $\dot{\bar{\phi}}$, then a small field displacement changes the local end time by

$$\delta t_{\text{end}} \simeq -\frac{\delta\phi}{\dot{\bar{\phi}}}. \tag{10.29}$$

A slower roll makes the same $\delta\phi$ correspond to a larger delay. During that extra time the earlier region expands and dilutes more. Up to order-one transfer factors, the curvature or density perturbation is therefore controlled by

$$\mathcal{R} \sim H\,\delta t_{\text{end}} \sim -\frac{H\,\delta\phi}{\dot{\bar{\phi}}}. \tag{10.30}$$

Combining this with equation (10.28) gives the familiar slow-roll scale

$$\mathcal{P}_{\mathcal{R}}^{1/2} \sim \frac{H^2}{2\pi|\dot{\bar{\phi}}|}, \tag{10.31}$$

with precise conventions and subsequent transfer to $\delta\rho/\rho$ requiring a fuller perturbation calculation. The slower the field rolls, the longer the time between neighboring cliff crossings and the greater the intervening dilution. The same field fluctuation therefore leaves a larger density imprint when $|\dot{\bar{\phi}}|$ is smaller.

Many modes cross the Hubble scale at different times, so many differently sized regions reheat at slightly different times. The resulting density field contains overdense and underdense pockets nested across scales. Its statistics depend on the height, slope, and curvature of the inflationary potential. Suitable slow-roll models naturally produce a nearly scale-invariant amplitude near

$$\frac{\delta\rho}{\rho} \sim 10^{-5},$$

and detailed CMB spectra allow a small number of cosmological parameters to be tested against many angular scales.

10.10 From Primordial Seeds Back to the Sky

We can now return to the observer at the center of a set of past-light shells. The last-scattering sphere cuts through the three-dimensional primordial density field. Directions in which that shell intersects different gravitational potentials produce slightly different microwave temperatures. WMAP displays the two-dimensional pattern on that sphere. The map is neither a photograph of time zero nor a projection of every structure inside the sphere; it is the temperature structure on the particular shell from which the microwave photons last scattered.

Inside the shell, the same primordial perturbations continue evolving. Gravity amplifies overdensities, caustic-like flows sharpen sheets and filaments, gas collects, and galaxies form. Thus the CMB pattern and the later cosmic web are two stages of one history: frozen inflationary fluctuations first appear as tiny density seeds and later become visible structure.

> **Question from the audience.** What epoch does the microwave background we see today represent?
>
> **Response.** The photons have traveled for roughly 13 billion years, but they were released a few hundred thousand years after the hot Big Bang, when electrons and protons combined into neutral hydrogen and the universe became transparent. Inflation ended vastly earlier. Reheating created the hot matter and radiation; the universe then expanded and cooled until last scattering. For this argument the robust statement is the order of magnitude—a few hundred thousand years—rather than a precise recombination date. [12]

10.11 One Sky and Cosmic Variance

> **Question from the audience.** Does inflation predict the reported preferred direction or north–south asymmetry in WMAP?
>
> **Response.** No such preferred direction follows from the simple inflationary picture developed here. The status of the reported large-angle anomalies was uncertain, foregrounds and statistical fluctuations were live concerns, and a responsible assessment required the specialist literature rather than a sensational summary. Inflation predicts statistical properties of an ensemble; deciding whether one unusual-looking sky feature is new physics is harder. [13]

Small angular features occur many times across the sky. They provide many samples, so their distribution and isotropy can be tested statistically. A quadrupole-sized feature is different: only a few independent large-angle modes fit on the observable sphere. Their measured power can therefore differ substantially from the ensemble mean even when the underlying theory is correct. This irreducible sampling uncertainty is *cosmic variance*.

[12] Lecture timestamp: 02:00:38–02:02:19.
[13] Lecture timestamp: 02:02:19–02:03:26.

Question from the audience. What about the apparent alignment of several low multipoles?

Response. It could be a clue, a foreground or analysis artifact, or an unlikely but allowed realization. The difficulty is that we cannot draw another observable universe from the same statistical ensemble. A coin giving four heads in six tosses need not be biased; repeated batches would decide, but cosmology supplies only one batch on the largest scales. The anomaly may deserve study, yet one sky cannot by itself provide the repeated trials needed for a decisive significance test. [14]

10.12 From Quantum Motion to Cosmic Structure

Three universal constants define the Planck scales and expose how far familiar particle physics lies from quantum gravity. The microwave sky supplies a precise cosmological target: near isotropy, a fluctuation amplitude around 10^{-5}, and structure spread over many scales. Quantum mechanics ensures that every field mode has zero-point fluctuations. Inflation stretches each inflaton mode until its physical wavelength is of order H^{-1}, at which point Hubble damping suppresses its oscillation and leaves an effectively frozen amplitude.

The homogeneous inflaton continues to roll beneath that frozen noise. When the field reaches the end of its plateau, neighboring regions do not reheat at exactly the same time. Earlier regions dilute longer; later regions remain denser. This converts $\delta\phi$ into a primordial density and curvature pattern. Gravity then amplifies the pattern into galaxies and the cosmic web, while the last-scattering shell preserves its earlier state in the microwave sky. Microscopic zero-point motion, cosmic expansion, and staggered reheating are therefore not separate stories. Together they explain how an inflationary universe can be both extraordinarily smooth and structured enough for us to exist.

[14] Lecture timestamp: 02:05:31–02:07:36.